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**Clinical-Conditioned LoRA: Leveraging Cardiac
MRI Foundation Models for Left Ventricular
Hypertrophy Differential Diagnosis**

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Abstract

Cardiac magnetic resonance imaging (CMR) is the gold standard for non-invasive evaluation of cardiac structure and function, and plays a central role in the differential diagnosis of left ventricular hypertrophy (LVH) etiologies - including hypertrophic cardiomyopathy, hypertensive cardiomyopathy, and cardiac amyloidosis - conditions that share overlapping imaging phenotypes yet require fundamentally different clinical management. Recent advances in vision foundation models pre-trained on large-scale CMR cohorts have opened new possibilities for automated cardiac image analysis. However, the integration of patient-level clinical information within these models remains relatively underexplored, particularly in the context of LVH differential diagnosis, where clinical context plays an important role in routine practice.

In this work, we leverage CineMA - a state-of-the-art CMR vision foundation model pre-trained on 74,916 UK Biobank studies - to address the task of LVH etiology classification, and propose Clinical-Conditioned LoRA (CC-LoRA), a novel method that makes the foundation model clinically aware by injecting patient clinical features directly into the image representation learning process rather than appending them at the output. CC-LoRA extends the Low-Rank Adaptation (LoRA) framework by introducing Feature-wise Linear Modulation (FiLM) conditioning into the LoRA adapter outputs of selected transformer blocks within the CineMA backbone. A shared clinical MLP maps patient features to a compact embedding that generates per-block scaling (γ) and shift (β) parameters, modulating how the frozen backbone processes imaging features during the forward pass. The backbone remains fully frozen throughout training, with only 0.258% of the total 89.5 million parameters updated in the best-performing 4-block configuration.

The proposed CC-LoRA framework was primarily evaluated on a private cardiac MRI cohort from Hospital de la Santa Creu i Sant Pau, Barcelona, comprising patients with different etiologies of left ventricular hypertrophy together with healthy control subjects, and including 27 structured clinical features per patient. To validate the implementation framework and facilitate reproducibility, additional experiments were conducted on the public ACDC and M&Ms cardiac MRI benchmarks. On the Sant Pau cohort, the 4-block CC-LoRA configuration achieved a mean AUC of $91.03 \pm 4.17\%$ and a balanced accuracy of $71.46 \pm 9.23\%$ across 5-fold cross-validation, outperforming image-only baselines by 15 - 17 AUC points, suggesting that clinical and imaging information provide complementary diagnostic value in this setting. In contrast, no consistent improvement was observed on ACDC and M&Ms, where only a limited number of clinical variables were available. This observation suggests that the effectiveness of the proposed approach may depend on the availability of informative patient-level clinical features.

These results suggest that clinical conditioning substantially benefits foundation model fine-tuning when rich patient-level data is available, and highlights CC-LoRA as a parameter-efficient and principled approach to multimodal cardiac image analysis for LVH differential diagnosis.

Resumen

La resonancia magnética cardíaca (RMC) es el estándar de referencia para la evaluación no invasiva de la estructura y función cardíaca, y desempeña un papel central en el diagnóstico diferencial de las etiologías de la hipertrofia ventricular izquierda (HVI) - incluyendo la miocardiopatía hipertrófica, la miocardiopatía hipertensiva y la amiloidosis cardíaca - condiciones que comparten fenotipos de imagen superpuestos pero requieren un manejo clínico fundamentalmente diferente. Los recientes avances en modelos de visión fundacionales preentrenados en grandes cohortes de RMC han abierto nuevas posibilidades para el análisis automatizado de imágenes cardíacas. Sin embargo, la integración de información clínica a nivel del paciente en estos modelos sigue siendo relativamente poco explorada, especialmente en el contexto del diagnóstico diferencial de la HVI, donde el contexto clínico desempeña un papel importante en la práctica habitual.

En este trabajo, aprovechamos CineMA - un modelo de visión fundacional de RMC de última generación preentrenado en 74.916 estudios del UK Biobank - para abordar la tarea de clasificación de etiologías de HVI, y proponemos Clinical-Conditioned LoRA (CC-LoRA), un método novedoso que dota al modelo fundacional de conciencia clínica mediante la inyección directa de las características clínicas del paciente en el proceso de aprendizaje de representaciones de imagen, en lugar de añadirlas a la salida. CC-LoRA extiende el marco de Adaptación de Bajo Rango (LoRA) introduciendo el condicionamiento de Modulación Lineal por Características (FiLM) en las salidas de los adaptadores LoRA de bloques transformadores seleccionados dentro del backbone de CineMA. Un MLP clínico compartido mapea las características del paciente hacia una representación compacta que genera parámetros de escalado (γ) y desplazamiento (β) por bloque, modulando cómo el backbone congelado procesa las características de imagen durante el paso hacia adelante. El backbone permanece completamente congelado durante el entrenamiento, actualizándose únicamente el 0,258% de los 89,5 millones de parámetros totales en la configuración de 4 bloques con mejor rendimiento.

El marco CC-LoRA propuesto fue evaluado principalmente en una cohorte privada de RMC del Hospital de la Santa Creu i Sant Pau, Barcelona, compuesta por pacientes con diferentes etiologías de hipertrofia ventricular izquierda junto con sujetos de control sanos, e incluyendo 27 características clínicas estructuradas por paciente. Para validar el marco de implementación y facilitar la reproducibilidad, se realizaron experimentos adicionales en los benchmarks públicos de RMC cardíaca ACDC y M&Ms. En la cohorte de Sant Pau, la configuración de CC-LoRA de 4 bloques alcanzó un AUC medio de $91,03 \pm 4,17\%$ y una exactitud balanceada de $71,46 \pm 9,23\%$ en validación cruzada de 5 pliegues, superando a las líneas base de solo imagen en 15 - 17 puntos de AUC, lo que sugiere que la información clínica y la de imagen aportan un valor diagnóstico complementario en este contexto. En cambio, no se observó una mejora consistente en ACDC y M&Ms, donde solo estaba disponible un número limitado de variables clínicas. Esta observación sugiere que la efectividad del enfoque propuesto puede depender de la disponibilidad de características clínicas informativas a nivel del paciente.

Estos resultados sugieren que el condicionamiento clínico beneficia sustancialmente el ajuste fino de modelos fundacionales cuando se dispone de datos clínicos ricos a nivel del paciente, y destacan CC-LoRA como un enfoque eficiente en parámetros y con base teórica sólida para el análisis multimodal de imágenes cardíacas en el diagnóstico diferencial de la HVI.

Resum

La resonància magnètica cardíaca (RMC) és l'estàndard de referència per a l'avaluació no invasiva de l'estructura i la funció cardíaca, i té un paper central en el diagnòstic diferencial de les etiologies de la hipertròfia ventricular esquerra (HVE) - incloent-hi la miocardiopatia hipertròfica, la miocardiopatia hipertensiva i l'amiloidosi cardíaca - condicions que comparteixen fenotips d'imatge superposats però que requereixen un maneig clínic fonamentalment diferent. Els avenços recents en models de visió fundacionals preentrenats en grans cohorts de RMC han obert noves possibilitats per a l'anàlisi automatitzada d'imatges cardíques. No obstant això, la integració d'informació clínica a nivell del pacient en aquests models continua sent relativament poc explorada, especialment en el context del diagnòstic diferencial de l'HVE, on el context clínic té un paper important en la pràctica habitual.

En aquest treball, aprofitem CineMA - un model de visió fundacional de RMC d'última generació preentrenat en 74.916 estudis del UK Biobank - per abordar la tasca de classificació d'etiologies de l'HVE, i proposem Clinical-Conditioned LoRA (CC-LoRA), un mètode nou que atorga consciència clínica al model fundacional mitjançant la injecció directa de les característiques clíniques del pacient en el procés d'aprenentatge de representacions d'imatge, en lloc d'afegir-les a la sortida. CC-LoRA estén el marc d'Adaptació de Baix Rang (LoRA) introduint el condicionament de Modulació Lineal per Característiques (FiLM) en les sortides dels adaptadors LoRA de blocs transformadors seleccionats dins del backbone de CineMA. Un MLP clínic compartit mapeja les característiques del pacient cap a una representació compacta que genera paràmetres d'escala (γ) i desplaçament (β) per bloc, modulant com el backbone congelat processa les característiques d'imatge durant el pas endavant. El backbone roman completament congelat durant l'entrenament, actualitzant-se únicament el 0,258% dels 89,5 milions de paràmetres totals en la configuració de 4 blocs amb millor rendiment.

El marc CC-LoRA proposat va ser avaluat principalment en una cohort privada de RMC de l'Hospital de la Santa Creu i Sant Pau, Barcelona, formada per pacients amb diferents etiologies d'hipertròfia ventricular esquerra juntament amb subjectes de control sans, i que inclou 27 característiques clíniques estructurades per pacient. Per validar el marc d'implementació i facilitar la reproductibilitat, es van dur a terme experiments addicionals en els benchmarks públics de RMC cardíaca ACDC i M&Ms. En la cohort de Sant Pau, la configuració de CC-LoRA de 4 blocs va assolir un AUC mitjà del $91,03 \pm 4,17\%$ i una exactitud balancejada del $71,46 \pm 9,23\%$ en validació creuada de 5 plects, superant les línies base d'imatge sola en 15 - 17 punts d'AUC, la qual cosa suggereix que la informació clínica i la d'imatge aporten un valor diagnòstic complementari en aquest context. En canvi, no es va observar cap millora consistent a ACDC i M&Ms, on només hi havia disponible un nombre limitat de variables clíniques. Aquesta observació suggereix que l'efectivitat de l'enfocament proposat pot dependre de la disponibilitat de característiques clíniques informatives a nivell del pacient.

Aquests resultats suggereixen que el condicionament clínic beneficia substancialment l'ajust fi de models fundacionals quan es disposa de dades clíniques riques a nivell del pacient, i destaquen CC-LoRA com un enfocament eficient en paràmetres i amb base teòrica sòlida per a l'anàlisi multimodal d'imatges cardíques en el diagnòstic diferencial de l'HVE.

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I dedicate this work to every patient whose cardiac MRI scan sits quietly in a hospital archive, waiting for a system intelligent enough to understand it fully. This thesis is a small step in that direction.

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Chapter 1

Introduction and Background

1.1 Motivation

Cardiovascular disease (CVD) remains the leading cause of death globally with an estimated 17.9 million annual deaths representing 32% of total deaths [37] with the highest absolute mortality occurring in East Asia and Europe. Cardiomyopathies and myocarditis are responsible for approximately 7% of deaths due to CVD the category of pathology that this thesis aims to address; Global Burden can be viewed in Figure 1.1 - GBD 2019 Cardiovascular Disease Collaborators, WHO Global Health Estimates 2020.

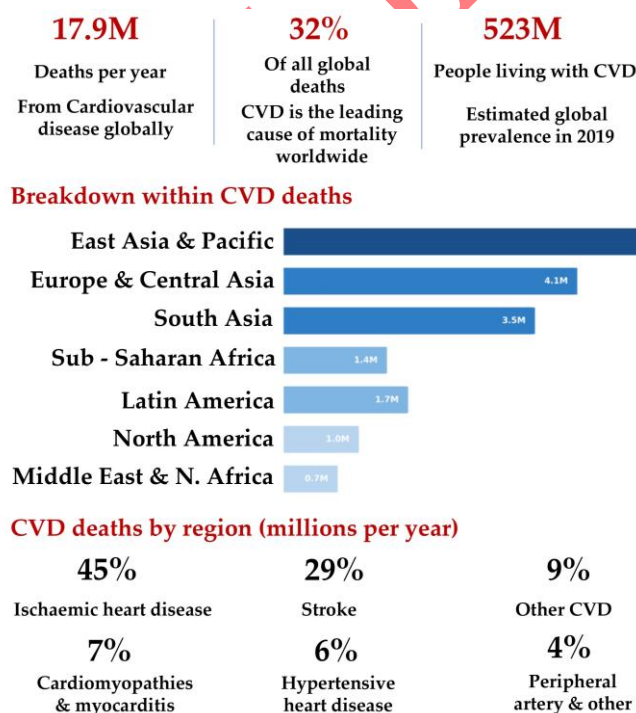


Figure 1.1: Global burden of cardiovascular disease

Within the spectrum of cardiovascular conditions, left ventricular hypertrophy (LVH) - characterized as an abnormal increase in left ventricular wall mass presents one of the most challenging diagnoses in clinical cardiology. LVH is not a disease in itself but a phenotypic consequence of a heterogeneous group of underlying conditions with fundamentally different

etiologies, prognoses, and treatment strategies. The most clinically relevant causes of LVH include hypertrophic cardiomyopathy primary (HCM), the primary genetic form driven by sarcomere mutations; hypertensive cardiomyopathy (HCM), resulting from sustained pressure overload due to systemic hypertension; and cardiac amyloidosis (AML), caused by deposition of misfolded proteins in the myocardium. Correctly identifying which of these etiologies underlies a given patient's LVH and distinguishing all three from a normal phenotype is of critical clinical importance, as each condition carries distinct prognostic implications and demands a different therapeutic approach. HCM may require implantable cardioverter-defibrillators or septal reduction therapy; hypertensive cardiomyopathy is managed through blood pressure control; cardiac amyloidosis increasingly benefits from disease-specific treatments such as tafamidis that are ineffective and potentially harmful in other LVH conditions [38, 39].

Cardiac magnetic resonance imaging (CMR) has emerged as the modality of choice for the non-invasive assessment of cardiac structure and function [40], and plays a central role in the differential diagnosis of LVH etiologies [38,39]. CMR provides high spatial resolution, excellent soft-tissue contrast, and quantitative evaluation of ventricular volumes, ejection fraction, wall thickness distribution, and myocardial tissue characterization through late gadolinium enhancement (LGE). These properties make it uniquely suited to LVH etiology classification - the spatial pattern and extent of hypertrophy, the distribution of LGE, and the biventricular functional profile together provide strong discriminative information. However, cine CMR alone is frequently insufficient to establish the etiology of LVH with confidence. The morphological overlap between HCM and hypertensive HCM is substantial: both present with increased wall thickness, and distinguishing primary genetic hypertrophy from hypertension-induced remodeling requires integration of imaging findings with the patient's clinical history, blood pressure profile, family history, biomarkers, and functional class. Similarly, cardiac amyloidosis can mimic concentric LVH on imaging without the characteristic diffuse LGE pattern that confirms the diagnosis - a pattern that may be absent or atypical in early disease. In clinical practice, the differential diagnosis of LVH therefore typically requires additional investigations - genetic testing, biomarker panels, nuclear imaging, or endomyocardial biopsy - which impose a significant burden on both patients and clinical workflows. Tools that can narrow the differential diagnosis from cine CMR alone, reducing the need for further testing and supporting clinical decision-making at an earlier stage, would represent a meaningful advance for cardiology practice.

From a machine learning perspective, the lack of labelled data makes it more difficult to classify LVH etiology in institutional datasets. Specialized cardiology centres such as the Hospital de la Santa Creu i Sant Pau accumulate disease-specific cohorts over many years, but these datasets are typically far smaller than the large-scale cohorts used to train general deep learning models, hundreds of patients rather than tens of thousands. Recent advances in cardiac CMR foundation models, pre-trained on large-scale cohorts such as the UK Biobank [1,18], offer a promising route to bridging this gap: pre-trained representations that encode rich cardiac morphology features can be efficiently adapted to smaller institutional datasets through parameter-efficient fine-tuning, without requiring the volume of annotated data that training from scratch demands.

However, a critical and underexplored question remains: how should clinical variables be integrated with imaging-derived representations to maximize diagnostic performance? In clinical practice, a cardiologist interpreting a CMR study for LVH etiology does not view images in isolation - they weigh imaging findings against the patient's age, sex, blood pressure history, family

history of cardiomyopathy, ejection fraction, LV mass index, and biomarkers such as troponin and BNP. Current deep learning approaches either discard clinical features entirely or incorporate them through late fusion - concatenating clinical embeddings to image representations only at the final classification layer. Late fusion is technically straightforward but has a fundamental limitation: the image representation is extracted without any awareness of the patient's clinical profile. Whether injecting clinical information directly into the image feature extraction process - conditioning how the model processes the CMR scan based on who the patient is - leads to more effective multimodal representations and improved LVH etiology classification is the central question this thesis investigates.

1.2 Objectives

The main objective of this thesis is to investigate whether cardiac MRI foundation models can be effectively adapted to the task of left ventricular hypertrophy (LVH) differential diagnosis in a small real-world dataset, and to evaluate the potential benefit of integrating clinical information into the classification process. To address this goal, the thesis pursues the following specific objectives:

O1 - Reproduce and validate the experimental framework, validate the implementation of the CineMA fine-tuning pipeline on established public cardiac MRI benchmarks [1, Fu et al. 2026], ensuring the reliability and reproducibility of the experimental framework before applying it to LVH differential diagnosis.

O2 - Investigate transferability to LVH differential diagnosis. Evaluate the transferability of the CineMA foundation model to the task of LVH etiology classification using the real-world clinical cohort from Hospital de la Santa Creu i Sant Pau, assessing whether representations pre-trained on large-scale UK Biobank data generalize to a small, institution-specific dataset with a diagnostically challenging four-class structure - Hypertrophic Cardiomyopathy Primary (HCM), Hypertensive Cardiomyopathy (HCM), Normal controls (NOR), Cardiac Amyloidosis (AML).

O3 - Compare full fine-tuning and parameter-efficient fine-tuning. Compare full fine-tuning of the CineMA backbone with Low-Rank Adaptation (LoRA) [6] as a parameter-efficient alternative, evaluating the trade-off between model capacity and regularization on the small Sant Pau cohort.

O4 - Investigate the added value of clinical information. Investigate whether incorporating patient-level clinical information improves disease classification performance in vision foundation model-based approaches, without requiring full model retraining from scratch.

O5 - Implement and evaluate multimodal fusion strategies. Implement and evaluate multimodal fusion strategies for combining CMR image representations with clinical information, including late fusion as a standard baseline and a novel approach based on LoRA and Feature-wise Linear Modulation (FiLM) [11] applied within selected transformer blocks of the CineMA encoder - referred to as Clinical-Conditioned LoRA (CC-LoRA).

1.3 State of the Art

1.3.1 Deep Learning for Cardiac MRI Analysis

Recent deep learning work applied to CMR analysis has developed at an unprecedented pace over the last decade. Early work successfully demonstrated that convolutional neural networks (CNNs) can serve as strong alternatives to expert analysis for ventricular segmentation [20, 21], establishing the foundations for automated CMR interpretation. The ACDC challenge [16] provided a standardised benchmark for the automated segmentation of the left ventricular cavity, right ventricular cavity, and myocardium, as well as multiclass pathology classification from cine CMR sequences, enabling systematic comparison of methods on a level playing field. The U-Net architecture [20] by Ronneberger et al. became the dominant approach for cardiac segmentation, subsequently extended into transformer-based variants such as TransUNet [23] and UNETR [22] that incorporate global context through long-range attention mechanisms.

Regarding disease classification from cine sequences, the specific task addressed in this thesis - CNN-based approaches have demonstrated strong performance on balanced, single-centre benchmarks. Several works have reported competitive classification results on ACDC [24, 25], and deep learning systems have been shown to approach or match expert-level accuracy for detecting structural and functional cardiac abnormalities from imaging data [24, 25, 27]. However, performance degrades substantially when evaluated on multi-centre, multi-vendor cohorts such as M&Ms [17], where variability in acquisition protocols across vendors introduces significant domain shift. While M&Ms was originally introduced as a segmentation challenge, its pathology annotations have since been used for classification benchmarking, revealing that models trained on single-centre data generalize poorly across scanner vendors. This robustness problem underscores the importance of learning representations that are invariant to acquisition variability - a property that large-scale pre-training is better positioned to provide than task-specific training from scratch.

ResNet architectures [27], originally developed for natural image classification, have been widely adopted as CNN baselines in medical imaging [31, 32]. Pre-trained on ImageNet and fine-tuned on CMR data [31, 32], these models provide powerful but task-specific baselines that nevertheless fall short of models designed with the specific temporal and volumetric properties of CMR in mind.

1.3.2 Vision Transformers and Self-Supervised Pre-training

The advent of Vision Transformer (ViT) [4] ignited a paradigm shift in image recognition, showing that attention-only models can reach competitive or better performance than CNNs, when provided enough data. The trick was to formulate image processing as sequential processing of non-overlapping image patches, so the Transformer was allowed to learn using the self-attention mechanism [34], developed for natural language processing. In medical imaging, the unavailability of large annotated datasets can be addressed with self-supervised pre-training approaches when applying Transformers. Masked Autoencoders (MAE) [3] work by trying to reconstruct randomly masked image patches, and therefore learn structural representations of the image from their

surroundings without any supervision. MAE-pre-trained representations can be then fine-tuned for various tasks and showed good results on many medical imaging modalities [33].

The concept of foundation models [2] - large models pre-trained on diverse, large-scale data and adapted to multiple downstream tasks - has become the dominant paradigm in both natural language processing and computer vision, and more recently in the medical imaging domain, including cardiac MRI [28, 29]. In radiology and pathology, foundation models have demonstrated remarkable label efficiency and cross-task generalization [13, 15]. Within the cardiac MRI domain specifically, several recent works have begun developing task-specific pre-trained backbones. Jacob et al. [29] proposed a vision foundation model for comprehensive CMR assessment pre-trained using self-supervised objectives on large cardiac datasets, and ViTa [28] introduced a transformer-based model adapted for whole-heart assessment that integrates both visual and tabular representations - both demonstrating that self-supervised pre-training on cardiac data produces representations that substantially outperform task-specific CNNs in low-data regimes.

1.3.3 CineMA: A Foundation Model for Cine Cardiac MRI

Among the foundation models developed for cine CMR, CineMA [1] represents the most comprehensive evaluation to date, building on earlier efforts such as Jacob et al. [29] and ViTa [28]. The model follows a convolutional-vision-transformer (ConvViT) architecture where convolutional layers capture local spatial patterns and transformer blocks aggregate global long-range dependencies. CineMA is pre-trained on 74,916 cine CMR studies - corresponding to approximately 15 million images - collected from the UK Biobank, using a masked autoencoder objective: masked patches of short-axis image sequences are reconstructed from the visible context. The encoder processes multi-frame SAX sequences through convolutional patch embedding layers followed by 12 transformer blocks. For downstream tasks, task-specific heads - UNet-style decoders for segmentation, linear layers for regression, or classifiers for disease detection - are appended to the frozen or fine-tuned encoder. CineMA has been evaluated across 23 tasks on 8 independent datasets, consistently outperforming CNN baselines in data-scarce settings. Under full fine-tuning, CineMA establishes a new upper bound for image-only methods, achieving 94.57% AUC on the ACDC classification benchmark.

In this thesis, CineMA is selected as the backbone model in preference to ViTa given its stronger reported performance on classification tasks and its more comprehensive multi-task, multi-dataset evaluation, making it better suited as a foundation for the multimodal adaptation investigated here.

1.3.4 Parameter-Efficient Fine-Tuning

Full fine-tuning of large pre-trained models, while capable of achieving strong downstream performance compared to task-specific training from scratch, incurs substantial computational cost and is prone to catastrophic forgetting when the downstream dataset is too small relative to the number of trainable parameters [31, 32]. Parameter-efficient fine-tuning (PEFT) methods overcome these challenges by updating only a small fraction of parameters while keeping the backbone frozen [7, 9].

Among PEFT approaches, adapter-based methods [7] insert small bottleneck modules

between transformer layers, training only the adapter weights while leaving all pre-trained parameters untouched. BitFit [9] demonstrates that fine-tuning only the bias terms of a pre-trained transformer can be surprisingly competitive in low-data settings. Among these approaches, Low-Rank Adaptation (LoRA) [6] has become one of the most widely adopted PEFT methods. It decomposes weight updates into the product of two low-rank matrices, constraining the update to a low-dimensional subspace of the original weight space. Mathematically, for a pre-trained weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA parameterises the update as $\Delta W = BA$, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$ is the rank. The adapted forward pass becomes:

$$h = Wx + \frac{\alpha}{r}BAx$$

where α is a scaling hyperparameter. During training, W is frozen and only A and B are updated. LoRA was originally proposed for large language models [6] but has since been applied successfully to vision transformers in both natural image [10] and medical imaging settings [8, 33]. The rank and number of adapted blocks used in this thesis ($r = 4$, last 2, 4, and 6 transformer blocks) were determined through experimental testing and are consistent with the findings reported in Med-DualLoRA [8], which applies LoRA to the CineMA backbone for 3D cardiac MRI adaptation and demonstrates that injecting LoRA into a small number of transformer blocks is sufficient to achieve strong downstream performance while preserving the generalization properties of the pre-trained representations.

1.3.5 Multimodal Fusion in Medical Imaging

Fusion of imaging data and clinical/tabular information remains an active problem in medical AI [12, 13]. Methods mainly fall into categories of three:

Early fusion: Concatenates modalities at the input layer, and all modalities need to be projected into the same feature space before further processing. However, it's not generally applied for the image-clinical fusion due to modal heterogeneity's.

Late fusion: Processes each modality separately and combines the extracted features at the prediction layer. It uses a clinical MLP to generate a condensed embedding from the tabular data. The resulting image and clinical embeddings are then concatenated at the classification head. It's interpretable, straightforward and widely used [13]. However, the image feature is extracted without clinical contextualization.

Intermediate fusion: Inject information from other modalities to intermediate layers in the image processing pipeline [14, 15] in order to facilitate cross-modal interaction. However, these models are much more complicated to implement and difficult to train with a limited amount of data.

1.3.6 Feature-wise Linear Modulation

Feature-wise Linear Modulation (FiLM) [11] provides a lightweight and differentiable framework for conditioning neural network computations on auxiliary inputs. Given a conditioning vector $\mathbf{c}$, FiLM applies an affine transformation to an intermediate feature map $\mathbf{x}$:

$$\text{FiLM}(\mathbf{x} \mid \mathbf{c}) = \gamma(\mathbf{c}) \odot \mathbf{x} + \beta(\mathbf{c})$$

where $\gamma(\mathbf{c})$ and $\beta(\mathbf{c})$ are learned functions of the conditioning vector. Originally designed for visual question answering - enabling language queries to condition visual feature extraction [11] - FiLM has since been applied in generative modelling [44], few-shot learning [45], and style transfer [46]. Its affine form is lightweight, differentiable, and amenable to identity initialization ($\gamma = 1, \beta = 0$), making it well-suited for training stability on small clinical datasets.

The combination of conditioning or modulation with low-rank adaptation has emerged as a cross-domain trend in recent work. MAIRA-2 [47] explores conditioned adapter approaches for grounded radiology report generation. TC-LoRA [48] proposes task-conditioned low-rank adaptation for continual learning in vision-language models, generating LoRA weights conditioned on task embeddings. More recently, modulation-based conditioning within parameter-efficient fine-tuning has been explored for medical image segmentation [49], demonstrating that feature modulation can guide adapter learning toward task-relevant representations. These works collectively establish that conditioning LoRA adapters on auxiliary signals is a promising and emerging direction across domains.

CC-LoRA, proposed in this thesis, applies this principle to the cardiac MRI domain for the first time, to the best of our knowledge. It is also, to our knowledge, the first method to perform LoRA modulation using structured patient-level clinical features rather than text or task embeddings - framing the conditioning explicitly as a patient personalization mechanism within a vision foundation model, rather than a task-switching mechanism.

1.3.7 Clinical Data Integration in Cardiac AI

The integration of clinical information with cardiac imaging data has a substantial history, particularly in radiomics-based approaches that combine hand-crafted imaging features with clinical variables for risk stratification, disease classification, and outcome prediction [24]. More recently, however, the challenge has shifted toward multimodal integration within transformer-based vision foundation models - a setting in which clinical conditioning remains largely unexplored. Within deep learning approaches, most existing work treats clinical measurements either as post-processing corrections or as inputs to external risk models operating independently of the imaging pipeline [24]. At the model level, the dominant strategy remains late fusion: concatenating demographic and clinical embeddings to image representations prior to the classification head. This approach, while simple and interpretable, does not allow clinical context to shape how the image representation itself is extracted - a fundamental limitation when clinical variables carry strong pathology-discriminative signal, as is the case in LVH etiology classification.

Motivated by these limitations, this thesis investigates multimodal integration strategies for combining cardiac MRI and clinical information within transformer-based vision foundation models, including parameter-efficient adaptation approaches based on Low-Rank Adaptation (LoRA) and feature modulation techniques for clinical conditioning, aiming to avoid the computational cost associated with full model fine-tuning while enabling clinically-aware image representation learning.

1.4 Document Organization

This thesis is structured into seven chapters, each addressing a distinct phase of the research pipeline from clinical motivation through experimental validation.

Chapter 1 - Introduction and Background establishes the clinical context of left ventricular hypertrophy differential diagnosis, motivates the need for multimodal approaches in cardiac MRI analysis, defines the research objectives, and surveys the relevant state of the art in vision foundation models, parameter-efficient fine-tuning, and clinical feature integration.

Chapter 2 - Datasets Information describes the three cardiac MRI datasets used throughout this work: the public ACDC and M&Ms benchmarks, and the private cohort from Hospital de la Santa Creu i Sant Pau. For each dataset, the clinical population, imaging protocol, available annotations, and class distribution are detailed, with particular attention to the richness of clinical tabular features in the Sant Pau cohort.

Chapter 3 - Data Preprocessing presents the full preprocessing pipeline applied to all datasets, covering end-diastolic and end-systolic frame extraction, spatial resampling, centre cropping and padding, intensity normalization, two-frame input construction, clinical feature preprocessing, data augmentation strategies, and class imbalance handling.

Chapter 4 - Development describes the complete set of architectures designed and implemented in this work. It covers the CineMA foundation model baseline, the ResNet-50 baseline, full fine-tuning, image-only LoRA adaptation, the late fusion architecture, and the proposed CC-LoRA method in full technical detail, followed by the training configuration and evaluation metrics used across all experiments.

Chapter 5 - Experiments and Results reports all experimental outcomes across the three datasets. It begins with a reproduction study validating the implementation framework against published CineMA results, then presents results on the public benchmarks with limited clinical features, followed by a comprehensive analysis of LVH differential diagnosis on Sant Pau, a cross-dataset analysis examining the relationship between clinical richness and CC-LoRA gain, and a summary of key findings.

Chapter 6 - Agreement of Results and Objectives maps the experimental results back to the five research objectives defined in Chapter 1, providing a structured discussion of the degree to which each objective was met and interpreting the broader implications of the findings.

Chapter 7 - Conclusions and Future Work synthesizes the contributions of this thesis, acknowledges the limitations of the current approach, and outlines directions for future research, including extensions to richer multimodal settings and alternative conditioning architectures.

Finally, the **Bibliography** collects all references cited throughout the document.

Chapter 2

Datasets Information

This thesis uses three datasets - 2 public and 1 private, chosen to cover a range of sizes, class distributions, clinical feature depth and acquisition modalities. The publicly available datasets are used for the initial development and validation of the proposed approaches, including the reproduction of previously reported findings in the literature, prior to transferring the framework to a private real-world LVH etiology classification cohort from Hospital de la Santa Creu i Sant Pau. Furthermore, they allow the evaluation of the generalizability of the proposed approaches to additional cardiovascular disease classification tasks beyond LVH etiology classification, while also facilitating the reproducibility of the proposed framework and experimental pipeline. Table 2.1 provides a summary overview of all three datasets after preprocessing.

Property	ACDC	M&Ms	Sant Pau
Total patients	150	271	292
Number of classes	5	4	4
Clinical features	3	2	27
Evaluation protocol	Fixed split, 3 seeds	Fixed split, 3 seeds	5-fold stratified CV
Primary role	Public benchmark	Robustness benchmark	Clinical conditioning target

Table 2.1 - Dataset Overview

Class Distribution across the three ACDC, M&Ms, and the Sant Pau Hospital datasets are shown in the Figure 2.1, ACDC contains of 5 pathology classes with no imbalance whereas M&Ms and Sant Pau hospital datasets have 4 pathology classes with imbalanced data.

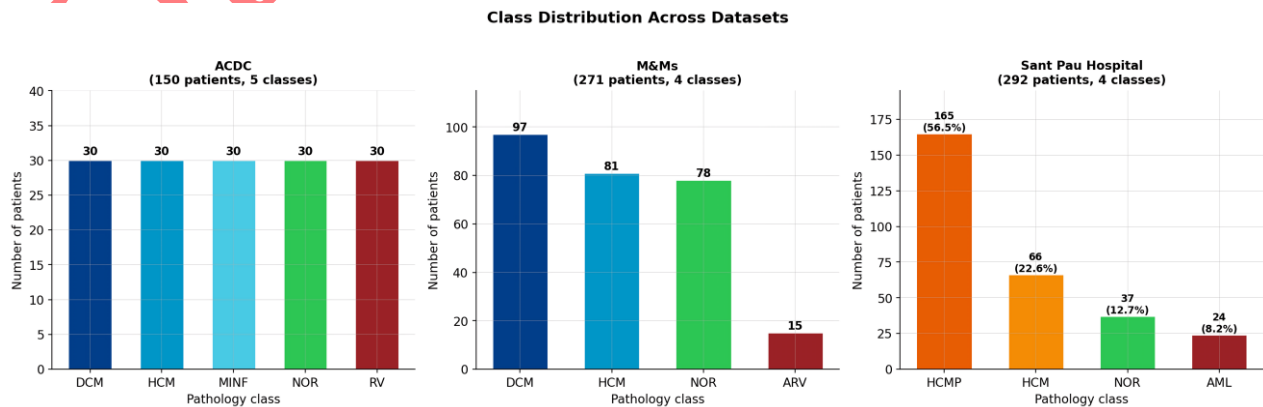


Figure 2.1: Class Distribution Across Datasets

2.1 ACDC - Automated Cardiac Diagnosis Challenge

2.1.1 Overview

The ACDC dataset [16] was originally proposed for the 2017 MICCAI Automated Cardiac Diagnosis Challenge, and is now by far the most cited benchmark for cardiac MRI segmentation and classification in the deep learning literature. The dataset comprises 150 patients uniformly distributed across five pathology classes - dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), myocardial infarction (MINF), normal controls (NOR), and right ventricular abnormality (RV) - with exactly 30 patients per class, making it perfectly balanced. All images were acquired at a single centre using a standardized protocol, and publicly available ground-truth segmentation masks are provided by multiple expert annotators for both end-diastolic and end-systolic frames. These characteristics - class balance, single-centre standardized acquisition, and expert-annotated ground truth - make ACDC an ideal controlled benchmark for systematic comparison of methods under well-defined conditions.

For the purposes of this thesis, ACDC serves as a clean, balanced public benchmark, allowing us to validate our implementation against previously reported results in the literature and to assess the behavior of clinical conditioning in a setting where the available clinical features - height, weight, and BMI - carry little pathology-discriminative signal. This makes ACDC a useful negative control for the clinical richness analysis: a method that benefits substantially from clinical conditioning on feature-rich datasets should show no improvement on ACDC, where clinical features are uninformative with respect to the five pathology classes.

2.1.2 Patient Cohort and Class Distribution

The dataset comprises 150 patients acquired at the University Hospital of Dijon, France. Patients are evenly distributed across five pathology classes with exactly 30 patients per class, making ACDC the only perfectly balanced dataset used in this thesis which as shown in the Table 2.2.

Class	Label	Count	Percentage	Clinical Description
Dilated Cardiomyopathy	DCM	30	20.0%	LV dilation, reduced systolic function
Hypertrophic Cardiomyopathy	HCM	30	20.0%	LV hypertrophy, preserved function
Myocardial Infarction	MINF	30	20.0%	Regional scar, wall motion abnormality
Normal	NOR	30	20.0%	No structural or functional abnormality
Right Ventricular Abnormality	RV	30	20.0%	Primary RV pathology
Total		150	100%	

Table 2.2 - ACDC Class Distribution

The five classes cover a broad range of cardiac pathology. Dilated cardiomyopathy is characterized by ventricular dilation and depressed systolic function. Hypertrophic cardiomyopathy is defined by ventricular hypertrophy without a secondary cause, while myocardial infarction is characterized by a ventricular scar resulting from previous ischemic disease (a regional wall motion abnormality and late gadolinium enhancement). RV abnormality refers to pathologies predominantly affecting the right ventricle. The highly balanced distribution

across the five classes, the ACDC dataset particularly suitable for model evaluation, thereby reducing the potential confounding effect of class imbalance.

2.1.3 Imaging Data

Images were acquired using a 1.5T Siemens scanner, with the standard cardiac MR protocol. Short axis cine sequences covering the entire cardiac cycle were acquired and frames at ED and ES were identified and labeled by cardiologists. Segmentation masks by cardiologists for the left and right ventricle cavity, and myocardium at both ED and ES are available. The data is provided in NIfTI format, with segmentation masks and the corresponding patient metadata files.

In this thesis, only ED and ES volumes were selected for classification according to the CineMA evaluation [1]. Input to the model for a patient will be a two-frame tensor stacking the ED and ES short axis volumes, representing the cardiac function estimation approach.

2.1.4 Clinical Features

Three clinical features exist for each patient: height(cm), weight(kg), and body mass index (BMI)(kg/m), available in the patient information files. These anthropometric measurements alone do not have any pathological discriminative information for the 5 ACDC classes - DCM, HCM, MINF, NOR, and RV as these diseases differ by the morphology and function of the heart, and not the anthropometric properties of the patients. Such a characteristic means that ACDC may serve as a negative control to assess the effectiveness of clinical conditioning: methods that learn informative signals from clinical features on data-rich tasks should not benefit from clinical features on ACDC. The 3 features are normalized using z-score normalization fit on the training set.

2.1.5 Data Splits and Evaluation Protocol

All experiments were conducted using the official training/validation/testing split defined by the ACDC challenge organizers: 100 patients for training, 10 for validation, and 50 for testing. This fixed split description shown in Table 2.3 enables direct comparison with the CineMA paper [1] and other published methods.

Split	Total	DCM	HCM	MINF	NOR	RV
Train	100	20	20	20	20	20
Validation	10	2	2	2	2	2
Test	50	10	10	10	10	10
Total	150	30	30	30	30	30

Table 2.3 - ACDC Dataset Split: Training, Validation, and Testing Sets

A limitation of this evaluation setup that it uses a very tiny validation set (10 patients, 2 per class). Due to the limited number of samples per class in the validation set (2 patients per class), even a single misclassification can produce substantial changes in per-class recall and lead to large fluctuations in validation AUC across checkpoints. This instability is reflected in the convergence plots presented in Chapter 5 and constitutes the primary reason why all ACDC experiments were

repeated across three random seeds, with the final performance reported as the mean $\pm$ standard deviation across seeds.

2.2 M&Ms - Multi-Centre, Multi-Vendor Cardiac Segmentation

2.2.1 Overview

The M&Ms dataset [17] was introduced to mitigate a crucial problem of single-centre benchmarks such as the ACDC challenge: the susceptibility of deep learning methods to differences across scanner vendors and acquisition protocols. It was distributed alongside the 2020 MICCAI Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Image Segmentation Challenge, and it is now the established benchmark to quantify how robust cardiac MRI methods are to a domain shift. The M&Ms dataset is used here as a second public benchmark, allowing us to validate our baseline implementations against published results and to further assess the proposed methods in a more challenging multi-vendor, class-imbalanced setting than ACDC.

2.2.2 Patient Cohort and Subset Selection

The whole M&Ms dataset consists of 375 patients collected using four different vendors in six different hospitals and three different countries (Spain, Canada and Germany). However, the 375 patients also include unlabeled cases that were used exclusively for the segmentation task during the challenge, as well as patients with uncertain class labels or incomplete clinical information. Therefore, this thesis focuses on a subset of 271 patients that consists of patients with verified class labels on the four classes that are relevant for classification and with valid clinical information according to the protocol used during the evaluation CineMA [1].

Class	Label	Train	Validation	Test	Total	Percentage
Dilated Cardiomyopathy	DCM	54	8	35	97	35.8%
Hypertrophic Cardiomyopathy	HCM	49	9	23	81	29.9%
Normal	NOR	38	9	31	78	28.8%
Abnormal Right Ventricle	ARV	8	1	6	15	5.5%
Total		149	27	95	271	100%

Table 2.4 - M&Ms Class Distribution (271 patients)

As shown in Table 2.4, the class distribution is unbalanced. Moreover, ARV contains only 15 patients overall (5.5% of the data) of which there are only 6 test and 8 training patients. ARV has so few training samples that it is challenging for a deep learning method to learn a decision boundary; therefore, the severe underrepresentation of the ARV class is likely to hinder robust learning and stable evaluation performance for this category. DCM is the class with the largest number of patients (35.8%).

2.2.3 Multi-Vendor Acquisition

The multi-vendor nature of the M&Ms introduces substantial domain variability that is not

present in ACDC. Each of the four vendors (Siemens, Philips, GE and Canon) uses a different approach to sample k-space and reconstruct images, resulting in images that vary significantly in texture, sharpness and noise while representing the same anatomical structures. The four clinical centres introduce additional variability in terms of acquisition protocol, with in-plane resolutions ranging from 0.68 to 1.37 mm and slice thicknesses ranging from 6 to 13 mm across centres and vendors, as well as differences in field strength (1.5T and 3T) [17]. The resulting M&Ms dataset is therefore considerably more challenging than ACDC, owing to its multi-vendor heterogeneity and increased domain variability. As such, it provides a suitable benchmark for evaluating how well a pre-trained model adapts to previously unseen domains. Per patient metadata in the M&Ms dataset consists of: vendor, center, index of ED frame, index of ES frame, age, sex, height, weight, and auto-derived measurements of cardiac function based on the supplied gold-standard segmentation masks - specifically left ventricular end-diastolic volume (LVEDV), left ventricular end-systolic volume (LVESV), left ventricular ejection fraction (LVEF), right ventricular end-diastolic volume (RVEDV), right ventricular end-systolic volume (RVESV), and right ventricular ejection fraction (RVEF).

2.2.4 Clinical Features

Two clinical features are used per patient: age (years, continuous, normalized with z-score) and sex (binary, 0/1). Although scanner vendor information is present in the metadata, we choose not to use it as a clinical conditioning feature; rather, it is an acquisition metadata and it represents patient-level clinical data only by using vendor as clinical feature would actually be domain adaptation, not clinical conditioning.

As with ACDC, age and sex alone have reduced discriminative power for distinguishing the four M&Ms pathology classes in the context of this task. While age and sex are known to carry meaningful signal in certain cardiovascular classification settings - for example, sex-related differences in cardiac morphology and age-related disease prevalence can improve classification in some tasks [12] - they are insufficient on their own to reliably differentiate DCM, HCM, NOR, and ARV in this cohort. This makes M&Ms our second negative control in the clinical richness evaluation: alongside ACDC, it allows us to assess the behaviour of CC-LoRA in a setting where the available clinical features provide limited independent discriminative signal beyond what the imaging already captures.

2.2.5 Evaluation Protocol

The fixed M&Ms split of 149 training, 27 validation, and 95 test patients is used throughout all experiments, consistent with the CineMA evaluation protocol [1], ensuring direct comparability with previously reported results. Due to the limited number of samples per class in the validation set, even a single misclassification can produce substantial changes in per-class recall and lead to large fluctuations in validation AUC across checkpoints - a similar instability to that observed on ACDC, though less severe given the larger validation set. For this reason, all M&Ms experiments were repeated across three random seeds, with the final performance reported as the mean $\pm$ standard deviation across seeds.

2.3 Sant Pau Hospital Dataset

2.3.1 Overview

The Sant Pau Hospital dataset represents the primary dataset used in this thesis and serves as the principal evaluation benchmark for the proposed CC-LoRA framework. It is a retrospective cohort retrieved from the Hospital de la Santa Creu i Sant Pau of Barcelona. Figure 2.2 illustrates the spatial organization of the data by showing representative slices from a patient diagnosed with hypertrophic cardiomyopathy (HCMP). A particularly relevant characteristic of this dataset for evaluating clinical conditioning is the availability of 27 structured clinical features for each patient, including echocardiographic measurements, ventricular volumes, functional indices, demographics, comorbidities etc. Local ethics committee approval was obtained and patient data were anonymized.

Patient: p1 · Class: HCMP · 12 slices · Resolution: 1.0×1.0 mm/px · Slice thickness: 10.0 mm · After preprocessing: 192×192 px

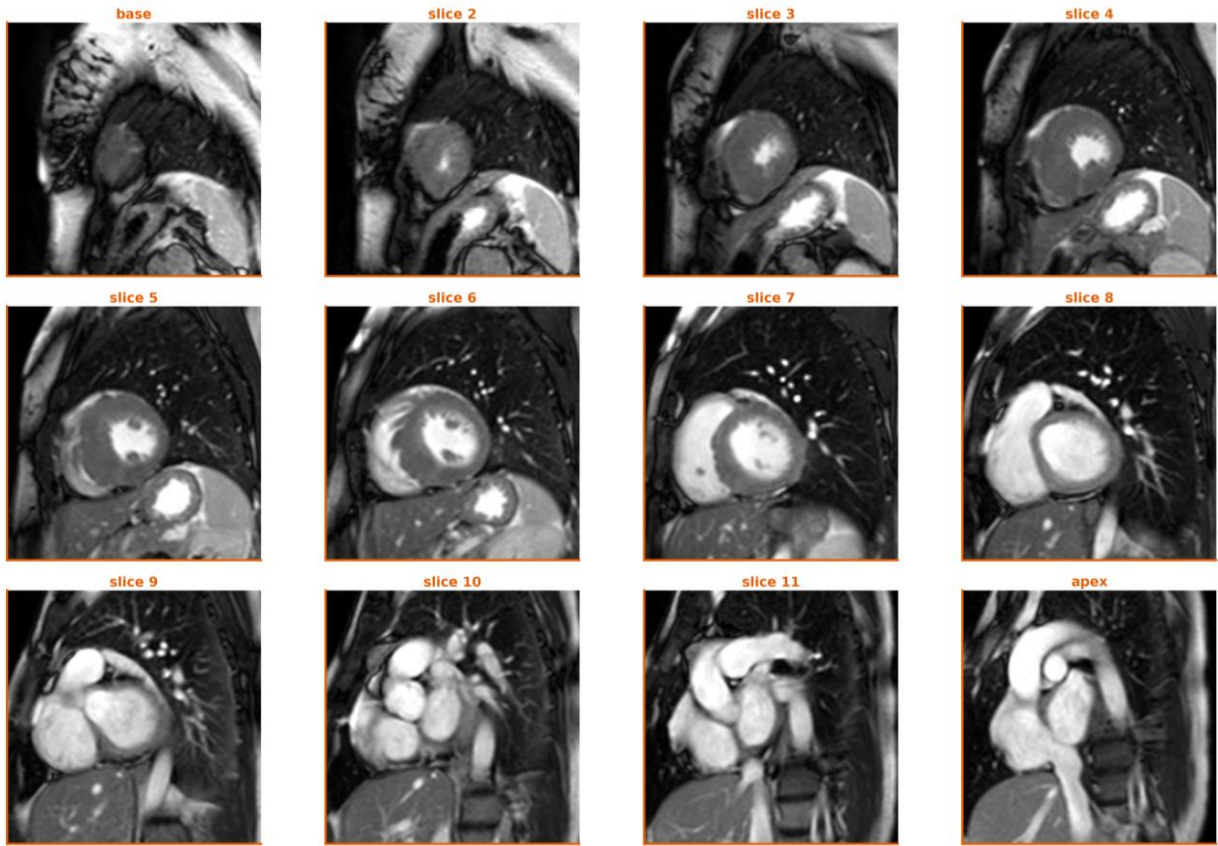


Figure 2.2: Sant Pau - Short-axis (SAX) slices at ED of HCMP class

2.3.2 Raw Dataset and Class Exclusion

The raw dataset comprised 294 patients with confirmed cardiac diagnoses, recorded via a numeric Etiology code in the clinical records. The raw class distribution is shown in Table 2.5.

Code	Diagnosis	Count	Action
9	NOR - Normal / Healthy	37	Retained

1	HCMP - Hypertrophic Cardiomyopathy Primary	165	Retained
0	HCM - Hypertensive Cardiomyopathy	66	Retained
2	AML - Cardiac Amyloidosis	24	Retained
4	Anderson Fabry Disease	2	Excluded

Table 2.5 - Raw Sant Pau Etiologia Distribution

The Anderson-Fabry disease category was excluded entirely from the study due to the extremely limited number of available cases, which prevented a reliable partition into training, validation, and test sets (Etiology code 4). Therefore, the final dataset used consists of 292 patients spread across 4 classes. All Sant Pau classification experiments use 5-fold stratified cross-validation applied to the full 292-patient dataset, with stratification on the pathology label to preserve class proportions across folds (random state 42). This produces approximately 234 training and 58 validation patients per fold. Performance is reported as mean $\pm$ standard deviation across the five validation folds.

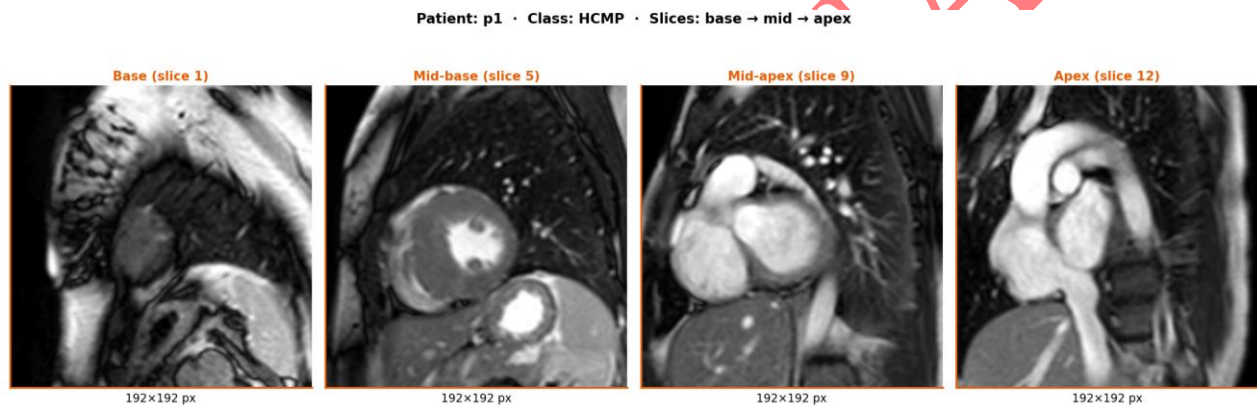


Figure 2.3: Sant Pau - Cardiac anatomy across SAX slices (ED Frame)

Cardiac structural changes for the short-axis slices of patient p1 can be seen in figure 2.3. These images are snapshots of ED. From these end-diastolic frames the structure can be observed going from the basal portion (slice 1) to the apex (slice 12) through the mid-ventricular portions. The mid-base and mid-apex frames clearly show that there is considerable myocardial wall thickening which is quite characteristic of the underlying Hypertrophic Cardiomyopathy (HCMP) condition.

2.3.3 Final Class Distribution

The four classes represent three clinically distinct etiologies of left ventricular hypertrophy and one negative control arm included to assess the model's ability to recognize the absence of disease. Understanding their clinical distinctions and the role of the negative control is essential for interpreting the experimental results. The final class distribution is presented in Table 2.6.

Class	Full Name	Train	Test	Total	Percentage
HCMP	Hypertrophic Cardiomyopathy Primary	132	33	165	56.5%
HCM	Hypertensive Cardiomyopathy	53	13	66	22.6%
NOR	Normal / Healthy	29	8	37	12.7%
AML	Cardiac Amyloidosis	19	5	24	8.2%

Total		233	59	292	100%
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Table 2.6 - Sant Pau Final Class Distribution

Hypertrophic Cardiomyopathy Primary (HCMP) is the dominant class at 56.5%. HCMP is used for primary genetic hypertrophic cardiomyopathy. Hypertrophic cardiomyopathy is an autosomal dominant disease characterized by an increased mass of the left ventricle without an obvious external cause (e.g. Hypertension). Asymmetric left ventricular hypertrophy, primarily at the septum, is typical. On CMR HCMP is characterized by increased wall thickness, hyperdynamic or normal LVSE, diastolic dysfunction, and mid-wall LGE. HCMP is a large group in our cohort and this is likely due to a referral pattern to a specialist cardiomyopathy clinic and the preponderance of hypertrophic phenotypes.

Hypertensive Cardiomyopathy (HCM) comprises 22.6% of patients. HCM in this context defines cardiomyopathy following the diagnosis of chronic hypertensive systemic heart disease, referring to concentric or asymmetric left ventricular hypertrophy secondary to sustained pressure overload. While morphologically similar to HCMP on imaging both present with a thickened left ventricular wall HCM is distinguished by its etiology: the hypertrophy is a secondary consequence of chronic elevated afterload rather than a primary genetic abnormality. This distinction is clinically critical but difficult to establish from imaging alone. Clinical history, particularly blood pressure history and family history of cardiomyopathy, becomes essential for reliable differentiation, making this class pair a key driver for the clinical conditioning approach investigated in this thesis, the goal being to integrate such contextual information directly into the image representation learning process rather than requiring invasive or additional investigations.

Normal controls (NOR) comprise 12.7% of the cohort (37 patients). These patients underwent CMR examination and were confirmed to have no structural or functional cardiac abnormality. NOR is included not as a fourth LVH etiology but as a negative control arm, serving a specific and essential methodological purpose: it forces the model to demonstrate that it can recognise the absence of pathology, not merely assign one of the three LVH etiologies to every patient. Without a healthy reference class, a model could achieve apparent success by always predicting some form of cardiomyopathy - a trivial strategy that would go undetected in a three-class setting. The inclusion of NOR therefore tests discriminative honesty: the model must learn to rule out disease, not only to distinguish between its etiologies. The small per-fold NOR training set (~30 patients in the 5-fold cross-validation setting) makes this a non-trivial requirement.

Cardiac Amyloidosis (AML) is the smallest and clinically most distinct LVH etiology at 8.2% of the cohort. It is caused by deposition of misfolded proteins transthyretin or immunoglobulin light chains in the myocardium, producing concentric hypertrophy, diastolic dysfunction, and a characteristic diffuse subendocardial LGE pattern that is morphologically distinct from the patchy focal LGE seen in HCMP and HCM. Despite this morphological distinctiveness relative to the other LVH etiologies, AML is by far the least represented class with only 24 patients in total (~19 per training fold), rendering it the most challenging class for deep learning approaches. As discussed in Chapter 5, AML consistently shows the highest classification variance across folds and the lowest per-class accuracy across all methods, a result attributable primarily to this severe underrepresentation rather than to inherent morphological ambiguity with respect to the other classes.

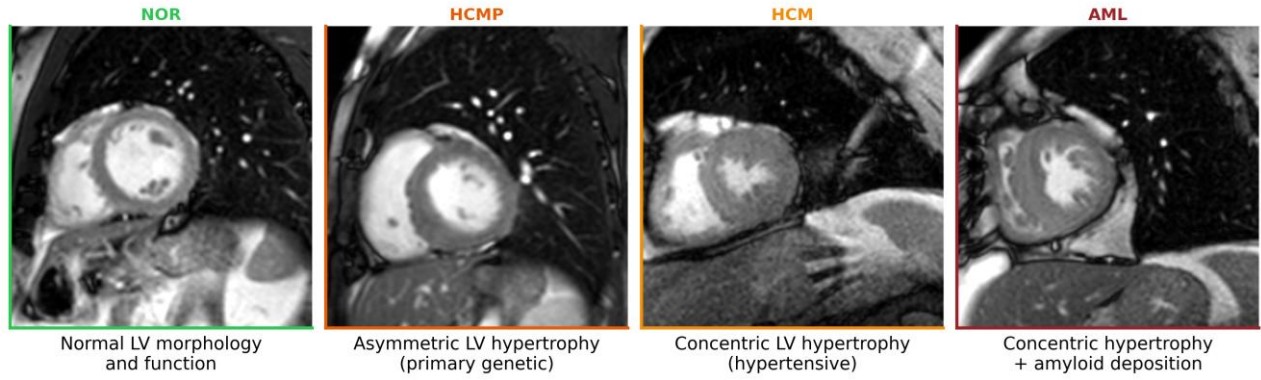


Figure 2.4: Sant Pau - Representative CMR images at end diastole, mid ventricular slice

Selected CMR images at end-diastole in mid-ventricular slice are provided for each of the four pathologies classes in Figure 2.4. The variations show difference among the different pathologies with regards to myocardial wall thickness, and the orientations of the cavities.

2.3.4 Imaging Data

The CMR images were acquired in a typical clinical protocol for cardiac MR at the Hospital Sant Pau. Cine images short axis was taken through the entire cardiac cycle in left ventricle from base to apex. Images were originally acquired in DICOM format and converted to NIfTI by Grzegorz Mrukwa as part of the dataset preparation and cleaning pipeline, prior to use in this thesis. All images were fully anonymized before transfer. Examination of the converted NIfTI files confirms consistent acquisition parameters across all 292 patients, as summarized in Table 2.7.

Property	Value
In-plane resolution	1.0×1.0 mm
Slice thickness	10.0 mm
Spatial dimensions (post-processing)	192×192 pixels
Number of short-axis slices	10–13 per patient (mean 12)
Frames used	ED + ES (2 frames)
Format	NIfTI (.nii.gz)

Table 2.7 - Sant Pau Image Properties

The uniform 1.0×1.0 mm in-plane resolution indicates standardized acquisition or resampling at the clinical center. The variable number of slices (10 - 13) reflects physiological differences in ventricular extent across patients and is handled during preprocessing by padding all volumes to a fixed depth of 16 slices, analysis on this is presented in Figure 2.5.

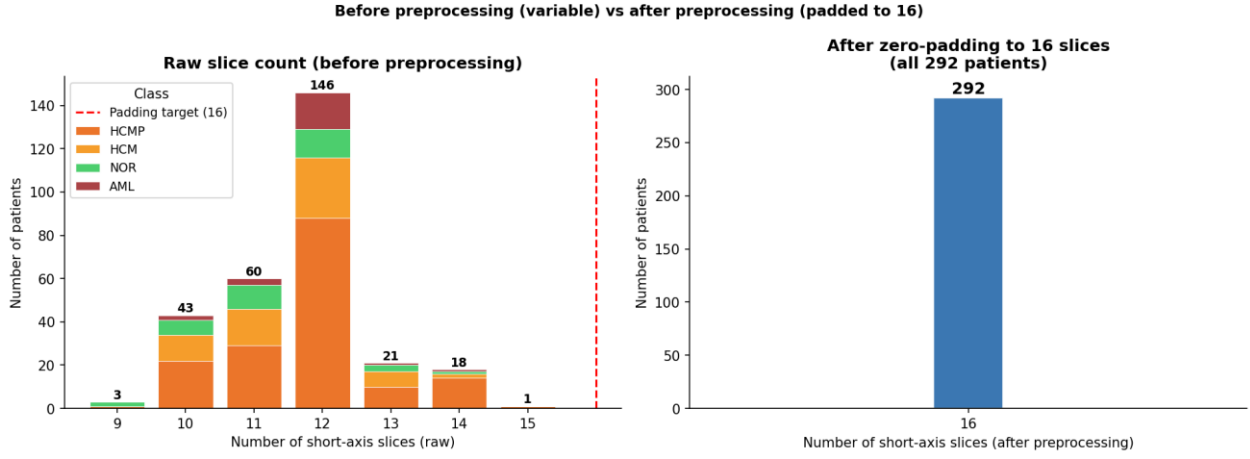


Figure 2.5: Sant Pau - SAX Slices count per patient

The spatial dimensions and resolution parameters of the CMR dataset, for example patient p1 are presented in Figure 2.6. Extracted from the end-diastolic frame across a total of 12 slices, the representative scans illustrate the physical scale at the base (slice 1), mid-ventricle (slice 7), and apex (slice 12).

In-plane resolution: 1.0×1.0 mm/px · After preprocessing: 192×192 px per slice · Slice thickness: 10.0 mm · 12 slices

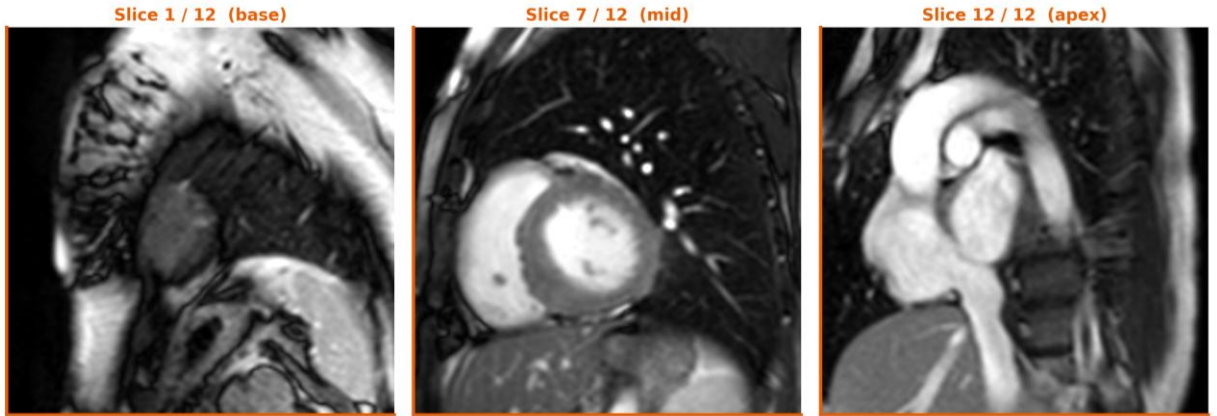


Figure 2.6: Sant Pau - Spatial Dimensions

As shown in the annotations, each image maintains an in-plane resolution of 192×192 pixels with a isotropic pixel spacing of 1.0×1.0 mm/px, corresponding to a physical field of view of 192×192 mm. This standardized spatial grid, combined with a slice thickness of 10.0 mm, establishes a consistent geometric framework for accurate volumetric evaluation and myocardial wall profiling.

2.3.5 Clinical Features

The special characteristic of the Sant Pau dataset is that for each patient 27 structured clinical features are available. The clinical information was derived from clinical records at the time of CMR acquisition and is composed of five groups of variables: the indices of the cardiac function, ventricular volumes and dimensions, patient characteristics, comorbidity variables and information about device/treatment. See Table 2.8.

Feature (abbreviation)	Category	Type	Clinical Meaning	Statistics
LV ejection fraction (FEVI)	Cardiac function	Continuous	LV ejection fraction (%)	63.1 ± 8.1%
Maximum LV wall thickness (Espesormximo1)	Cardiac structure	Continuous	Maximum LV wall thickness (mm)	16.9 ± 4.9 mm
LV end-diastolic volume index (VTDVI)	Ventricular volumes	Continuous	LV end-diastolic volume index (mL/m ²)	145.2 ± 39.6 mL/m ²
LV end-systolic volume index (VTSVI)	Ventricular volumes	Continuous	LV end-systolic volume index (mL/m ²)	54.5 ± 21.4 mL/m ²
LV stroke volume index (VS_VI)	Ventricular volumes	Continuous	LV stroke volume index (mL/m ²)	90.7 ± 24.6 mL/m ²
LV mass index (Masa_VI)	Cardiac structure	Continuous	LV mass index (g/m ²)	133.7 ± 51.0 g/m ²
RV end-diastolic volume (VTDVD)	Ventricular volumes	Continuous	RV end-diastolic volume (mL)	134.8 ± 38.4 mL
RV end-systolic volume (VTSVD)	Ventricular volumes	Continuous	RV end-systolic volume (mL)	50.8 ± 20.2 mL
RV stroke volume (VS_VD)	Ventricular volumes	Continuous	RV stroke volume (mL)	84.0 ± 23.8 mL
RV ejection fraction (FEVD)	Cardiac function	Continuous	RV ejection fraction (%)	62.9 ± 8.4%
NYHA functional class (Clasefuncional)	Functional status	Continuous	NYHA functional class (1–4)	1.4 ± 0.6
Body mass index (IMC)	Demographics	Continuous	Body mass index (kg/m ²)	26.6 ± 4.4 kg/m ²
Age (Edad)	Demographics	Continuous	Patient age (years)	56.9 ± 17.5 years
Weight (Peso)	Demographics	Continuous	Weight (kg)	75.8 ± 14.9 kg
Height (Talla)	Demographics	Continuous	Height (cm)	168.6 ± 9.4 cm
Body surface area (Sup_Corporal)	Demographics	Continuous	Body surface area (m ²)	1.9 ± 0.2 m ²
Sex (sex_bin)	Demographics	Binary	Sex (1=male, 0=female)	65.4% male
LGE presence (LGE_bin)†	Cardiac imaging	Binary	Late gadolinium enhancement present	54.8%
Dyslipidaemia (DLP_bin)	Comorbidities	Binary	Dyslipidaemia	48.3%
Hypertension (HTA_bin)	Comorbidities	Binary	Hypertension	49.3%
Atrial fibrillation (FA_bin)	Comorbidities	Binary	Atrial fibrillation (chronic or paroxysmal)	21.2%
Diabetes mellitus (DBT_bin)	Comorbidities	Binary	Diabetes mellitus (type 1 or 2)	14.7%
Pacemaker or ICD (DAI_bin)	Device history	Binary	Pacemaker or ICD implanted	2.4%
Family history of cardiomyopathy (Antecedfam_bin)	Family history	Binary	Family history of cardiomyopathy (HCM or amyloidosis)	20.2%
Prior HF hospitalisation (InternICC_bin)	Clinical history	Binary	Prior hospitalisation for heart failure	33.6%
Smoking history (Tabaco_bin)	Lifestyle	Binary	Smoking history (current or former)	41.8%
Repeat HF hospitalisation (InternacinporICC_bin)	Clinical history	Binary	Repeat hospitalisation for heart failure	5.5%

Table 2.8 - Sant Pau Clinical Features

LGE presence (LGE_bin) is derived from the same CMR acquisition used for classification

and therefore represents imaging-derived rather than fully independent clinical information. However, in standard clinical practice, LGE assessment is performed as part of the same CMR examination and its result is available to the reporting cardiologist at the time of diagnosis. Nevertheless, the potential circularity between LGE as a conditioning feature and the imaging-based classification task is acknowledged as a limitation of this work. A systematic ablation study evaluating model performance with and without LGE_bin is proposed as a direction for future work to precisely quantify its contribution to the observed performance gains.

There are a series of features directly related to the pathological processes of these four classes that are expected to carry strong discriminative signal. Maximum LV wall thickness (Especormximo1) is one of the parameters that define the hypertrophic phenotype; a value greater than 15 mm is one of the widely used diagnostic criteria for HCM [38, 39]. LV ejection fraction helps to distinguish the hypertrophic phenotype (HCM and HCM), which typically presents with preserved or hyperdynamic systolic function, from cardiac amyloidosis, in which LV function may be impaired in more advanced disease [40]. The presence of LGE in the myocardium indicates myocardial fibrosis or protein deposition, with distinct spatial distribution patterns characteristic of each pathology [40]. LV mass index quantifies the degree of LV hypertrophy and is elevated across all three LVH aetiologies relative to normal controls, though the extent and pattern differ between conditions [38]. Atrial fibrillation is found more commonly in cardiac amyloidosis than in the remaining classes, reflecting the infiltrative disruption of atrial architecture and conduction system by amyloid deposits [39]. Family history of cardiomyopathy is a highly sensitive feature of the hereditary form HCM, given its autosomal dominant inheritance pattern with high penetrance [38, 39].

2.3.6 Evaluation Protocol

Each classification trial is trained on all 292 patients using 5-fold stratified cross-validation (stratified by pathology label so that class distributions are maintained in each fold) with random state 42. Compared with a static split, this strategy provides a more reliable evaluation for small datasets by enabling estimation of both the mean performance and the standard deviation across folds. On each fold approximately 234 patients are used for training and 58 for validation.

Chapter 3

Data Preprocessing

This chapter presents the preprocessing pipeline used to all 3 datasets prior to training any model. For the public datasets ACDC and M&Ms, the preprocessing pipeline follows the protocol provided by the CineMA authors [1], ensuring that the inputs are consistent with the pre-training format and enabling direct reproduction of previously reported results. For the Sant Pau dataset, the same pipeline was adopted and adapted to handle the raw DICOM-derived NIfTI files prepared by Grzegorz Mrukwa, including automatic ED/ES frame detection from the 4D cine sequences, as no pre-annotated frame indices were available for this cohort. Therefore, these processing steps consist of the extraction of ED and ES frames, spatial re-sampling, center cropping, intensity normalization, and standardization of clinical features.

3.1 Overview

The preprocessing pipeline used for all three datasets is identical in structure, differing only in the parameters applied based on original image size and spacing. The complete pipeline is shown in Figure 3.1.

The preprocessing pipeline uses the SimpleITK library and utilities from the `cinema.data.sitk` module as provided by the CineMA authors [1], ensuring full consistency with the pre-training pipeline for ACDC and M&Ms. For the Sant Pau dataset, an additional preprocessing step was added to automatically extract the end-diastolic (ED) and end-systolic (ES) frames from the raw 4D cine NIfTI sequences, as no pre-annotated frame indices were available for this cohort. This step was implemented in a dedicated preprocessing notebook and applied prior to the standard CineMA pipeline.

3.2 ED/ES Frame Extraction

3.2.1 ACDC and M&Ms

For ACDC and M&Ms, ED and ES frames were predefined in the dataset's metadata files. The metadata for each patient contains `ed_index` and `es_index` fields, that store indices for the frame positions within the 4D cine sequence for the end-diastole and end-systole. These indices were manually annotated by the dataset provider. The respective 3D volumes are extracted from

the 4D NIfTI sequence by use of the corresponding index, one for end-diastole volume and one for end-systole volume for each patient.

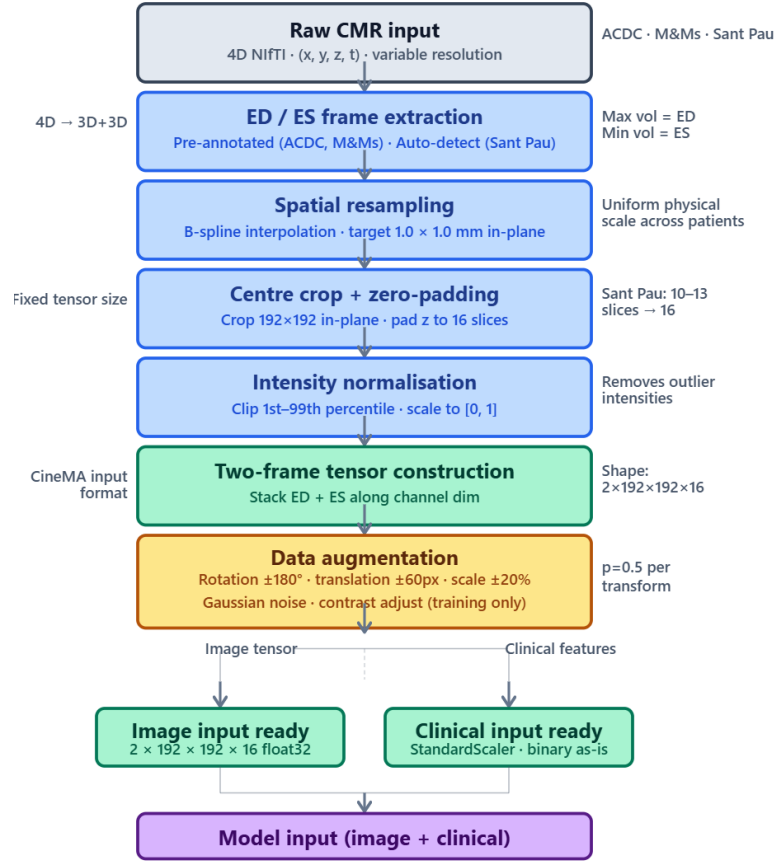


Figure 3.1: Preprocessing pipeline

3.2.2 Sant Pau

Raw imaging data at Sant Pau restored as 4D NIfTI volumes with dimensions (x, y, z, t), where t, where t is the number of temporal frames in the cine series. The ED and ES frame indices are not already labelled within the Sant Pau metadata. An automated detection step was applied prior to the standard preprocessing pipeline. ED and ES frames were identified using an intensity-based heuristic. The ED frame was taken as the temporal frame with the highest mean voxel intensity across the short-axis volume, and the ES frame as the frame with the lowest mean voxel intensity. This serves as an approximation to true volumetric ED/ES detection, which would require per-frame LV cavity segmentation masks and is acknowledged as a limitation of the Sant Pau preprocessing pipeline. A segmentation-based approach is proposed as a more principled alternative for future work.

Heuristics were run separately for each patient. Quality control plots were created for a representative sample of patients to confirm that the detected ED and ES frames are indeed at the correct points in the cardiac cycle. Expected frames were detected for all patients. A comparison of the contractility and appearance of the mid-ventricular slice at end-diastole (ED) and end-systole (ES) is shown in Figure 3.2.

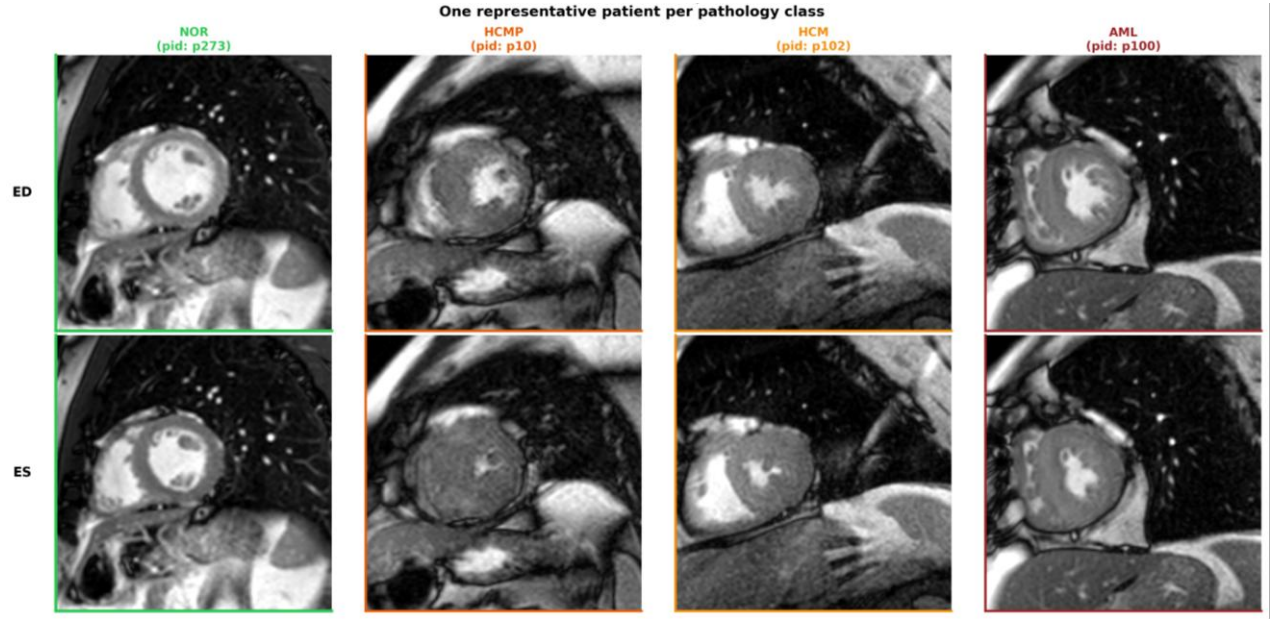


Figure 3.2: Sant Pau - ED vs ES at middle slice

3.3 Spatial Resampling

After ED/ES extraction, each 3D volume is resampled to a target in-plane spacing of 1.0×1.0 mm using third-order B-spline interpolation for image intensities, implemented from the CineMA preprocessing library. The slice thickness (z-spacing) is not resampled, as all three datasets already have slice thicknesses close to the target of 10.0 mm.

Resampling to 1.0 mm in-plane resolution at a uniform level eliminates the problem of relating the measured features to physical units when pixel size differences are solely based on the acquisition protocol (which impacts the actual physical sizes of wall thickness, ventricular volumes and scar extents differently depending on patient and protocol).

The visualization of contractility using absolute difference maps for all four pathology classes is displayed in Figure 3.3. Subtracting the end-systolic (ES) frame from the end-diastolic (ED) frame for one representative middle slice results in a localized $|ED-ES|$ wall motion map where brighter areas correspond to areas of higher physical displacement and deformation. We see that the normalized control (NOR) produces a clear uniform, bright ring representing healthy, symmetric thickening of the left ventricle wall as it contracts. Each of the pathology classes, however, displays impaired mechanics with both HCMP and HCM showing distinct regions of limited brightening where highly hypertrophied and fibrotic muscle does not participate, and AML revealing weakened, diffuse contraction.

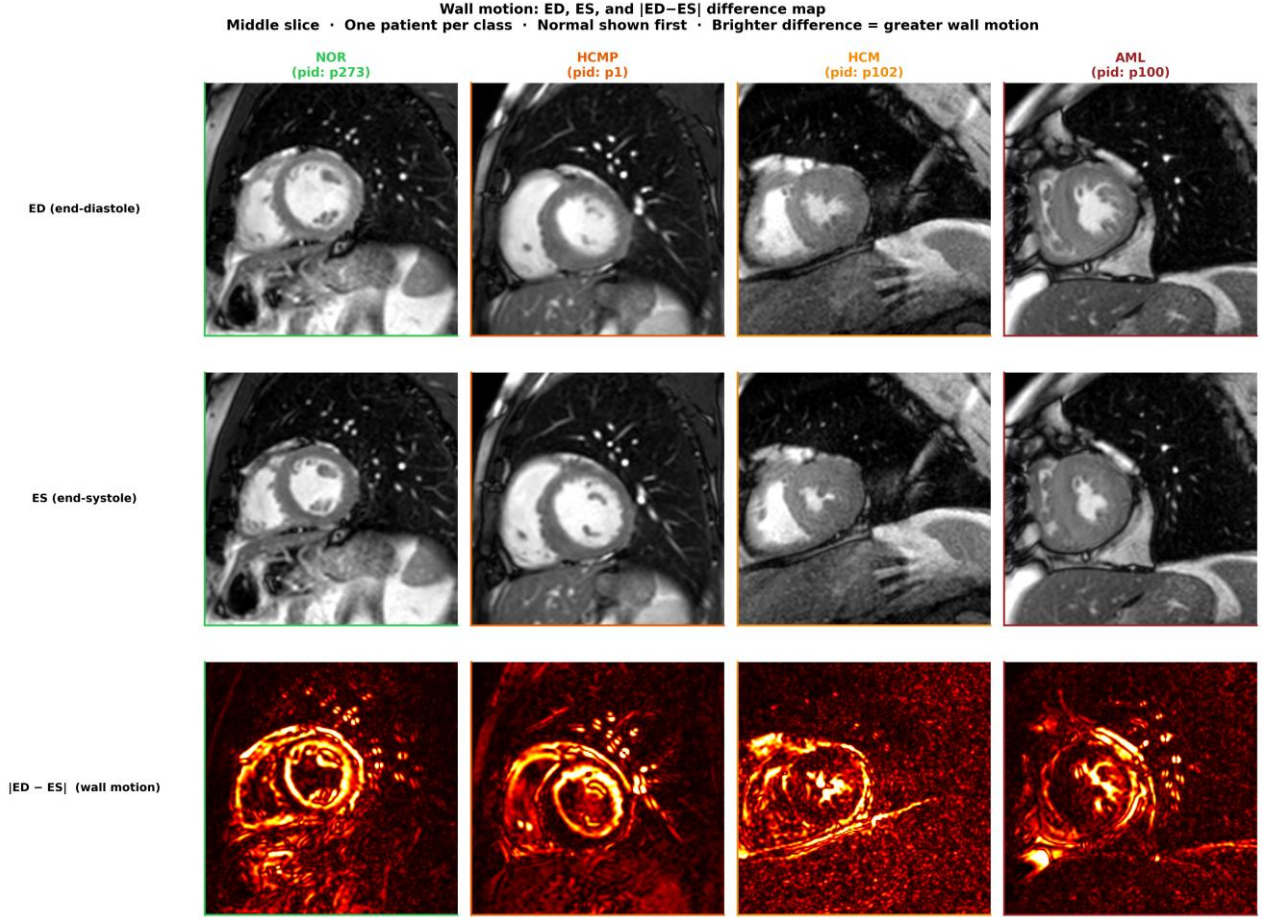


Figure 3.3: Wall motion ED, ES and $|ED-ES|$ difference map

3.4 Centre Cropping and Padding

All volumes across ACDC, M&Ms, and Sant Pau are resampled to a uniform in-plane resolution of 1.0×1.0 mm and produce consistent 192×192 in-plane dimensions following the CineMA preprocessing pipeline. Zero-padding is applied only in the slice (z) dimension to reach the target depth of 16 slices, as the number of short-axis slices varies across patients - 10 slices in the ACDC sample, 10–13 in Sant Pau, and up to 14 in M&Ms.

- Volumes with fewer than 16 slices (all Sant Pau patients have 10 - 13 slices) are padded with zero slices at the inferior end
- Volumes with more than 16 slices are center-cropped in the z dimension

The target size of $192 \times 192 \times 16$ is chosen to match the CineMA pre-training input format, ensuring that the patch embedding convolutional layers receive inputs of the expected spatial dimensions. Using a mismatched input size would require architectural modifications to the patch embedding or produce positional embedding interpolation errors.

3.5 Intensity Normalization

Intensity normalisation is applied independently per volume using the CineMA preprocessing pipeline:

1. Clips voxel intensities to the 1st and 99th percentile range to remove outliers caused by imaging artefacts or reconstruction errors
2. Scales the clipped intensities linearly to the range (0, 1)

This normalization on per-volume is performed on ED volumes and ES volumes separately. The procedure reduces intensity variability arising from differences in image acquisition across patients and scanner vendors, thereby standardizing the intensity distributions across studies. As a result, the model is encouraged to learn clinically relevant anatomical and intensity-related patterns rather than scanner-specific absolute intensity ranges. Normalized volumes are cast to uint8 to optimize space before saving as NIfTI files.

While training the model, an extra on-line intensity augmentation is performed for training volumes only by randomly applying contrast modification (gamma range [0.5,1.5], prob 0.5) and random Gaussian noise (prob 0.5) to create higher intensity variations and prevent over-fitting.

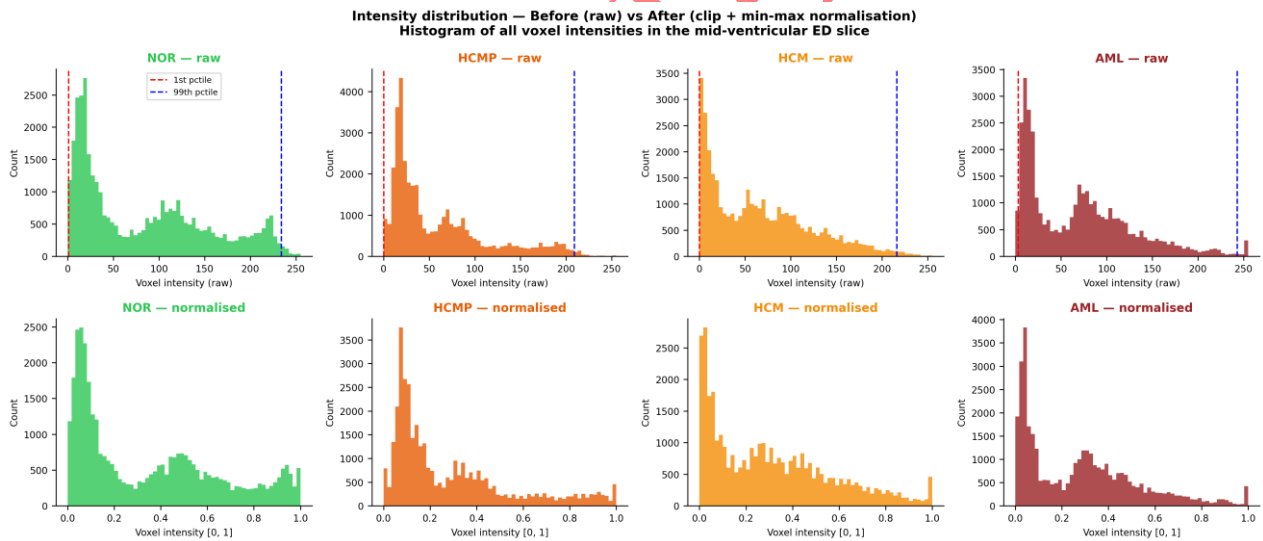


Figure 3.4: Voxel intensity distributions

All pathology class voxel intensity distributions are represented in Figure 3.4. Top panel of the figure shows the histograms of the raw voxel intensities as recorded in the mid-ventricular end-diastolic (ED) slices, where the 1st (red) and 99th (blue) percentile are represented by the dashed vertical lines indicating the values for outlier clipping. The effectiveness of clipping by high-intensity artifact followed by min-max scaling of all recorded values onto the range [0, 1] is evident in the bottom panel, where it transforms the extremely skewed raw data into uniformly distributed bimodal/multimodal patterns depending on the cohort's tissue type (HCMP, HCM, NOR, and AML).

3.6 Two-Frame Input Construction

After preprocessing, the ED and ES volumes for each patient are stacked along a new channel dimension to produce the final input tensor:

$$\mathbf{X} \in \mathbb{R}^{2 \times 192 \times 192 \times 16}$$

Where the first channel represents ED volume and the second represents ES volume. This two-frame sequence encapsulates maximum and minimum volume per cycle, which provides enough information to compute ejection fraction, wall motion and to classify pathologies. ED/ES frames were considered sufficient.

The CineMA backbone was pre-trained using two frame ED/ES inputs [1], so this format is readily available for using the pre-trained patch embedding weights. In order to transfer the representation learning of patch embeddings from pre-training to our task it is important to use the same format as used during pre-training, as using other formats would require us to train the patch embedding layer from scratch or to use a frame duplication heuristic, both of which would lessen the quality of the transferred representation learning.

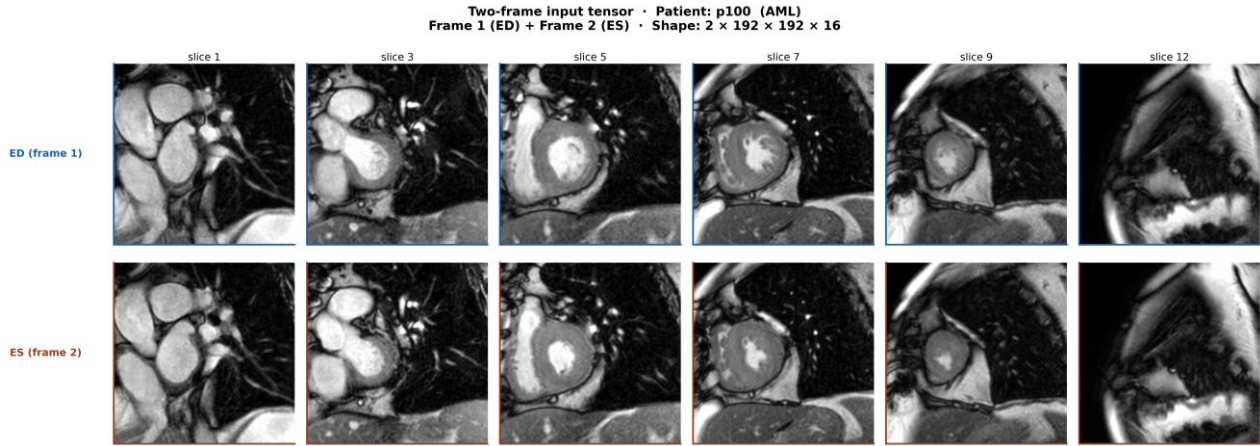


Figure 3.5: Two-frame Input Tensor

The layout of the multi-dimensional network input format is illustrated in Figure 3.5. This indicates the compositional layout of a two-frame input tensor for an example patient of the dataset suffering from HCM. A fully constituted 3D spatial volume for both ED (frame 1) and ES (frame 2) phases is directly appended together. Through representative slice slices (slice 1, 3, 5, 7, 9, 12) breakdown of panels, it can be observed that this setup maintains coherent spatial registration while at the same time, embedding motion throughout the time course within the structure of the data itself. The arrangement of the cardiac cycles into a unified tensor format allows the deep learning models to access the anisotropic spatial shape and regional wall motion contraction simultaneously in the ventricular space.

3.7 Clinical Feature Preprocessing

Clinical feature preprocessing differs from image preprocessing in that it must be performed in a fold-aware manner to prevent data leakage.

3.7.1 Continuous Feature Normalization

The 13 continuous clinical features in Sant Pau (and 3 in ACDC, 1 in M&Ms) are normalized using z-score normalization, which subtracts the feature mean and divides by the standard deviation:

$$x_{\text{norm}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

The mean (μ_{train}) and standard deviation (σ_{train}) are computed exclusively from the training data and then applied to normalize both the training and validation sets without refitting. This ensures that no information from the validation set influences the normalization parameters.

For Sant Pau, which uses 5-fold cross-validation with only a training and validation partition per fold (no separate test set), a new z-score normalization is fitted independently on the training partition of each fold k and then applied to normalize both the training and validation patients of that same fold k . This per-fold fitting is essential: fitting the scaler on the full dataset prior to splitting would leak validation fold statistics into the normalisation parameters, introducing a subtle but real form of data leakage that would lead to optimistic performance estimates.

3.7.2 Categorical or Categorical Clinical Feature Handling

The 14 binary clinical features in Sant Pau are passed directly to the model without any scaling, as binary features encoded as 0/1 are already on a consistent scale that does not benefit from standardization. For M&Ms, 1 binary feature (sex) is used alongside the single continuous feature (age), following the same convention.

3.7.3 Missing Value Handling

If a clinical feature was not recorded for a specific patient, zero imputation is performed after normalization. Mean imputation for numerical and most frequent for categorical would have been more appropriate. Imputing with 0 for clinical features that have not been recording inputting will artificially lower your dataset's average and can lead to inaccurate models. Zero imputation rarely occurred in the Sant Pau dataset, as the clinical data is collected in a standardized manner in a specialist center.

3.8 Data Augmentation

Online data augmentation is used solely on the training set during the model training phase. No data augmentation is used for the validation and test set. This augmentation pipeline is

built using MONAI transforms and executed randomly at each iteration of the training with no training epoch seeing the same augmented version of the same patient.

Transform	Probability	Parameters	Purpose
Random contrast adjustment	0.5	$\gamma \in [0.5, 1.5]$	Intensity diversity
Random Gaussian noise	0.5	σ sampled per call	Noise robustness
Random affine transformation	0.5	rotation $\pm 180^\circ$, translation ± 60 px, scale $\pm 20\%$	Geometric invariance
Random spatial crop	1.0	target size $192 \times 192 \times 16$	Positional invariance
Spatial padding	1.0	zero-pad to $192 \times 192 \times 16$	Fixed size guarantee
Intensity scaling	1.0	min-max to $[0, 1]$	Consistent range

Table 3.1 - Data Augmentation Pipeline

Table 3.1 shows the pipeline of data augmentation; the random affine transformation deserves specific comment. Rotation of $\pm 180^\circ$ is applied in the short-axis plane (around the z-axis), exploiting the rotational symmetry of the heart - a rotation of the short-axis image does not change the underlying pathology. Translation of ± 60 pixels allow the heart to be off-center within the 192×192 field of view, improving robustness to variations in patient positioning. The 20% scale variation accounts for differences in effective heart size due to acquisition geometry.

Clinical features are not augmented. Since clinical measurements are not images, but rather patient-level attributes (e.g., EF, wall thickness, age), their values remain constant when the image is augmented. In other words, when the image of patient p is randomly rotated in a particular training iteration, the clinical features associated with that particular patient p remain the same; rotating p 's MRI doesn't change his BMI. This is the desired behavior in the case of multimodal learning on tabular clinical data.

3.9 Class Imbalance Handling

In the Sant Pau dataset, we can observe the class imbalance of the samples - HCMP 56.5%, HCM 22.6%, NOR 12.7%, and AML only 8.2%. During the training phase, two methods are combined in order to avoid the model collapsing into the majority class:

Weighted random sampling. Each training patient is assigned a sampling weight inversely proportional to its class frequency within the training fold:

$$w_i = \frac{1}{n_{c(i)}}$$

where $n_{c(i)}$ is the number of training patients in the class of patient i . During training, weighted random sampling is applied by assigning each patient a sampling weight inversely proportional to its class frequency within the training fold. Batches are then drawn with replacement using these weights, ensuring that each training batch contains approximately equal proportions of all four classes regardless of their actual prevalence in the dataset. The sampling weights are recomputed for each fold from the fold-specific training set class counts.

Label smoothing. A label smoothing parameter of $\varepsilon = 0.1$ is applied to the cross-entropy loss,

replacing one-hot encoded labels with softened targets:

$$\tilde{y}_k = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{K} & \text{if } k = y \\ \frac{\varepsilon}{K} & \text{if } k \neq y \end{cases}$$

Where $K=4$ is the number of classes. Label smoothing can help overcome the overconfidence problems with imbalanced datasets, and can improve calibration [46]. This is an important consideration for the AML minority class, as it is very likely to yield near zero probability for AML when applied to uncertain patients. Both methods are only used in validation and test evaluation, where we restore the natural class distribution for an unbiased performance estimation.

Chapter 4

Development

This chapter presents the full methodological development of this thesis, covering the complete set of models, architectures, and training strategies evaluated across the three datasets.

4.1 CineMA Foundation Model Baseline

4.1.1 Architecture Overview

CineMA [1] is a multi-view convolutional-vision-transformer (ConvViT) masked autoencoder foundation model for cine cardiac MRI. It is pre-trained on 74,916 cine MRI studies from the UK Biobank using a masked autoencoder objective. CineMA combines two processing stages: a convolutional front-end to learn local spatial features and a Vision Transformer encoder to aggregate global context.

The convolutional front-end processes the two-frame ED/ES input tensor. It applies convolutional downsampling blocks that gradually decrease spatial resolution and increase feature dimensionality. In the short-axis view, two convolutional downsampling blocks precede the patch embedding stage, where a stride-2 convolution with 64 then 128 channels are applied before projecting the resulting feature map to the transformer embedding dimension 768. The hybrid architecture ensures local spatial patterns such as wall boundaries, chamber borders and scar texture are captured in the convolutional stage, prior to aggregation in the global transformer. The transformer encoder is a 12-block sequential transformer, where each block follows the standard architecture: a multi-head self-attention mechanism followed by a position-wise feed-forward network. There are 12 attention heads, each operating on query, key, and value projections of dimension 64, so the overall attention dimension is $12 * 64 = 768$, matching the embedding dimension.

The CineMA backbone contains approximately 89.5M parameters in total. The convolutional front-end accounts for approximately 1.8M parameters, while the remaining 87.7M comprise the 12 transformer blocks, positional embeddings, CLS token, and task-specific classification head. Throughout this thesis, the figure of 87.7M refers to the full set of parameters available for fine-tuning when the convolutional front-end is frozen, as reported by the CineMA training pipeline.

4.1.2 Pre-training Objective

CineMA was pre-trained using a masked autoencoder objective [3]. A randomly chosen proportion (commonly 75%) of patch tokens is masked, and the model is then tasked with reconstructing the masked patches using visible context. This encourages the encoder to learn generalisable cardiac image representations based on both the global structure and the local texture of the cardiac image. By reconstructing the masked patch, the encoder is forced to encode the global cardiac shape as well as local textures.

When pre-trained on a dataset comprising 74,916 cine CMR studies, i.e. subjects. This corresponds to 15 million CMR images from the UK Biobank, CineMA encounters variety of cardiac geometries, imaging sequences, and patient demographics. Through the masked autoencoder objective, the encoder is encouraged to learn rich structural representations of cardiac anatomy that may capture features relevant to downstream tasks such as chamber geometry, wall thickness, and myocardial texture, among others. Such representations have been shown to improve generalisation to downstream clinical tasks while reducing the amount of labelled data required for task-specific adaptation [3, 33].

4.1.3 Downstream Task Adaptation

For down-stream classification tasks, a task-specific head is attached on top of the frozen or fine-tuned CineMA encoder. A 768-dimensional global feature vector is obtained by taking an average of the CLS token embedding and the mean of all the patch token embeddings from the last transformer block:

$$\mathbf{f} = \frac{1}{2} \left(\mathbf{z}_{\text{CLS}} + \frac{1}{N_{\text{patches}}} \sum_{i=1}^{N_{\text{patches}}} \mathbf{z}_i \right) \in \mathbb{R}^{768}$$

The aggregated features incorporate the global, holistic, cardiac information present in the CLS token and the spatially separated information in the patch tokens to create a single feature vector representing global state and local morphological detail. This feature vector is fed into a linear classification head.

4.1.4 Input Format Compatibility

The two-frame ED/ES two-channel volume format CineMA was pre-trained on is employed by all the methods that this thesis investigated. This two-channel volume is a tensor of shape $\mathbf{X} \in \mathbb{R}^{2 \times 192 \times 192 \times 16}$, with the first and second channel representing short-axis volumes for ED and ES states. Consistency with the format that pre-trained network weights were optimized to is extremely important for transfer learning to succeed; when the input format is different, either a portion of the pre-trained patch embedding layer weights has to be reinitialized or the transferred representation quality will degrade significantly.

4.2 ResNet-50 Baseline

4.2.1 Architecture

ResNet-50 [1, 27] acts as a reference point against which all three networks are compared, using a non-pretrained CNN baseline. The network is a 50-layer residual convolutional network which is made up of four stages of bottleneck residual blocks that reduce the spatial dimension of feature maps and increase the number of channels as depth increases. We replace the ImageNet classification head of ResNet-50 with a linear layer whose size is the same as the number of classes being targeted for each dataset (5, 4, and 4 for ACDC, M&Ms and Sant Pau respectively).

The 2-frame input (ED + ES) is concatenated along the channel dimension prior to the first convolutional layer, resulting in a 2-channel input. Weights for the first convolutional layer are randomly initialized using Kaiming uniform initialization; there is no ImageNet pre-training as domain mismatch with respect to cardiac MRI slices makes ImageNet weight initialization not useful.

ResNet-50 has a high parameter count (46.2 million), all of which are randomly initialized, and therefore lies on the high parameter count, low data-efficiency extreme among methods tested. Its purpose is to indicate how well a method can do without pre-trained representations; the benefit of using CineMA pre-training can be seen in the difference between this and the CineMA-based methods.

4.2.2 Role in the Evaluation

There are a number of purposes the ResNet-50 baseline serves within our experimental design. It acts as a performance lower-bound for each data set against which the gains attributed to the pre-trained CineMA backbone are compared. It illustrates how challenging classification without domain specific pre-training can be: on Sant Pau where only ~186 training patients are available per fold, trying to learn 46.2 million parameters without an explicit mechanism for doing so results in high variance across folds. Comparing the ResNet-50 and CineMA + LoRA setups allow us to evaluate solely the contribution of the pre-trained representations.

4.3 CineMA Full Fine-Tuning

4.3.1 Method Description

CineMA full fine-tuning (Full FT) uses pre-trained CineMA backbone weights for initialization and fine-tunes the entire backbone parameters. The pre-trained weights enable excellent initialization with rich representations for cardiac images, and the full fine-tuning can adjust all the layers of the backbone according to the new domain and task.

The full fine-tuning configuration follows the two-phase training protocol used in the CineMA paper [1]:

Phase 1 - Head warm-up (20 epochs): The backbone is fully frozen and only the classification head is trained at a learning rate of 1×10^{-3} . This phase allows the randomly initialised head to learn a basic mapping from the pre-trained features to the target classes before the backbone weights are updated, preventing the large gradients from an untrained head from disrupting the pre-trained representations at the start of training.

Phase 2 - Full fine-tuning (180 epochs): Both the backbone and head are trained simultaneously. The backbone learning rate is reduced to 1×10^{-5} - two orders of magnitude below the Phase 1 head learning rate - to ensure that the pre-trained representations are adapted gradually rather than overwritten. The head learning rate is also reduced from Phase 1 to 1×10^{-4} .

The trainable parameters of CineMA are shown in Table 4.1. With approximately 87.7 million trainable parameters, full fine-tuning represents the highest-capacity adaptation strategy evaluated in this thesis. On the balanced and moderately sized ACDC benchmark, full fine-tuning achieves the strongest image-only performance, reaching approximately 95% validation AUC, a result consistent with the findings reported in the original CineMA paper [1]. However, on Sant Pau, where only approximately 234 training patients are available per fold, the large number of trainable parameters makes full fine-tuning highly susceptible to overfitting, resulting in an AUC of $73.48 \pm 9.54\%$ - below both the parameter-efficient LoRA image-only baseline ($74.13 \pm 6.05\%$) and all CC-LoRA configurations ($89.65\% - 91.03\%$). This confirms that parameter efficiency is not merely a computational advantage in the low-data regime but a meaningful quality improvement, and that the primary contribution of this thesis - CC-LoRA - substantially outperforms full fine-tuning on the primary evaluation dataset despite training only 0.215% of the backbone parameters.

Component	Parameters	Trainable
Convolutional front-end	~1.8M	Phase 2
Transformer blocks 1-12	~85.9M	Phase 2
Classification head	~3K	Both phases
Total (Phase 2)	~87.7M	100%

Table 4.1 - CineMA Full FT Trainable Parameters

4.4 CineMA + LoRA (Image-Only)

4.4.1 LoRA Background

Low-Rank Adaptation (LoRA) [6] is a parameter-efficient fine-tuning method first introduced for large language models and later adopted for vision and medical imaging fields. LoRA builds on the idea that fine-tuning weight updates are intrinsically low-rank, allowing them to be approximated by a low-rank decomposition rather than updating the full weight matrix. For a pre-trained weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA constrains the weight update ΔW to be the product of two low-rank matrices:

$$\Delta W = BA$$

where $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $A \in \mathbb{R}^{r \times d_{\text{in}}}$, with $\text{rank } r \ll \min(d_{\text{out}}, d_{\text{in}})$. The adapted forward pass computes:

$$h = Wx + \frac{\alpha}{r}BAx = Wx + \frac{\alpha}{r}\Delta Wx$$

where α is a scaling hyperparameter that controls the magnitude of the LoRA update relative to the pre-trained weights. During training, W is completely frozen and only A and B are updated. The matrices A and B are initialized such that the LoRA update is zero at the start of training: A is initialized with Kaiming uniform initialization and B is initialized to zero, guaranteeing $\Delta W = BA = 0$ at epoch 0. This ensures training stability by starting from the pre-trained model's behaviour and introducing the adaptation gradually.

At inference time, the LoRA update can be merged directly into the pre-trained weight matrix - $W_{\text{merged}} = W + \frac{\alpha}{r}BA$ - adding zero computational overhead compared to the unmodified pre-trained model.

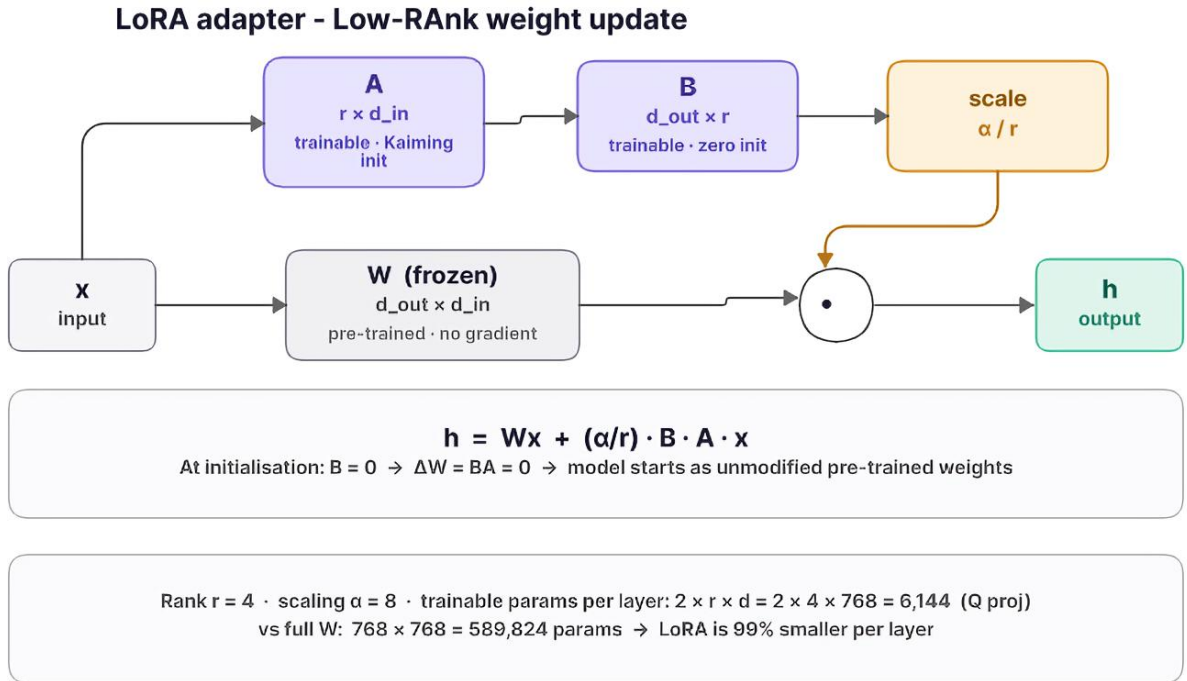


Figure 4.1: LoRA Adapter

Figure 4.1 shows the LoRA adapter decomposes the weight update as $\Delta W = BA$ where B and A are low-rank matrices. W is frozen throughout training. Zero initialization of B ensures the adapter output is zero at epoch 0, preserving the pre-trained model's behavior.

4.4.2 Application to CineMA

LoRA adapters are injected into the query (Q) and key-value (KV) projection matrices of the multi-head self-attention mechanism within the last six transformer blocks (blocks 6–11). We adopt the design of applying modification on the last blocks only, as shown in Med-DualLoRA [8], it is only necessary to tune the later blocks (higher level semantic representation) for downstream adaption and the lower-level spatial feature which is common to all cardiac imaging tasks could be maintained without modification.

The projection matrices targeted are:

- **Q projection:** $W_Q \in \mathbb{R}^{768 \times 768}$, adapted with LoRA rank $r = 4$, producing $A_Q \in \mathbb{R}^{4 \times 768}$ and $B_Q \in \mathbb{R}^{768 \times 4}$
- **KV projection:** $W_{KV} \in \mathbb{R}^{1536 \times 768}$, adapted with LoRA rank $r = 4$, producing $A_{KV} \in \mathbb{R}^{4 \times 768}$ and $B_{KV} \in \mathbb{R}^{1536 \times 4}$

The LoRA scaling factor is set to $\alpha = 8$, following the hyperparameter choices of Med-DualLoRA [8] which demonstrated these values work well for the CineMA backbone. A LoRA dropout rate of 0.05 is applied to the input before the low-rank multiplication to provide mild regularisation. All backbone parameters outside the LoRA matrices - layer norms, MLP layers, output projections, and the convolutional front-end are frozen throughout training. Only the LoRA matrices A and B are updated alongside the classification head.

4.4.3 Parameter Count

The hyperparameter efficiency that LoRA has compared to fine-tuning the full backbone on a small dataset (only 0.127% of the backbone parameters are trained) is its biggest benefit. With less than 115 K training parameters and ~234 patients per fold on Sant Pau, the risk of over-fitting compared to 87.7M parameters using full fine-tuning becomes significantly less (Table 4.2).

Component	Parameters	Percentage
LoRA A matrix, Q projection (per block)	3,072	—
LoRA B matrix, Q projection (per block)	3,072	—
LoRA A matrix, KV projection (per block)	6,144	—
LoRA B matrix, KV projection (per block)	6,144	—
LoRA total per block	18,432	—
LoRA total (6 blocks)	110,592	0.123%
Classification head	~3,076	0.003%
Total trainable	~113,668	~0.127%
Frozen backbone	~89.4M	99.87%

Table 4.2 - CineMA + LoRA Trainable Parameters

4.5 Late Fusion Architecture

4.5.1 Motivation and Design

Late Fusion is one of the most widely used in combining imaging and clinical data in medical AI [12, 13]. Each modality is processed separately; a global image embedding is extracted from the backbone and a clinical embedding is extracted from an MLP that is trained independently from the image backbone. Late Fusion combines the two modalities only in the final classification stage through concatenation. Its relative simplicity and interpretability along with requiring no change to the image processing pipeline makes it a natural, and frequently employed, baseline.

In this thesis, two variations on Late Fusion are explored: Full FT + Late Fusion (where the image backbone is fully fine-tuned along with the clinical MLP and head) and parameter-efficient fine-tuning using LoRA (LoRA + Late Fusion), where the backbone weights remain frozen and only the LoRA adapters, clinical MLP, and classification head are trained. Both variations utilize the same fusion architecture with differing image backbone adaptation strategies.

4.5.2 Architecture

The late fusion architecture consists of three components:

Image branch: The CineMA backbone (either fully fine-tuned or LoRA-adapted) extracts a 768-dimensional image embedding using the same CLS + mean-patch aggregation described in Section 4.1.3:

$$\mathbf{f}_{\text{img}} \in \mathbb{R}^{768}$$

Clinical branch: A two-layer MLP maps the n_{tab} -dimensional clinical feature vector to a 32-dimensional clinical embedding:

$$\text{ClinicalMLP: } \mathbf{x}_{\text{tab}} \xrightarrow{W_1 \in \mathbb{R}^{64 \times n_{\text{tab}}}} \text{ReLU} \xrightarrow{W_2 \in \mathbb{R}^{32 \times 64}} \text{ReLU} \rightarrow \mathbf{c} \in \mathbb{R}^{32}$$

The MLP is intentionally small - two layers with hidden dimension 64 to prevent the clinical branch from dominating the fused representation on small datasets. Dropout (rate 0.1) is applied after the first layer.

Fusion and classification head: The image and clinical embeddings are concatenated to form an 800-dimensional fused representation, which is normalized and passed through a three-layer classification head:

$$\mathbf{f}_{\text{fused}} = [\mathbf{f}_{\text{img}} \parallel \mathbf{c}] \in \mathbb{R}^{800}$$

$$\hat{\mathbf{y}} = \text{Head}(\text{LN}(\mathbf{f}_{\text{fused}})): 800 \xrightarrow{\text{GELU}} 256 \xrightarrow{\text{GELU}} 64 \rightarrow n_{\text{classes}}$$

where LN denotes layer normalization applied to the fused representation before the head, which stabilizes the scale of the concatenated embeddings. Dropout (rate 0.3) is applied after the first two head layers.

4.5.3 Fundamental Limitation of Late Fusion

One weakness of the late fusion approach is that the image representation is learned completely in the blind of patient clinical data. In fact, the backbone network computes the same feature transformation on all patients independent of their clinical information. A 35-year-old athlete vs a 75-year-old patient with diabetes, sharing the same pathology, are associated with different image features solely due to anatomy, but a system trained in the late fusion regime has no way to "re-weight" how the features are extracted based on age, comorbidity profile.

This issue is critical for any classification task in which ambiguous image features can only be resolved with clinical context. In LVH etiology classification, differentiating between HCMP, HCM, and AML is particularly challenging from imaging alone, as all three conditions present with increased left ventricular wall thickness and overlapping morphological features. Knowing that a patient has a family history of cardiomyopathy increases the prior probability of the primary genetically-based form HCMP and in this context, a borderline wall thickness measurement should shift the model's prediction toward HCMP rather than remaining ambiguous.

Similarly, a history of long-standing hypertension supports the diagnosis of hypertensive HCM, while the combination of concentric hypertrophy with atrial fibrillation and elevated troponin raises suspicion for AML. In each of these scenarios, clinical context does not merely supplement the image - it fundamentally changes how the image should be interpreted. In late fusion, clinical information is made available to the classification head, but the head receives an image embedding computed in complete ignorance of the patient's clinical profile and therefore cannot influence how the image features themselves were extracted.

This is precisely what CC-LoRA is designed to circumvent as discussed in Section 4.6.

4.6 CC-LoRA: Clinical-Conditioned LoRA

4.6.1 Core Idea

This thesis introduces Clinical-Conditioned LoRA (CC-LoRA), a method that extends the standard LoRA fine-tuning process by conditioning the LoRA adapter's output on the patient's clinical features through Feature-wise Linear Modulation (FiLM) [11]. This allows the clinical information to modulate image feature extraction directly within selected transformer blocks of the CineMA encoder, as shown in Figure 4.2.

The main difference from late fusion is the level at which clinical information is fused. In late fusion, Clinical features are concatenated with the image embedding at the end of the whole transformer calculations. The image encoder works purely clinically-agnostic through the whole stage. Instead, in CC-LoRA, clinical features are injected in the middle of the transformer block, conditioning the LoRA adapter outputs on different levels of the encoder hierarchy. Hence, clinical information guides the backbone how to attend to the cardiac images and hence generating clinically-aware images representations instead of clinically-agnostic images representations followed by a late fusion.

4.6.2 Standard LoRA Review

In standard LoRA, the adapted output of a linear projection layer is:

$$h_{\text{LoRA}} = Wx + \frac{\alpha}{r}BAx$$

where W is the frozen pre-trained weight, B and A are the low-rank adapter matrices, and $\frac{\alpha}{r}$ is the scaling factor. The adapter output $\frac{\alpha}{r}BAx$ is identical for all patients with the same input x - it is a

fixed linear transformation that does not depend on any patient-specific information.

4.6.3 FiLM Conditioning

Feature-wise Linear Modulation (FiLM) [11] provides the mechanism for making the adapter output patient-specific. Given a conditioning vector $\mathbf{c}$ (the clinical embedding), FiLM applies an affine transformation to an intermediate feature:

$$\text{FiLM}(\mathbf{f} \mid \mathbf{c}) = \gamma(\mathbf{c}) \odot \mathbf{f} + \beta(\mathbf{c})$$

where $\gamma(\mathbf{c})$ and $\beta(\mathbf{c})$ are element-wise scaling and shift vectors generated as linear functions of the conditioning vector $\mathbf{c}$, and $\odot$ denotes element-wise multiplication. The scaling vector γ modulates the amplitude of individual feature dimensions, effectively emphasizing or suppressing specific features based on clinical context. The shift vector β translates the feature distribution along each dimension, shifting the operating point of the subsequent computation.

4.6.4 CC-LoRA Forward Pass

CC-LoRA integrates FiLM conditioning directly into the LoRA adapter output. The adapted forward pass for a CC-LoRA linear layer is:

$$h_{\text{CC-LoRA}} = Wx + \frac{\alpha}{r} [(1 + \Delta\gamma(\mathbf{c})) \odot BAx + \beta(\mathbf{c})]$$

where:

- $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ is the frozen pre-trained weight matrix
- $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $A \in \mathbb{R}^{r \times d_{\text{in}}}$ are the LoRA matrices
- $\mathbf{c} \in \mathbb{R}^{32}$ is the clinical embedding generated by the shared ClinicalFiLM module
- $\Delta\gamma(\mathbf{c}) = W_\gamma \mathbf{c} \in \mathbb{R}^{d_{\text{out}}}$ is the patient-specific FiLM scaling deviation
- $\beta(\mathbf{c}) = W_\beta \mathbf{c} \in \mathbb{R}^{d_{\text{out}}}$ is the patient-specific FiLM shift
- $\frac{\alpha}{r}$ is the LoRA scaling factor

The use of $1 + \Delta\gamma$ rather than γ directly is a critical design choice. At initialization, $\Delta\gamma = 0$ and $\beta = 0$, so the CC-LoRA forward pass reduces exactly to standard LoRA:

$$h_{\text{CC-LoRA}}|_{\text{init}} = Wx + \frac{\alpha}{r} [(1 + 0) \odot BAx + 0] = Wx + \frac{\alpha}{r} BAx = h_{\text{LoRA}}$$

This identity initialization property ensures that at epoch 0, CC-LoRA behaves identically to plain LoRA, providing a stable starting point that does not disrupt the pre-trained representations.

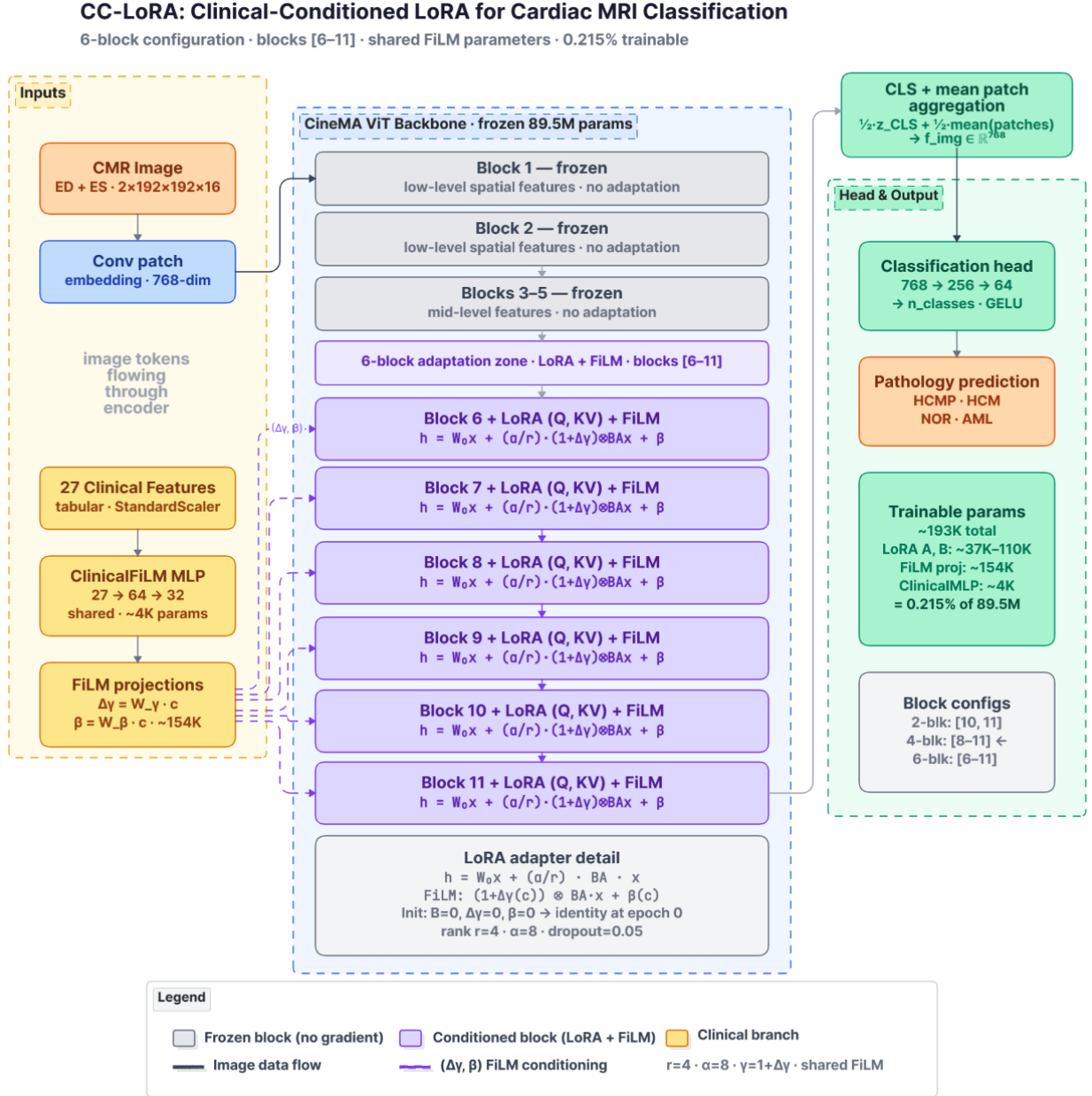


Figure 4.2: CC-LoRA Architecture

As training proceeds, $\Delta\gamma$ and β gradually diverge from zero, learning to apply patient-specific modulations that improve the classification loss. This gradual introduction of clinical conditioning is especially important on small datasets such as Sant Pau, where a large initial perturbation to the feature space could destabilize training.

4.6.5 Clinical FiLM Module

The Clinical FiLM module is the shared clinical conditioning component of CC-LoRA. It maps the input clinical feature vector to the conditioning parameters used across all conditioned transformer blocks.

The module architecture is:

$$\mathbf{c} = \text{ReLU}(W_2 \text{ReLU}(W_1 \mathbf{x}_{\text{tab}})) \in \mathbb{R}^{32}$$

where $W_1 \in \mathbb{R}^{64 \times n_{\text{tab}}}$ and $W_2 \in \mathbb{R}^{32 \times 64}$. The 32-dimensional embedding $\mathbf{c}$ is then linearly projected to the FiLM parameters for each conditioned layer:

$$\Delta\gamma = W_\gamma \mathbf{c}, \beta = W_\beta \mathbf{c}$$

where $W_\gamma, W_\beta \in \mathbb{R}^{d_{\text{out}} \times 32}$ and d_{out} is the output dimension of the corresponding LoRA layer (768 for Q projections, 1536 for KV projections).

Crucially, the FiLM parameters are shared across all conditioned blocks. There is only one Clinical FiLM module, and thus one conditioning vector $\mathbf{c}$, which is sent to every conditioned layer in every conditioned block. This reduces the clinical conditioning parameter count from about 590K (if we were to have one FiLM projection per block, per projection) down to about 154K (shared across all), which is significant over a 292-patient dataset. Another effect of the shared structure is a global interpretation of the clinical condition; we use the same clinical embedding to influence all levels of abstraction in the hierarchy, rather than having one specific clinical feature be weighted heavily in one block and a different clinical feature be weighted heavily in another.

The forward pass at inference time is:

1. Compute clinical embedding: $\mathbf{c} = \text{ClinicalMLP}(\mathbf{x}_{\text{tab}})$
2. Compute FiLM parameters: $(\Delta\gamma, \beta) = (W_\gamma \mathbf{c}, W_\beta \mathbf{c})$
3. Store the computed $(\Delta\gamma, \beta)$ parameters in the model's internal state, making them accessible to each conditioned layer during the subsequent backbone forward pass.
4. Run the backbone forward pass - each CC-LoRA Linear layer reads $(\Delta\gamma, \beta)$ from state
5. Clear FiLM state after the forward pass

4.6.6 Block Selection

CC-LoRA is applied to the Q and KV projections of the attention mechanism in the last K transformer blocks of the CineMA encoder. Three configurations are evaluated in Table 4.3.

Configuration	Conditioned Blocks	Description
CC-LoRA 2 blocks	(10, 11)	Last 2 blocks only - most conservative
CC-LoRA 4 blocks	(8, 9, 10, 11)	Last 4 blocks - intermediate
CC-LoRA 6 blocks	(6, 7, 8, 9, 10, 11)	Last 6 blocks - maximum evaluated

Table 4.3 - CC-LoRA Block Configurations

The rationale to only condition later blocks is because transformer blocks later in the output sequence are higher semantic representations, the abstract structural features of the heart that are most sensitive to the context of the patient. The earlier blocks in the sequence represent lower-level features, such as edges, textures, or local shapes, that are invariant across different cardiac

abnormalities and would therefore not be greatly improved by conditioning with the patient-specific features. The choice of which transformer blocks to condition was determined empirically through the three configurations evaluated in this thesis (2, 4, and 6 blocks), with results presented in Chapter 5. This design space is consistent with the findings of Med-DualLoRA [8], which also applies LoRA to the CineMA backbone and demonstrates that adapting only a small number of the final transformer blocks is sufficient for effective downstream adaptation.

The three conditions provided for a way to systematically evaluate how conditioning depth interacts with performance, and to explore the hypothesis that a conditioning depth of two blocks would capture the main clinical to image conditioning effect.

4.6.7 Comparison with Late Fusion

Table 4.4 summarizes the architectural differences between late fusion and CC-LoRA across three key dimensions.

Property	Late Fusion	CC-LoRA
Clinical integration point	After all encoder blocks (output level)	Inside selected encoder blocks
Image representation	Clinically agnostic	Clinically conditioned
Backbone modification	None (only head + clinical MLP)	LoRA + FiLM conditioning
Clinical influence on attention	None	Direct - modulates Q and KV projections
Parameter count (clinical)	~3K (MLP) + ~200K (head)	~154K (shared FiLM)
Initialisation	Random head	Identity (reduces to plain LoRA)

Table 4.4 - Late Fusion vs CC-LoRA Comparison

The critical difference is the layer at which the clinical information influences the computation. In late fusion, the full transformer forward pass is performed, and then the clinical features affect it. This model produces an image embedding and then adds a clinical modification to it using the clinical branch. In CC-LoRA, clinical conditioning affects the transformer forward pass at multiple layers to influence the attention patterns and the features that are used to construct the image embedding. This allows the model, in principle, to learn attention patterns that are responsive to clinical context - for example, attending more strongly to the septal wall when the clinical embedding encodes high wall thickness, or weighting LGE-related image regions differently when the LGE feature is positive though these mechanisms are not explicitly analyzed in this thesis and are proposed as a direction for future interpretability work.

4.6.8 Parameter Count

Component	Parameters	Percentage of 89.5M
LoRA A matrix, Q (per block)	3,072	—
LoRA B matrix, Q (per block)	3,072	—
LoRA A matrix, KV (per block)	6,144	—
LoRA B matrix, KV (per block)	6,144	—
LoRA total (4 blocks)	73,728	0.082%
ClinicalFiLM MLP ($\mathbf{W}_1, \mathbf{W}_2$)	~3,872	0.004%
FiLM projections ($\mathbf{W}_\gamma, \mathbf{W}_\beta$)	~150,528	0.168%
Classification head	~3,076	0.003%
Total trainable	~231,204	0.258%

Frozen backbone	~89.3M	99.742%
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Table 4.5 - CC-LoRA Trainable Parameters (4-block configuration)

The FiLM projection parameters (W_γ, W_β) shown in Table 4.5 account for the majority of the trainable parameter budget because they map from the 32-dimensional clinical embedding to the full output dimension of each conditioned layer (768 for Q, 1536 for KV), producing a large projection matrix that is shared across all conditioned blocks. The 4-block and 6-block configurations add 36,864 LoRA parameters per additional block pair but keep the FiLM parameters identical (shared), resulting in totals of approximately 229,674 (4 blocks) and 266,538 (6 blocks) - all well below 0.30% of the full backbone.

4.7 Training Configuration

4.7.1 Optimizer

All methods are trained using Adam W [43] with weight decay $\lambda = 0.05$ and momentum parameters $\beta_1 = 0.9, \beta_2 = 0.95$. Adam W decouples the weight decay from the gradient update, applying it directly to the weights rather than incorporating it into the gradient step, which is particularly beneficial for transformer architectures where the weight magnitudes vary significantly across layers.

4.7.2 Learning Rate Schedule

A warmup-cosine schedule is applied to all methods. The learning rate increases linearly from 0 to the base learning rate over 10 warmup epochs, then decays following a cosine annealing schedule to a minimum of 10^{-6} . The warmup phase prevents large gradient steps during the first epochs when the randomly initialized head produces large errors, which could destabilize the pre-trained backbone representations. The cosine decay then gradually reduces the learning rate as the model approaches convergence, allowing fine-grained optimization in the later training stages.

4.7.3 Layer-wise Learning Rate Decay

For LoRA and CC-LoRA, a layer-wise learning rate decay factor of $\eta_{\text{decay}} = 0.75$ is applied across the conditioned transformer blocks. Shallower blocks (further from the output) receive lower learning rates:

$$\eta_{\text{block}_i} = \eta_{\text{max}} \cdot \eta_{\text{decay}}^{(K-1-i)}$$

where K is the total number of conditioned blocks and i is the zero-indexed block position within the conditioned range. For the 6-block configuration, the learning rates range from $\eta_{\text{max}} \cdot 0.75^5 = 0.237 \cdot \eta_{\text{max}}$ for block 6 to η_{max} for block 11. This decay preserves low-level representations in shallower blocks while allowing more aggressive adaptation of the high-level semantic representations in the final blocks.

4.7.4 Loss Function

Cross-entropy loss with label smoothing ($\varepsilon = 0.1$) is used for all classification experiments:

$$\mathcal{L} = - \sum_{k=1}^K \tilde{y}_k \log \hat{y}_k$$

where $\tilde{y}_k = (1 - \varepsilon) \cdot \mathbb{1}[k = y] + \frac{\varepsilon}{K}$ is the smoothed target distribution and $\hat{y}_k$ is the predicted probability for class k . Label smoothing prevents the model from becoming overconfident on the training set and has been shown to improve calibration on imbalanced datasets [46].

4.7.5 Regularization

Several regularization strategies are applied in combination to prevent overfitting on the small Sant Pau dataset:

Gradient clipping: The global gradient norm is capped at a maximum of 5.0 prior to each parameter update, as the randomly initialized head could lead to high magnitudes for gradients early in training (before the head learns, and thus could potentially cause explosion)

Weight decay: Adam W weight decay of 0.05 is applied to all trainable parameters, providing L2 regularization that penalizes large weight magnitudes and promotes sparse, generalizable solutions.

Dropout: Dropout with rate 0.05 is applied within the LoRA forward pass (on the input to the low-rank multiplication) and within the classification head (rate 0.3 after each hidden layer).

Early stopping: Training is stopped when no improvement is observed in validation AUC for 5 consecutive evaluation steps (evaluation is performed at every 10 epochs), within a limit of 200 epochs. Best checkpoint- the model state that gets best validation AUC, is saved and used for test evaluation.

4.7.6 Mixed Precision Training

All training is performed using automatic mixed precision (AMP), which selectively computes operations in 16-bit floating point where numerically safe while maintaining 32-bit precision for sensitive operations such as loss computation and weight updates. This reduces GPU memory consumption and accelerates training without sacrificing numerical stability. To prevent underflow of small gradient values that can occur when using reduced precision, dynamic loss scaling is applied: the loss is multiplied by a large-scale factor before backpropagation and the resulting gradients are unscaled before the parameter update, with the scale factor adjusted automatically throughout training based on whether gradient overflow is detected.

This decreases VRAM usage by roughly 40-50%, compared to training entirely in float32, to a larger effective batch size on the RTX 2080 Ti (11GB VRAM) used for the Sant Pau, ACDC, and M&Ms trials.

4.7.7 Batch Size and Weighted Sampling

Hyperparameters for training are presented in Table 4.6, We use batch size of 8 for all datasets. For Sant Pau, we employ the weighted random sampling with inverse class frequency weights to roughly balance the class representation in each training batch, so to alleviate the class imbalance problem with majority class HCMP (56.5%).

Hyperparameter	Value	Applies to
Base learning rate	1×10^{-3}	All methods
Backbone LR (Full FT Phase 2)	1×10^{-5}	Full FT only
AdamW β_1, β_2	0.9, 0.95	All methods
Weight decay	0.05	All methods
Warmup epochs	10	All methods
Max epochs	200	All methods
Batch size	8	All datasets
Label smoothing ϵ	0.1	All methods
Gradient clip norm	5.0	All methods
Early stopping patience	5 checks $\times$ 10 epochs	All methods
Layer-wise LR decay	0.75	LoRA, CC-LoRA
LoRA rank r	4	LoRA, CC-LoRA
LoRA scaling α	8	LoRA, CC-LoRA
LoRA dropout	0.05	LoRA, CC-LoRA
AMP	Enabled	All methods

Table 4.6 - Training Hyperparameters

Figure 4.3 trainable parameters shows counts of all methods compared to 89.5M CineMA backbone. CC-LoRA trains only <0.3% parameters of the backbone, hence being the most parameter efficient among the tested multimodal approaches.

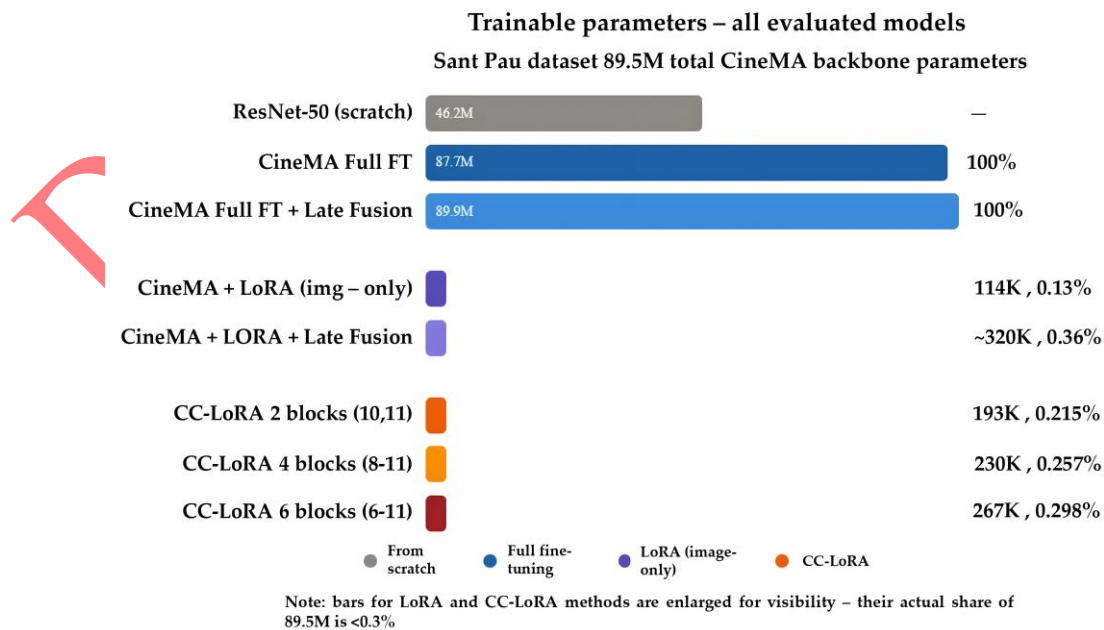


Figure 4.3: Parameter Efficiency Chart

4.8 Evaluation Metrics

4.8.1 ROC AUC

The area under the receiver operating characteristic curve (ROC AUC), computed using a one-vs-rest (OvR) strategy with macro averaging across classes, is one of the primary evaluation metrics used throughout this thesis. While AUC is threshold-independent and robust to class imbalance when macro-averaged across classes, it can be misleading in severely imbalanced settings where high discriminability for the majority class dominates the aggregate score. For this reason, balanced accuracy and macro F1 are reported alongside AUC as complementary metrics that provide a more complete picture of per-class classification performance.

$$AUC_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K AUC(k \text{ vs rest})$$

4.8.2 Balanced Accuracy

Balanced accuracy is defined as the arithmetic mean of per-class sensitivity (recall) [53]:

$$\text{BalAcc} = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FN_k}$$

where TP_k and FN_k are the true positives and false negatives for class k in the OvR decomposition. Balanced accuracy provides a class-imbalance-corrected accuracy measure that assigns equal weight to each class regardless of prevalence. A balanced accuracy of 25% corresponds to random chance for a 4-class problem.

4.8.3 Macro F1 Score

The macro-averaged F1 score computes the harmonic mean of precision and recall for each class and averages across all classes:

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot \text{Prec}_k \cdot \text{Rec}_k}{\text{Prec}_k + \text{Rec}_k}$$

F1 penalizes methods for too high recall (by adding too many predicted positive cases which then reduces precision) or for too high precision (by being too conservative which then reduces recall). Macro F1 for imbalanced data set, more than accuracy.

4.8.4 Per-class Accuracy

Individual class accuracy (sensitivity) per class k is reported along with aggregate results to pinpoint failures at a per-class level. This is especially critical for the AML minority class (8.2% of Sant Pau) where aggregate metrics can obscure a complete failure to predict the minority class.

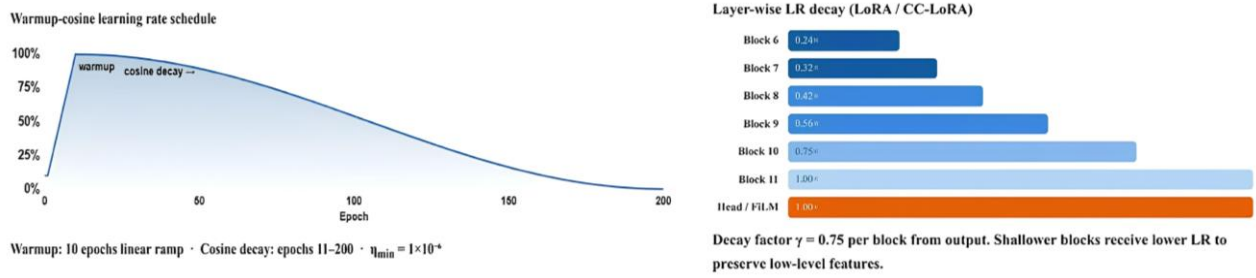


Figure 4.4: Training and regularization parameters. Optimization incorporates AdamW ($wd=0.05$), gradient clipping ($norm=5.0$), and label smoothing ($\epsilon=0.1$). Regularization includes early stopping ($patience=5$), LoRA dropout (0.05), and head dropout (0.3). AMP float16 precision.

Figure 4.4 shows, the **Left:** warmup-cosine learning rate schedule over 200 epochs with 10-epoch linear warmup. **Right:** layer-wise learning rate decay across conditioned blocks ($\gamma=0.75$), assigning lower learning rates to shallower blocks. Bottom: full regularization configuration.

4.8.5 Reporting Protocol

For ACDC and M&Ms experiments, all metrics are computed on the fixed test set and reported as mean $\pm$ standard deviation across three random seeds. For Sant Pau experiments, all metrics are computed on the validation fold and reported as mean $\pm$ standard deviation across five cross-validation folds. The best model checkpoint - selected by validation AUC - is used for test evaluation in both cases.

Chapter 5

Experiments and Results

In this chapter, we report the full experimental results across all three datasets and all eight evaluated methods. These experiments were designed to build the thesis argument incrementally: beginning with reproduction of previously reported results on public benchmarks to validate the implementation framework, followed by evaluation on the public benchmarks with limited clinical features, then the primary LVH differential diagnosis task on Sant Pau, and finally a cross-dataset analysis of when clinical conditioning provides benefit.

All methods used the same preprocessing (Chapter 3) and training pipeline (Chapter 4), ensuring that performance differences reflect the methods themselves rather than differences in data handling or optimization. Results are reported as mean $\pm$ standard deviation across 5-fold stratified cross-validation for Sant Pau, and across 3 random seeds on the fixed held-out test set for ACDC and M&Ms. ROC AUC, balanced accuracy, and macro F1 are reported throughout as complementary metrics - ROC AUC provides a threshold-independent measure of discriminability, while balanced accuracy and macro F1 are more informative under class imbalance as they weight each class equally regardless of prevalence. Per-class accuracy is additionally reported to identify class-specific failure patterns. All figures throughout this chapter were generated by the results plotting pipeline described in Chapter 4.

Sant Pau is the main evaluation benchmark, as CC-LoRA was developed primarily for this dataset which provides 27 clinically relevant patient-level features. ACDC and M&Ms play a supporting role: they were used to develop and validate the pipeline by reproducing baseline results reported in the literature, and they enable the cross-dataset analysis of the conditions under which clinical conditioning provides benefit.

5.1 Reproduction - Validating the Implementation Framework

Before reporting results on the primary Sant Pau dataset, we first validate that the implementation framework correctly reproduces previously reported results from the literature on the public ACDC and M&Ms benchmarks. This section focuses on ResNet-50, CineMA Full FT, and CineMA + LoRA - the methods for which published reference numbers are available.

5.1.1 ACDC Reproduction

ACDC uses a fixed split of 100 train / 10 validation / 50 test patients across five balanced classes (DCM, HCM, MINF, NOR, RV). All experiments are repeated across three random seeds.

CineMA full fine-tuning was successfully reproduced on ACDC, achieving $94.57 \pm 0.56\%$ AUC - in the same range as the 97.98% AUC reported in the original CineMA paper [1] shown in Table 5.1. The small difference is attributable to differences in experimental setup, specifically the use of a fixed 3-seed protocol rather than the original paper's training configuration, and confirms that the preprocessing pipeline and fine-tuning framework are correctly implemented.

Method	AUC	BalAcc	F1	Paper [1] AUC
ResNet-50	$77.53 \pm 2.14\%$	$42.67 \pm 3.40\%$	$40.51 \pm 3.26\%$	81.05%
CineMA Full FT	$94.57 \pm 0.56\%$	—	$75.04 \pm 2.38\%$	97.98%
CineMA Full FT + LF	$79.42 \pm 2.27\%$	$52.00 \pm 2.83\%$	$50.99 \pm 3.30\%$	—
CineMA + LoRA	$80.85 \pm 11.80\%$	$47.33 \pm 19.75\%$	$41.61 \pm 24.98\%$	—

Table 5.1 - ACDC Reproduction Results (3 Seeds)

Our ResNet-50 result (77.53%) is broadly consistent with the 81.05% reported in the original paper [1], with the difference attributable to architectural and training configuration differences. The validated framework was subsequently used as the foundation for all LoRA, late fusion, and CC-LoRA experiments across all three datasets.

5.1.2 M&Ms Reproduction

M&Ms uses a fixed split of 149 train / 27 validation / 95 test patients across four classes (DCM, HCM, NOR, ARV). All experiments are repeated across three random seeds shown in Table 5.2.

Method	AUC	BalAcc	F1	Paper [1] AUC
ResNet-50	$64.56 \pm 1.64\%$	$36.78 \pm 0.73\%$	$33.98 \pm 0.83\%$	63.34%
CineMA Full FT	$83.87 \pm 3.01\%$	—	$63.16 \pm 1.48\%$	77.41%
CineMA Full FT + LF	$76.32 \pm 2.16\%$	$52.29 \pm 3.34\%$	$55.04 \pm 3.31\%$	—
CineMA + LoRA	$71.65 \pm 2.71\%$	$39.91 \pm 1.71\%$	$34.55 \pm 5.14\%$	—

Table 5.2 - M&Ms Reproduction Results (3 Seeds)

Our M&Ms Full FT result 83.87% AUC is higher than the 77.41% reported in the original paper [1]. This discrepancy is attributable to the use of a filtered subset of the M&Ms dataset - only the 271 patients with complete pathology labels were used rather than the full cohort - which reduces class ambiguity and likely inflates performance relative to the full dataset evaluation. Our ResNet-50 result (64.56%) closely matches the paper's 63.34% , confirming that the baseline implementation is consistent. The validated framework is confirmed to be correctly implemented and is used consistently across all subsequent experiments.

5.2 Public Benchmarks with Limited Clinical Features

Having validated the implementation framework, this section reports the full results on ACDC and M&Ms including CC-LoRA, to assess the behavior of clinical conditioning in settings where the available clinical features carry limited pathology-discriminative signal.

5.2.1 ACDC - Full Results

ACDC splits all 100 train / 10 validation / 50 test patients among 5 balanced classes (DCM, HCM, MINF, NOR, RV). All runs are performed across 3 different random seeds. Because there are only 10 patients in the validation set (2 per class), any single checkpoint AUC may have large variance and seed averaging is essential for reliable performance estimation. Performance and convergence for all methods are provided in Table 5.3 and Figure 5.1.

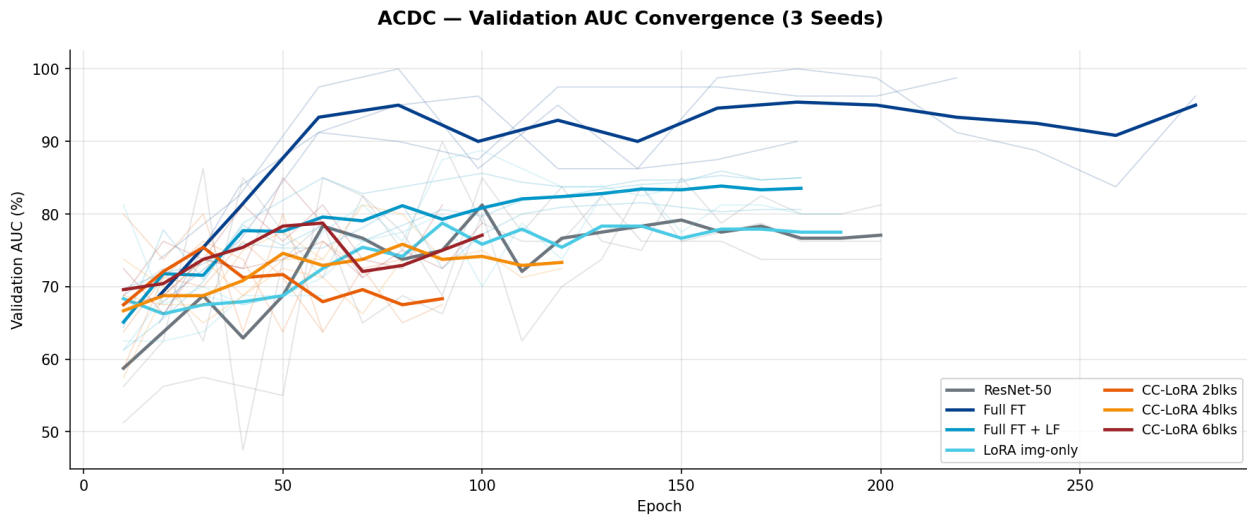


Figure 5.1: ACDC Convergence curves

CineMA Full FT has the best performance on ACDC with validation AUC reaching approximately 95%, consistent with the published CineMA results as reported in Section 5.1. For ACDC - a clean, balanced, single-centre dataset - full fine-tuning of the backbone is the most effective strategy, as the dataset size and class balance support full backbone adaptation without severe overfitting.

Method	AUC	BalAcc	F1	Params
ResNet-50	$77.53 \pm 2.14\%$	$42.67 \pm 3.40\%$	$40.51 \pm 3.26\%$	46.2M
CineMA Full FT	$94.57 \pm 0.56\%$	—	$75.04 \pm 2.38\%$	87.7M
CineMA Full FT + LF	$79.42 \pm 2.27\%$	$52.00 \pm 2.83\%$	$50.99 \pm 3.30\%$	89.9M
CineMA + LoRA	$80.85 \pm 11.80\%$	$47.33 \pm 19.75\%$	$41.61 \pm 24.98\%$	114K
CC-LoRA 2 blocks	$77.50 \pm 1.77\%$	$30.00 \pm 8.16\%$	$22.23 \pm 9.56\%$	193K
CC-LoRA 4 blocks	$76.25 \pm 3.54\%$	$33.33 \pm 9.43\%$	$26.16 \pm 14.76\%$	193K
CC-LoRA 6 blocks	$80.83 \pm 3.58\%$	$46.67 \pm 9.43\%$	$33.98 \pm 8.50\%$	193K

Table 5.3 - ACDC Full Results (3 Seeds, Test Set)

CC-LoRA cannot outperform LoRA image-only on ACDC. The best CC-LoRA result is

achieved with the 6-block configuration (80.83%), approximately equivalent to LoRA image-only (80.85%), while the 2-block and 4-block configurations perform below LoRA image-only. The three ACDC clinical features - height, weight, and BMI - do not contain pathology-discriminative signal for the 5-class classification task, and the FiLM conditioning therefore has no meaningful information to learn from. It should also be noted that BMI is directly derived from height and weight and therefore does not constitute an independent third feature - effectively only two independent clinical measurements are available for ACDC. The seed variance of LoRA image-only on ACDC is notably high (64.35%–91.25% AUC across seeds), reflecting the instability of early stopping decisions on the small 10-patient validation set. Notably, Full FT + Late Fusion (79.42% AUC) underperforms Full FT image-only (94.57% AUC) on ACDC, consistent with the finding on Sant Pau, that using uninformative clinical features with late fusion may be detrimental rather than beneficial.

Figure 5.1 convergence curves shows, the CineMA Full FT curve reaches approximately 95% validation AUC quickly and plateaus there, a markedly different pattern from Sant Pau where full FT was unstable. The stability difference reflects the dataset: ACDC's balanced classes and sufficient data points per class (100 total patients, 20 per class) support stable full fine-tuning, whereas the limited and imbalanced Sant Pau does not.

The CC-LoRA curves plateau at approximately 70–80% validation AUC and are prone to early stopping. After a certain point there is no further progression, indicating there is no longer any clinically useful conditioning signal for the network to extract - a consequence of the input feature limitations rather than the architecture.

5.2.2 M&Ms - Full Results

M&Ms uses a fixed split of 149 train / 27 validation / 95 test patients across four classes (DCM, HCM, NOR, ARV). There are only 8 training patients for the ARV class, rendering reliable classification of this class intractable for all methods. All results are averaged across 3 random seeds. Table 5.4 lists all method results and Figure 5.2 shows the convergence plots.

CineMA Full FT again achieves the strongest performance of $83.87 \pm 3.01\%$ AUC. Full FT + Late Fusion again underperforms Full FT image-only (76.32 % vs 83.87%), consistent with both ACDC and Sant Pau findings.

Method	AUC	BalAcc	F1	Params
ResNet-50	$64.56 \pm 1.64\%$	$36.78 \pm 0.73\%$	$33.98 \pm 0.83\%$	46.2M
CineMA Full FT	$83.87 \pm 3.01\%$	—	$63.16 \pm 1.48\%$	87.7M
CineMA Full FT + LF	$76.32 \pm 2.16\%$	$52.29 \pm 3.34\%$	$55.04 \pm 3.31\%$	89.9M
CineMA + LoRA	$71.65 \pm 2.71\%$	$39.91 \pm 1.71\%$	$34.55 \pm 5.14\%$	114K
CC-LoRA 2 blocks	$61.78 \pm 0.80\%$	$38.77 \pm 1.34\%$	$39.10 \pm 1.53\%$	193K
CC-LoRA 4 blocks	$64.43 \pm 2.25\%$	$43.29 \pm 4.27\%$	$42.35 \pm 5.62\%$	193K
CC-LoRA 6 blocks	$65.35 \pm 1.72\%$	$39.12 \pm 1.56\%$	$37.94 \pm 1.27\%$	193K

Table 5.4 - M&Ms Results (3 Seeds, Test Set)

CC-LoRA again underperforms LoRA image-only on M&Ms. The best CC-LoRA result is the 6-block configuration (65.35%), still 6.3% below LoRA image-only (71.65%). The two available

clinical features for M&Ms - age and sex are not sufficiently discriminative for the FiLM conditioning to learn meaningful patient-specific modulations, and further clinical information would be necessary for clinical conditioning to provide benefit in this setting.

The ARV class achieves uniformly low accuracy across all seeds and all CC-LoRA configurations, representing an inability to learn a reliable decision boundary from only 8 training patients - a dataset limitation rather than a methodological failure.

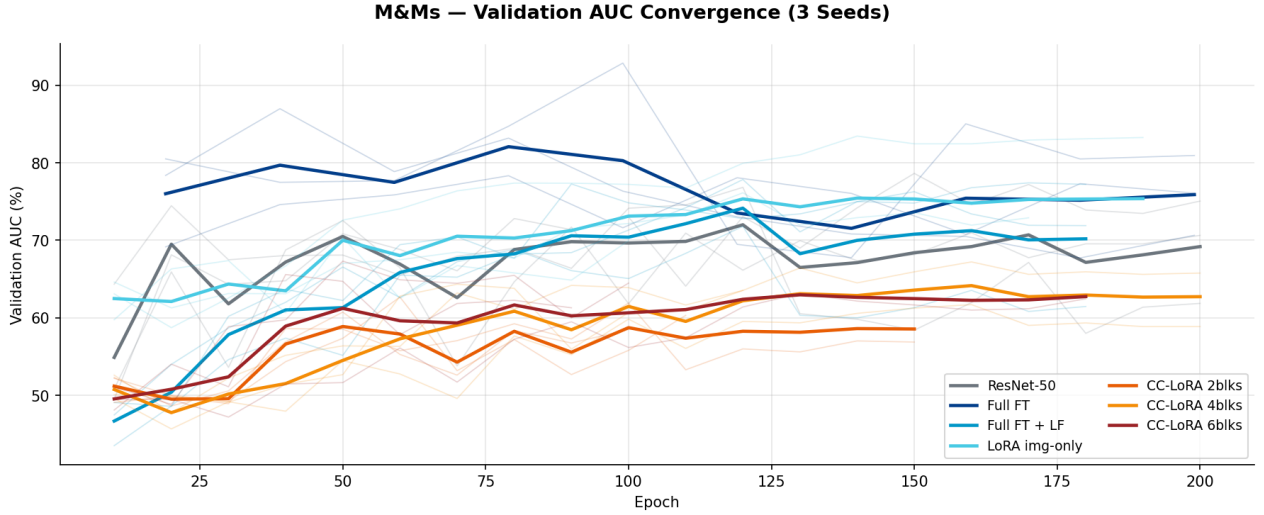


Figure 5.2: M&Ms Convergence curves

The CC-LoRA methods plateau at 60–65% validation AUC with consistent inter-seed behavior. The consistent low-AUC convergence indicates that while the method converges reliably, it can only reach a suboptimal solution because the FiLM conditioning is not discovering clinically relevant modulations from the available weak features.

5.3 LVH Differential Diagnosis - Sant Pau Hospital

5.3.1 Experimental Setup

All Sant Pau experiments use 5-fold stratified cross-validation on the full 292-patient dataset (HCMP=165, HCM=66, NOR=37, AML=24). Stratification preserves the class proportions across folds, producing approximately 234 training and 58 validation patients per fold. Weighted random sampling is applied during training to address the 56.5% HCMP class imbalance, and label smoothing ($\epsilon=0.1$) is applied to the cross-entropy loss.

CC-LoRA achieves $91.03 \pm 4.17\%$ AUC on Sant Pau, representing a gain of approximately 17 percentage points over the strongest image-only baseline (CineMA + LoRA, $74.13 \pm 6.05\%$). This demonstrates that clinical conditioning substantially improves the discriminative power of the foundation model representations beyond what imaging alone can provide. The clinical-only XGBoost baseline (88.00% AUC) is included as a secondary reference to confirm that the two modalities are complementary - CC-LoRA exceeds both the image-only and clinical-only baselines, demonstrating that neither source of information alone is sufficient and that their integration within the transformer produces a more powerful representation than either individually. Figure

5.3 provides the complete visual comparison across all methods and metrics.

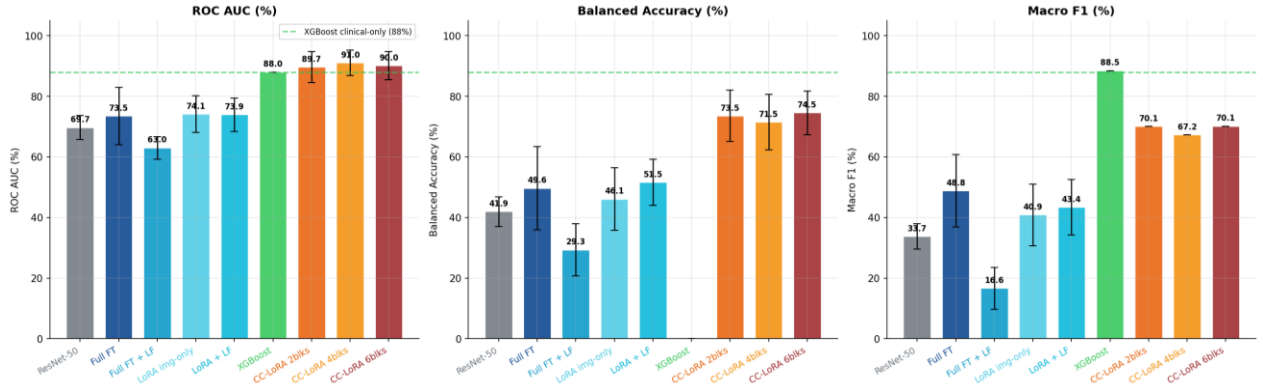


Figure 5.3: Sant Pau Hospital Data - Complete Method Comparison (5-Fold CV)

Figure 5.4 shows the per-fold AUC distribution revealing fold-level stability differences between methods.

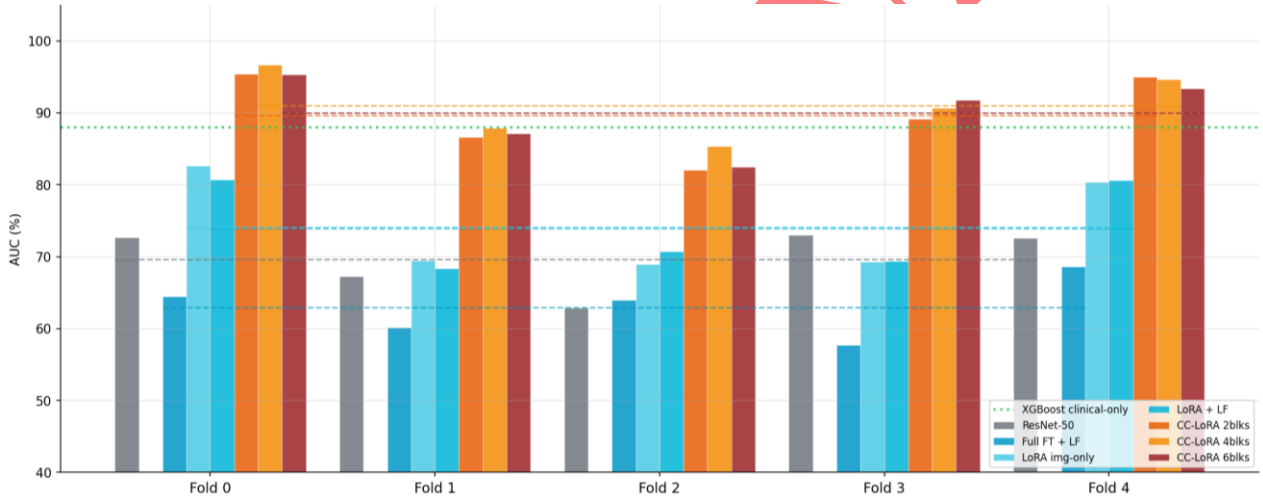


Figure 5.4: Sant Pau Hospital Data - Per Fold AUC of all methods (5-Fold CV)

5.3.2 ResNet-50 Baseline

Following the evaluation approach of the original CineMA publication, a randomly initialized ResNet-50 is used for comparison purposes, as shown in Table 5.5.

Metric	Mean	Std
ROC AUC	69.66%	±3.99%
Balanced Accuracy	41.90%	±4.94%
Macro F1	33.69%	±4.19%
HCMP accuracy	44.24%	±24.81%
HCM accuracy	40.00%	±25.00%
NOR accuracy	40.36%	±30.25%
AML accuracy	43.00%	±25.22%

Table 5.5 - ResNet-50 Sant Pau Results (5-Fold CV)

As a baseline performance, ResNet-50 represents what can be achieved without any pre-

trained representations on this dataset. It achieves an AUC of 69.66%, substantially above random chance (50%).

All four per-class accuracy values exhibit notably high standard deviations (>24%), indicating that performance is sensitive to the random selection of patients in each fold. Training from scratch with approximately 186 training patients per fold and 46.2 million parameters causes difficulties in learning discriminative features, as shown in the convergence plots in Figure 5.5 - the ResNet-50 mean curve is the lowest and climbs slowest of all methods.

5.3.3 CineMA Full Fine-Tuning

CineMA Full FT, initialized from the pre-trained weights and fine-tuned with all 87.7 million backbone parameters trainable, achieves the results shown in Table 5.6.

Metric	Mean	Std
ROC AUC	73.48%	$\pm 9.54\%$
Balanced Accuracy	49.62%	$\pm 13.74\%$
Macro F1	48.81%	$\pm 11.96\%$
HCMP accuracy	76.97%	$\pm 10.43\%$
HCM accuracy	36.37%	$\pm 15.67\%$
NOR accuracy	52.14%	$\pm 29.93\%$
AML accuracy	33.00%	$\pm 23.58\%$

Table 5.6 - CineMA Full FT Sant Pau Results (5-Fold CV)

Full fine-tuning results in an improvement of +3.82% AUC over ResNet-50 and shows that CineMA pre-training transfers successfully to the Sant Pau domain, though by a small margin when compared with the performance achievable on larger and more balanced datasets such as ACDC (~95% AUC). Full fine-tuning is susceptible to overfitting due to the large number of trainable parameters relative to the limited training data - with 87.7M trainable parameters and approximately 234 patients per fold, the parameter-to-data ratio is extremely unfavorable. The high standard deviation of $\pm 9.54\%$ compared to $\pm 3.99\%$ for ResNet-50 indicates severe fold-to-fold inconsistency, with the HCM class achieving a mean accuracy of only 36.37%, clearly confused with the morphologically similar HCMP. HCMP achieves the highest per-class accuracy (76.97%), reflecting its numerical dominance in training batches despite weighted sampling.

Interestingly, Full FT performs very low than CineMA + LoRA image-only in AUC (73.48% vs 74.13%), despite using approximately 770 times more trainable parameters. This is consistent with overfitting under Full FT and the regularization benefit of LoRA's parameter efficiency on a small dataset. However, given the high standard deviations and the small number of folds, formal statistical testing would be required to determine whether this difference is statistically significant - the observed difference should be interpreted as a trend rather than a definitive conclusion.

5.3.4 CineMA Full Fine-Tuning + Late Fusion

The performance of Full FT + Late Fusion - the concatenation of an MLP-based clinical embedding to the image embedding upstream of the classification head, applied on top of a fully fine-tuned backbone - is notably poor. With an AUC of $62.97 \pm 3.74\%$, it falls below even ResNet-50 ($69.66 \pm 3.99\%$). Four of the five folds exhibit a balanced accuracy of precisely 25%, the minimum

value for a 4-class classification task, equivalent to predicting the same class for every patient regardless of image content or clinical data. The model has effectively collapsed to predicting HCMP for essentially every patient in the cohort.

One potential explanation for this collapse is the interaction between full fine-tuning and late fusion on a small dataset. The fully fine-tuned backbone is simultaneously attempting to adapt approximately 87.7 million parameters for appropriate image feature extraction while the randomly initialized clinical MLP and fusion head are attempting to learn a meaningful combination of both modalities. The resulting optimization signals compete: backbone gradients influenced by the classification loss are themselves shaped by the poorly initialized clinical branch, potentially destabilizing the pre-trained representations before they can adapt appropriately - a form of catastrophic interference between the two learning processes. A dedicated warm-up phase for the clinical branch prior to full backbone fine-tuning may have mitigated this collapse and is acknowledged as a methodological limitation. The single non-collapsed fold (fold 4, AUC=68.57%, BalAcc=46.43%) suggests the failure is stochastic rather than deterministic, consistent with the gradient interference hypothesis. It should be noted that without formal statistical testing across repeated runs, this explanation remains a plausible interpretation rather than a demonstrated conclusion.

An important finding from this result is that late fusion applied on top of a fully fine-tuned backbone is not a safe default for multimodal learning on small medical imaging datasets and may, in certain cases, perform substantially worse than single-modality training alone. This is consistent with broader observations in multimodal learning that naive fusion of modalities with heterogeneous training dynamics can destabilize optimization, particularly in low-data regimes [12, 13]. This provides strong motivation for the CC-LoRA design choice of keeping the backbone frozen and introducing clinical conditioning gradually through FiLM parameters initialised to identity, which ensures a stable training trajectory from the first epoch.

5.3.5 CineMA + LoRA (Image-Only)

LoRA image-only fine-tuning, with the backbone frozen and LoRA adapters injected into blocks 6–11, produces the results shown in Table 5.7.

Fold	AUC	BalAcc	F1	HCMP	HCM	NOR	AML
0	82.60%	65.85%	58.10%	55%	54%	75%	80%
1	69.43%	45.50%	38.00%	33%	46%	62%	40%
2	68.97%	38.20%	26.54%	12%	69%	71%	0%
3	69.28%	37.96%	39.07%	88%	15%	29%	20%
4	80.36%	42.83%	42.55%	61%	43%	43%	25%
Mean	74.13 ± 6.05%	46.07 ± 10.29%	40.85 ± 10.16%	49.7%	45.5%	56.1%	33.0%

Table 5.7 - CineMA + LoRA Image-Only Sant Pau Results (5-Fold CV)

LoRA image-only achieves 74.13% AUC, the best among all image-only approaches on Sant Pau, clearly indicating that parameter-efficient fine-tuning outperforms full fine-tuning in the small-data regime. The frozen backbone preserves all pre-trained cardiac representations while a small LoRA matrix adapts to the new domain - a much more favorable parameter-to-data ratio than full fine-tuning.

Class-specific failures can be identified from the per-class analysis. AML achieves 0% accuracy in fold 2, and HCMP accuracy varies greatly from 12% in fold 2 to 88% in fold 3 depending on patient distribution within the folds. These instabilities suggest that incorporating clinical information during image feature extraction may provide benefit motivating the CC-LoRA approach.

The convergence plots in Figure 5.5 show that the LoRA image-only mean AUC curve slowly rises from approximately 65% to 74% over the full 200 epochs with a broad spread among individual fold curves. The training loss stabilizes at approximately 1.0–1.1, considerably higher than the approximately 0.4 minimum achieved by CC-LoRA, suggesting that image information alone is insufficient to drive the loss as low as the combination of image and clinical data.

5.3.6 CineMA + LoRA + Late Fusion

LoRA + Late Fusion, combining frozen backbone LoRA with a clinical MLP at the output, produces the results shown in Table 5.8.

Fold	AUC	BalAcc	F1
0	80.67%	62.97%	53.18%
1	68.35%	42.20%	31.01%
2	70.71%	51.68%	35.03%
3	69.33%	44.60%	44.00%
4	80.57%	56.22%	53.60%
Mean	73.93 $\pm$ 5.52%	51.54 $\pm$ 7.59%	43.36 $\pm$ 9.20%

Table 5.8 - CineMA + LoRA + Late Fusion Sant Pau Results (5-Fold CV)

LoRA + Late Fusion achieves 73.93% AUC, nearly equivalent to LoRA image-only (74.13%), indicating that the late addition of clinical features at the output of a frozen backbone provides no substantial added value. The clinical MLP receives clinical features and the classification head receives both embeddings, yet the image embedding does not take into account the patient's clinical context, and the head cannot exploit the interaction between the two modalities.

Crucially, as shown in Table 5.9, LoRA + Late Fusion substantially underperforms CC-LoRA across all three metrics.

Metric	LoRA + LF	CC-LoRA 4blks	Difference
AUC	73.93%	91.03%	+17.10%
Balanced Accuracy	51.54%	71.46%	+19.92%
Macro F1	43.36%	67.25%	+23.89%

Table 5.9 - Performance Difference: LoRA + Late Fusion vs CC-LoRA

It should be noted that LoRA + Late Fusion and CC-LoRA are trained with an identical backbone, identical LoRA configuration, and identical clinical features. The only variation is the point at which clinical data is introduced: either after the completed encoder forward pass (late fusion) or integrated within the transformer blocks during the forward pass (CC-LoRA). The gap of +17.10% AUC between these methods is thus solely attributable to in-transformer clinical conditioning, providing direct empirical support for the central thesis hypothesis.

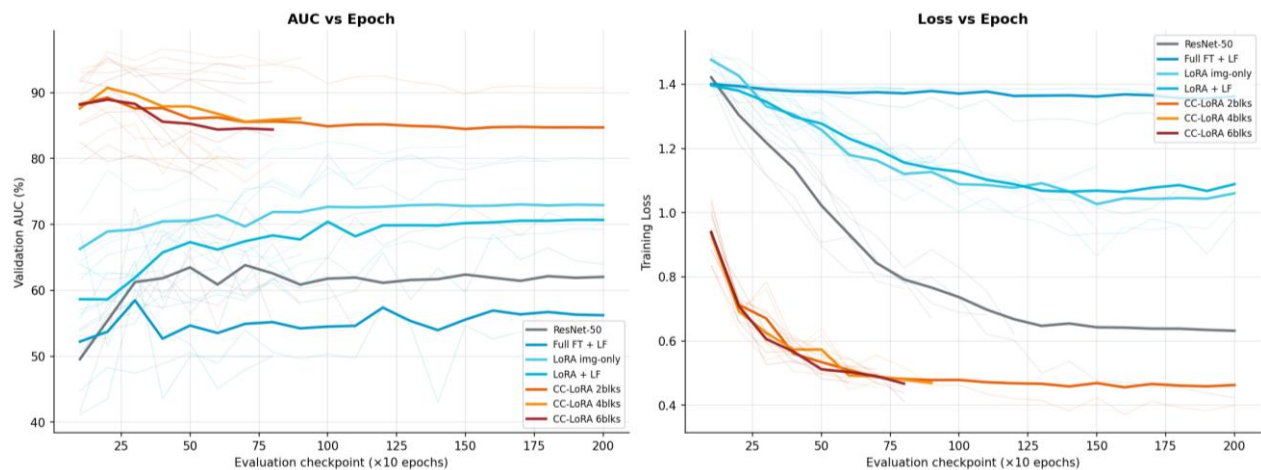


Figure 5.5: Sant Pau - Training Convergence plot of all methods. (5-Fold CV)

5.3.7 CC-LoRA - 2 Blocks (10-11)

CC-LoRA with the last two transformer blocks conditioned produces the results shown in Table 5.10.

Fold	AUC	BalAcc	F1	HCMP	HCM	NOR	AML
0	95.41%	87.76%	81.19%	82%	69%	100%	100%
1	86.58%	66.32%	63.51%	45%	92%	88%	40%
2	82.08%	64.38%	57.67%	61%	77%	100%	20%
3	89.15%	71.52%	71.54%	76%	85%	86%	40%
4	95.01%	77.46%	76.54%	85%	50%	100%	75%
Mean	89.65 ± 5.08%	73.49 ± 8.46%	70.09%	69.6%	74.6%	94.7%	55.0%

Table 5.10 - CC-LoRA 2 Blocks Sant Pau Results (5-Fold CV)

Conditioning only the final two transformer blocks produces a large and immediate gain over all image-only and late fusion baselines. CC-LoRA effectively combines imaging and clinical features to yield a representation more discriminative than imaging alone.

NOR classification achieves a near-perfect mean accuracy of 94.7%, confirming that the negative control is functioning as intended - the model successfully identifies the absence of LVH rather than defaulting to a disease label. This is consistent with normal controls being the most morphologically and clinically distinct group once patient-level clinical information is available. In principle, FiLM conditioning allows the model to respond to clinical features indicative of a normal cardiac profile, though the specific features driving this discrimination were not formally analysed through explainability methods and this is acknowledged as a direction for future work. AML achieves a mean accuracy of 55.0%, representing a substantial gain over LoRA image-only (33.0%). However, fold-level variance remains high (20% - 100%), reflecting the small number of AML patients available per training fold (~19 patients), which makes reliable decision boundary learning inherently difficult regardless of the method used.

A striking feature observable in the convergence plots in Figure 5.5 is that CC-LoRA AUC reaches the 85–90% range immediately from the first evaluation checkpoint at epoch 10 - a level that LoRA image-only never reaches even by epoch 200. This rapid convergence reflects the frozen

backbone providing high-quality cardiac features from epoch 0 with FiLM conditioning adding patient-specific modulation that immediately enables the classification head to exploit the provided features. The fold-level curves are closely clustered around the mean, indicating low fold-level variance relative to image-only methods.

5.3.8 CC-LoRA - 4 Blocks (8-11)

The 4-block configuration achieves the highest AUC of all evaluated methods, as shown in Table 5.11.

Fold	AUC	BalAcc	F1	HCMP	HCM	NOR	AML
0	96.64%	81.26%	77.17%	73%	92%	100%	60%
1	87.90%	69.29%	68.57%	73%	77%	88%	40%
2	85.32%	57.37%	52.26%	55%	69%	86%	20%
3	90.66%	67.41%	66.82%	73%	77%	100%	20%
4	94.62%	81.98%	71.42%	64%	64%	100%	100%
Mean	91.03 $\pm$ 4.17%	71.46 $\pm$ 9.23%	67.25%	67.3%	75.9%	94.6%	48.0%

Table 5.11 - CC-LoRA 4 Blocks Sant Pau Results (5-Fold CV)

This 4-block configuration (8-11) achieves the highest overall AUC of any method evaluated in this thesis at 91.03 $\pm$ 4.17%, training only 0.258% of the backbone parameters (~231K of 89.5M). Within this configuration, fold 0 produces the highest per-fold AUC of any individual experiment across the entire thesis at 96.64%. This configuration also achieves the best mean accuracy for the HCM class at 75.9%, suggesting that conditioning blocks 8 and 9 in addition to blocks 10 and 11 provides additional benefit for distinguishing between the morphologically similar HCMP and HCM etiologies. The slightly lower balanced accuracy compared to the 2-block configuration (71.46% vs 73.49%) indicates that the 4-block configuration optimizes more strongly for AUC at the cost of per-class balance. AML accuracy reflects the same trade-off, dropping from 55.0% with 2 blocks to 48.0% with 4 blocks, consistent with the model prioritizing the more represented classes when additional conditioning capacity is available.

5.3.9 CC-LoRA - 6 Blocks (6-11)

The 6-block configuration achieves the best-balanced accuracy and F1, as shown in Table 5.12.

Fold	AUC	BalAcc	F1	HCMP	HCM	NOR	AML
0	95.36%	78.93%	75.90%	79%	77%	100%	60%
1	87.12%	70.49%	69.29%	73%	69%	100%	40%
2	82.51%	64.77%	55.92%	61%	38%	100%	60%
3	91.79%	72.60%	72.68%	67%	92%	71%	60%
4	93.40%	85.77%	76.81%	79%	64%	100%	100%
Mean	90.04 $\pm$ 4.65%	74.51 $\pm$ 7.23%	70.12%	71.5%	68.2%	94.3%	64.0%

Table 5.12 - CC-LoRA 6 Blocks Sant Pau Results (5-Fold CV)

The 6-block configuration achieves the best-balanced accuracy at 74.51% and jointly the highest F1 at 70.12%. AML accuracy reaches its maximum at 64.0% - conditioning more blocks

appears to provide additional benefit for the most difficult minority class, perhaps because deeper clinical conditioning influences more levels of the feature hierarchy and provides richer patient-specific modulation. NOR is classified at 100% accuracy in four of five folds. The AUC standard deviation of 4.65% is comparable to the 4-block configuration (4.17%), confirming that increasing the number of conditioned blocks does not add fold-level instability.

5.3.10 Sant Pau Complete Summary

Table 5.13 presents the full Sant Pau comparison across all methods. Figure 5.3 provides the visual comparison across AUC, balanced accuracy, and F1. Figure 5.6 groups methods by fusion strategy to highlight the key thesis findings.

Method	Clinical	AUC	BalAcc	F1	Params
ResNet-50	No	69.66 ± 3.99%	41.90%	33.69%	46.2M
CineMA Full FT	No	73.48 ± 9.54%	49.62%	48.81%	87.7M
CineMA Full FT + LF	Yes	62.97 ± 3.74%	29.29%	16.61%	89.9M
CineMA + LoRA	No	74.13 ± 6.05%	46.07%	40.85%	114K
CineMA + LoRA + LF	Yes	73.93 ± 5.52%	51.54%	43.36%	~320K
XGBoost (clinical-only)	Yes	88.00%	—	88.46%	—
CC-LoRA 2 blocks	Yes	89.65 ± 5.08%	73.49%	70.09%	193K
CC-LoRA 4 blocks	Yes	91.03 ± 4.17%	71.46%	67.25%	231K
CC-LoRA 6 blocks	Yes	90.04 ± 4.65%	74.51%	70.12%	267K

Table 5.13 - Sant Pau Complete Results (5-Fold CV)

Three key observations emerge from Table 5.13. First, all CC-LoRA configurations (89.65%–91.03% AUC) substantially outperform all image-only baselines (69.66%–74.13%) and both late fusion variants (62.97%–73.93%), demonstrating that in-transformer clinical conditioning is the critical ingredient for strong performance on this dataset. Second, CC-LoRA exceeds the clinical-only XGBoost baseline (88.00%), confirming that cardiac MRI provides complementary information that CC-LoRA successfully integrates with the clinical features, and that neither modality alone is sufficient. Third, the collapse of Full FT + Late Fusion (62.97%) demonstrates that naive combination of full backbone fine-tuning with output-level clinical fusion can be severely counterproductive, while the frozen backbone and identity-initialized FiLM conditioning of CC-LoRA provide a safe and effective mechanism for clinical integration.

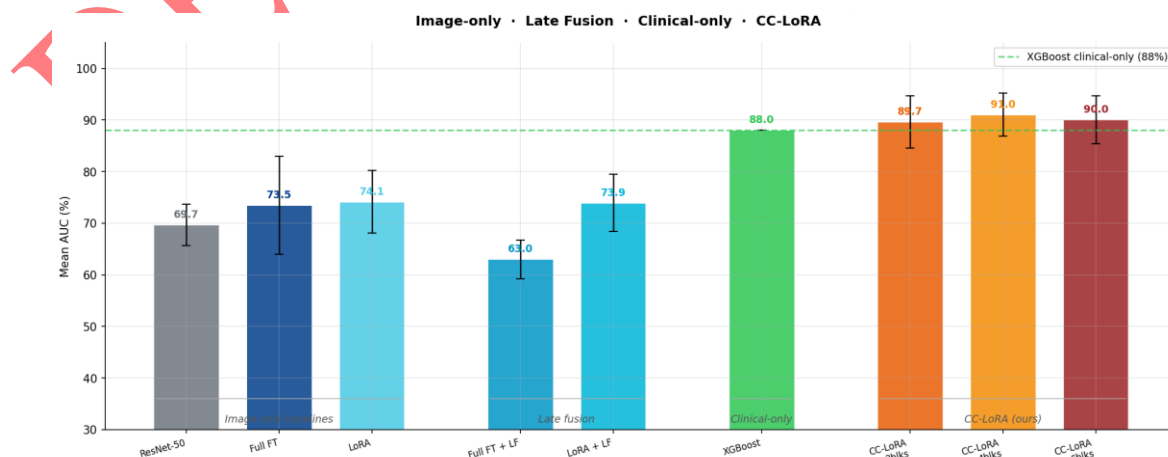


Figure 5.6: Sant Pau – Fusion Strategy Comparison: AUC by Method Group

5.3.11 Block Count Analysis

Table 5.14 contains the ordered comparison of CC-LoRA block configurations, also shown in Figure 5.7. The three configurations achieve closely comparable AUC scores, with a maximum difference of 1.38% points between the best (4-block, 91.03%) and worst (2-block, 89.65%) configurations.

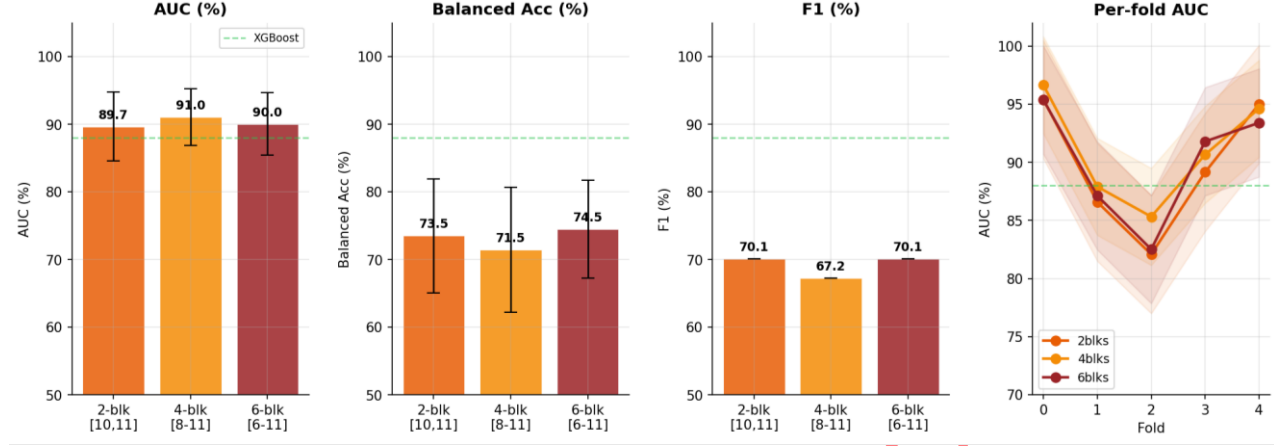


Figure 5.7: CC-LoRA Block Configuration Comparison on Sant Pau

This suggests that even conditioning only the last two transformer blocks captures the majority of the benefit from clinical conditioning. The 4-block configuration (8–11) is selected as the primary reported result throughout this thesis as it achieves the highest AUC ($91.03 \pm 4.17\%$) while remaining highly parameter-efficient at 0.258% of the backbone parameters (~231K trainable).

Config	AUC	BalAcc	F1	Best for
2 blocks (10-11)	$89.65 \pm 5.08\%$	73.49%	70.09%	AML accuracy (55%)
4 blocks (8-11)	$91.03 \pm 4.17\%$	71.46%	67.25%	Highest AUC, HCM (75.9%)
6 blocks (6-11)	$90.04 \pm 4.65\%$	74.51%	70.12%	BalAcc, F1, AML (64%)

Table 5.14 - Systematic Comparison of CC-LoRA Block Configurations

This matches the theoretical understanding that the highest-level semantic representations -most readily modulated by clinical conditioning are encoded in the later transformer blocks [4, 6], while earlier blocks capture lower-level spatial and structural features that are less directly influenced by high-level conditioning signals. Conditioning earlier transformer blocks therefore offers successively smaller marginal benefits for AUC.

On the other hand, the 6-block configuration consistently yields better balanced accuracy, F1, and AML minority class performance, suggesting that more conditioned blocks improve class-level balance at the price of a small reduction in AUC. In clinical deployment scenarios where false negatives for AML carry high clinical consequence, the 6-block configuration may be preferable despite its marginally lower AUC.

5.4 Cross-Dataset Analysis: Clinical Richness and CC-LoRA Gain

5.4.1 Clinical Feature Informativeness

The discriminative ability of the 27 Sant Pau clinical features is assessed using a clinical-only XGBoost model [41] trained solely on the tabular features without any image data. This model yields an AUC of 88.00%, F1 of 88.46%, precision of 85.19%, recall of 92.00%, and accuracy of 82.86% on the static test set. This high performance with only tabular data demonstrates that the 27 clinical features together contain substantial pathology-discriminative information. Crucially, CC-LoRA (91.03%) exceeds this clinical-only baseline, confirming that cardiac MRI provides complementary spatial and morphological information that clinical measurements alone cannot capture, and that the two modalities are genuinely complementary rather than redundant.

5.4.2 Clinical Richness vs CC-LoRA Gain

One of the key observations arising from this thesis is the systematic relationship between clinical feature richness and CC-LoRA performance gain when results across the three datasets are compared. Figure 5.8 plots clinical feature richness versus CC-LoRA gain, as presented in Table 5.15.

Dataset	Features	XGBoost AUC	LoRA img-only	Best CC-LoRA	Gain
ACDC	3 (height, weight, BMI)	~50% (est.)	80.85%	80.83%	~0%
M&Ms	2 (age, sex)	~50% (est.)	71.65%	65.35%	-6.30%
Sant Pau	27 (rich, pathology-specific)	88.00%	74.13%	91.03%	+16.90%

Table 5.15 - Clinical Richness vs CC-LoRA Performance

The pattern is clear. CC-LoRA shows a gain of +16.90% AUC over image-only LoRA on Sant Pau, where 27 clinical features demonstrate strong independent discriminative power. On ACDC and M&Ms, where clinical features are weakly informative, CC-LoRA yields no improvement and slightly degrades performance on M&Ms.

This behavior confirms the core hypothesis: CC-LoRA does not intrinsically benefit all tasks but provides substantial benefit when informative clinical features are present. When clinical features carry meaningful discriminative signal as confirmed by a strong tabular baseline - CC-LoRA successfully integrates both imaging and clinical information to outperform either modality alone. When clinical features are weak, the FiLM conditioning has no meaningful signal to learn from and the method reduces to standard LoRA. This establishes a practical screening criterion: a strong tabular baseline (XGBoost AUC substantially above chance) is a reliable indicator of likely CC-LoRA benefit.

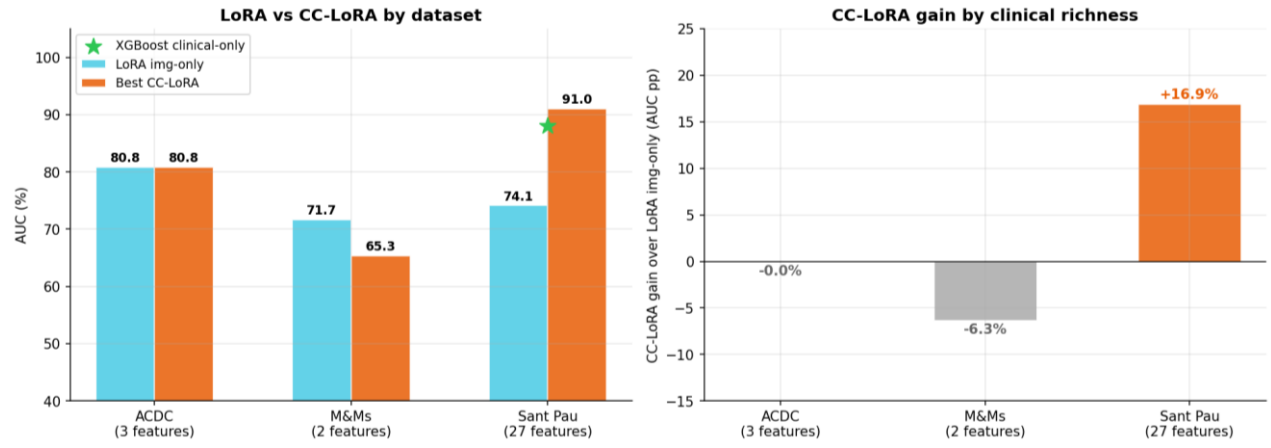


Figure 5.8: Clinical Feature Richness vs CC-LoRA Performance Gain

The performance of CC-LoRA (91.03%) exceeding XGBoost (88.00%) demonstrates that cardiac MRI provides complementary information - including spatial morphology, tissue texture, wall motion patterns, and other imaging-derived features that cannot be captured by clinical measurements alone. CC-LoRA successfully integrates both information sources to produce a predictive model that outperforms either modality individually.

5.4.3 Full FT + Late Fusion Collapse - Cross-Dataset Evidence

The consistent underperformance of Full FT + Late Fusion relative to Full FT image-only across all three datasets deserves special mention as a cross-dataset finding. Table 5.16 shows the AUC results across datasets.

Dataset	Full FT img-only AUC	Full FT + LF AUC	AUC Difference
ACDC	94.57%	79.42%	-15.15%
M&Ms	83.87%	76.32%	-7.51%
Sant Pau	73.48%	62.97%	-10.51%

Table 5.16 - Full FT + Late Fusion Low Performance Across All Datasets

The approximate between 7% to 16% AUC reduction across all three datasets suggests that this represents a systematic weakness of the Full FT + Late Fusion combination rather than an artefact specific to a particular dataset or class distribution. The Sant Pau balanced accuracy drop is even more striking a 20.33 percentage point reduction confirming near-complete collapse to the majority class. Indirect evidence for the gradient interference hypothesis described in Section 5.3.4 can be observed in the convergence plots (Figure 5.5): the Full FT + Late Fusion training loss plateaus at a substantially higher value than all other methods and fails to decrease meaningfully beyond the first few epochs, while the validation balanced accuracy remains at or near 25% throughout training in four of the five Sant Pau folds - indicating that the model collapses to the majority class from the very beginning of training rather than converging to a suboptimal solution after initially learning. This pattern is consistent with a fundamental training dynamics failure in the early epochs rather than overfitting or capacity limitations, supporting the gradient interference interpretation. The consistency of this behavior across three datasets of different sizes, class distributions, and clinical feature sets constitutes a notable cautionary observation for practitioners developing multimodal medical imaging pipelines.

5.5 Summary of Key Findings

The experimental results across all three datasets establish the following key findings:

Finding 1 - Pre-trained foundation models provide a consistent benefit over training from scratch. Across all three datasets, CineMA-based approaches outperformed the randomly initialised ResNet-50 baseline, confirming that representations learned from large-scale cardiac pre-training transfer effectively beyond the UK Biobank distribution. Image-only LoRA fine-tuning achieved gains of +4.47% AUC over ResNet-50 on Sant Pau, +3.32% on ACDC, and +7.09% on M&Ms. Notably, this benefit extends to the private Sant Pau cohort with a different pathology distribution and acquisition protocol from the UK Biobank pre-training data demonstrating that CineMA representations generalize meaningfully to institution-specific clinical datasets without requiring pre-training on in-domain data.

Finding 2 - Parameter efficiency outperforms full fine-tuning on small datasets. CineMA + LoRA outperforms CineMA Full FT on Sant Pau (74.13% vs 73.48%), supporting that parameter-efficient fine-tuning provides better regularisation than full fine-tuning in low-data settings. However, given the high standard deviations and small number of folds, formal statistical testing would be required to confirm this as a statistically significant difference - the observed pattern should be interpreted as a trend.

Finding 3 - Full FT + Late Fusion is consistently harmful. Full fine-tuning combined with output-level clinical fusion underperforms full fine-tuning without clinical features by approximately 10% AUC on all three datasets. The consistency of this effect across datasets suggests a systematic phenomenon rather than a dataset-specific artefact. A detailed investigation of its underlying cause is left to future work.

Finding 4 - Late fusion with frozen backbone provides negligible benefit. LoRA + Late Fusion (73.93%) performed almost identically to LoRA image-only (74.13%), confirming that simply appending clinical features at the output of a frozen backbone is insufficient for effective clinical conditioning of the image representation.

Finding 5 - CC-LoRA substantially outperforms all image-only and late fusion baselines when clinical features are informative. The 4-block configuration (8-11) achieves the highest AUC of any method evaluated in this thesis at $91.03 \pm 4.17\%$ on Sant Pau, representing a gain of +16.90% over LoRA image-only and +17.10% over LoRA + Late Fusion, while training only 0.258% of the backbone parameters (~231K of 89.5M). This is the central result of the thesis, demonstrating that in-transformer clinical conditioning via FiLM-modulated LoRA adapters produces substantially more effective multimodal representations than either image-only fine-tuning or output-level clinical fusion, at negligible additional parameter cost.

Finding 6 - CC-LoRA benefit depends on clinical feature richness. The +16.90% gain on Sant Pau (27 features, XGBoost AUC=88%) contrasts with approximately 0% gain on ACDC (3 features) and - 6.30% on M&Ms (2 features), establishing a clear and practically useful relationship between clinical informativeness and CC-LoRA effectiveness.

Finding 7 - Block count has diminishing returns for AUC but may benefit minority class

performance. The 2-block, 4-block, and 6-block CC-LoRA configurations achieve closely comparable AUC scores on Sant Pau (89.65%, 91.03%, 90.04%), with a maximum difference of 1.38 percentage points. However, the 6-block configuration achieves the highest AML minority class accuracy (64.0% vs 48.0% vs 55.0% for 4-block and 2-block respectively), suggesting that conditioning a greater number of transformer blocks may benefit underrepresented classes more than it improves overall AUC. This implies that deeper clinical conditioning might provide more consistent per-class discrimination, though this hypothesis would require evaluation across a broader range of block configurations and datasets to confirm.

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Chapter 6

Agreement of Results and Objectives

This chapter reviews the five objectives outlined in section 1.2, discussing the extent to which they have been accomplished from the entirety of the experimental results found in chapter 5.

In addition, it summarizes the experimental results, placing them in context within the overall research area of multimodal medical imaging.

6.1 Five Objectives

O1 - Reproduce and validate the CineMA foundation model.

Objective: Reproduce the results reported for the CineMA cardiac MRI foundation model [1] on the public ACDC and M&Ms benchmarks in order to validate the implementation framework and confirm that the fine-tuning pipeline behaves as expected before applying it to the primary dataset.

CineMA full fine-tuning was successfully reproduced on ACDC, achieving 94.57% AUC, which is in the same range as the 97.98% reported in the original CineMA paper [1]. The small difference is attributable to differences in experimental setup specifically the use of a fixed 3-seed protocol rather than the original paper's training configuration and that the preprocessing pipeline and fine-tuning framework are correctly implemented. On M&Ms, our reproduction achieves 83.87% AUC compared to 77.41% reported in [1], a higher result likely reflecting differences in the patient subset used, we used the 271 patients with complete pathology labels of M&Ms cohort. The validated framework was subsequently used as the foundation for all LoRA, late fusion, and CC-LoRA experiments across all three datasets.

O2 - Investigate transferability to LVH differential diagnosis.

Objective: Evaluate the transferability of the CineMA foundation model to LVH etiology classification using the real-world clinical cohort from Hospital de la Santa Creu i Sant Pau, assessing whether representations pre-trained on large-scale UK Biobank data generalize to a small, institution-specific dataset with a diagnostically challenging four-class structure (HCMP, HCM, NOR, AML).

CineMA-based methods consistently outperform the randomly initialized ResNet-50

baseline on Sant Pau, with LoRA image-only achieving +4.47% AUC over ResNet-50 (74.13% vs 69.66%). This confirms that UK Biobank pre-trained representations transfer meaningfully to cohort with a different pathology distribution and acquisition protocol. The transferability extends to all four LVH etiology classes including the negative control NOR, which achieves near-perfect classification accuracy (94.3%–94.7%) across all CC-LoRA configurations, confirming that the model successfully learns to distinguish the absence of LVH from its presence across three distinct etiologies.

O3 - Compare full fine-tuning and parameter-efficient fine-tuning.

Objective: Compare full fine-tuning of the CineMA backbone with LoRA-based parameter-efficient fine-tuning, evaluating the trade-off between model capacity and regularisation on the small Sant Pau cohort.

LoRA image-only fine-tuning ($74.13 \pm 6.05\%$ AUC, 114K trainable parameters) outperforms CineMA Full FT ($73.48 \pm 9.54\%$ AUC, 87.7M trainable parameters) on Sant Pau despite using approximately 750 times fewer trainable parameters. On ACDC, the relationship reverses - Full FT (~95% val AUC) substantially outperforms LoRA ($80.85 \pm 11.80\%$) - reflecting the dataset's size and balanced class distribution, which can support full backbone adaptation without severe overfitting. These contrasting results across datasets confirm that parameter efficiency is not merely a computational advantage but provides a genuine regularization benefit in the low-data regime, and that the appropriate fine-tuning strategy depends critically on dataset size and balance.

O4 - Investigate the added value of clinical information.

Objective: Investigate whether incorporating patient-level clinical information improves disease classification performance in vision foundation model-based approaches across public and real-world clinical datasets, without requiring full model retraining from scratch.

The added value of clinical information is strongly dataset-dependent. On Sant Pau, where 27 clinically relevant patient-level features are available, CC-LoRA achieves $91.03 \pm 4.17\%$ AUC - a gain of +16.90% over LoRA image-only and exceeding the clinical-only XGBoost baseline, demonstrating that both modalities are complementary and that their integration produces a more powerful representation than either alone. On ACDC and M&Ms, where clinical features are limited to three or fewer variables with weak pathology-discriminative signal, CC-LoRA provides no improvement over image-only fine-tuning. This cross-dataset pattern establishes a clear and practically useful relationship: clinical conditioning adds value when clinical features carry independent discriminative signal, as confirmed by a strong tabular baseline performance, and provides no benefit otherwise.

O5 - Implement and evaluate multimodal fusion strategies.

Objective: Implement and evaluate multimodal fusion strategies for combining CMR image representations with clinical information, including late fusion as a standard baseline and CC-LoRA - a novel approach based on LoRA and FiLM conditioning within selected transformer blocks of the CineMA encoder.

Both late fusion variants and all three CC-LoRA block configurations have been

implemented and evaluated across all three datasets. The results establish a clear hierarchy of fusion strategies on Sant Pau: CC-LoRA (91.03%) $\gg$ LoRA + Late Fusion (73.93%) $\approx$ LoRA image-only (74.13%) $\gg$ Full FT + Late Fusion (62.97%). The +17.10% AUC gap between CC-LoRA and LoRA + Late Fusion - using identical backbones, identical LoRA configurations, and identical clinical features - directly validates the central thesis hypothesis that in-transformer clinical conditioning is substantially more effective than output-level clinical concatenation. The unexpected low performance of Full FT + Late Fusion across all three datasets adds a further practically important finding: full backbone fine-tuning combined with output-level clinical fusion is systematically detrimental on small datasets, providing a cautionary result for multimodal medical imaging pipeline design.

Summary

All five objectives are fully achieved. The experimental results confirm that CC-LoRA successfully integrates patient-level clinical information into the CineMA foundation model through parameter-efficient FiLM-conditioned LoRA adaptation, achieving $91.03 \pm 4.17\%$ AUC on the primary Sant Pau dataset while training only 0.258% of the backbone parameters - substantially outperforming all image-only and late fusion baselines and demonstrating that in-transformer clinical conditioning is a more effective multimodal integration strategy than output-level clinical concatenation.

6.2 Result Discussion

6.2.1 Why CC-LoRA Works: The Mechanism of Clinical Conditioning

The main finding of this thesis is a +16.90% AUC and +25.39% balanced accuracy improvement over LoRA image-only on Sant Pau requires a mechanistic explanation. Understanding why CC-LoRA succeeds is as important as demonstrating that it does.

In standard LoRA fine-tuning, the backbone learns a single set of low-rank weight updates that must produce useful representations for all patients regardless of their clinical profile. The frozen pre-trained weights W encode general cardiac morphology features, and the LoRA matrices A and B adapt these features toward the target classification task. This adaptation is patient-agnostic, every patient passes through identical transformer computations, and the model must learn decision boundaries that are invariant to the heterogeneity of clinical presentations within each class.

CC-LoRA relaxes this constraint. By modulating the LoRA adapter output with patient-specific $(\Delta\gamma, \beta)$ parameters derived from clinical features, the model learns different feature transformations for different patients. The FiLM conditioning $(1 + \Delta\gamma(\mathbf{c})) \odot BAx + \beta(\mathbf{c})$ scales and shifts the LoRA output based on the clinical embedding $\mathbf{c}$, effectively allowing the transformer to compute personalised feature representations. A patient with high wall thickness and positive LGE receives different intermediate representations from a patient with normal wall thickness and no LGE, even if their images have similar raw pixel values in ambiguous cases. The convergence behavior provides evidence for this interpretation. CC-LoRA variants reach 85–90% AUC from the very first evaluation checkpoint, a level that LoRA image-only never achieves even at the end of

training. This suggests that the FiLM parameters quickly learn meaningful clinical modulations, the clinical features contain sufficient discriminative information that even early, imperfect FiLM conditioning substantially improves the classification quality. The pre-trained backbone already extracts rich cardiac features from the first epoch; the clinical conditioning immediately directs the attention of the classification pathway toward the relevant features for each specific patient.

The identity initialization ($\Delta\gamma = 0$, $\beta = 0$ at epoch 0) is critical for this rapid convergence. Because the model starts identically to plain LoRA, the first few training epochs update the LoRA matrices A and B from a stable, well-defined starting point. The FiLM parameters then gradually diverge from zero, learning to personalise the LoRA output based on the clinical signals that most reduce the training loss. This sequential learning, first LoRA adaptation, then clinical conditioning, prevents the catastrophic interference that causes the Full FT + Late Fusion collapse described in Section 6.2.2. The shared FiLM parameter design also contributes to the method's effectiveness on small datasets. A single ClinicalFiLM MLP ($27 \rightarrow 64 \rightarrow 32$, ~4K parameters) and shared FiLM projections (~154K parameters) are broadcast to all conditioned blocks. This sharing enforces a consistent clinical interpretation across the encoder hierarchy; the same clinical features modulate features at all conditioned levels. While keeping the total clinical conditioning parameter count at ~154K. This is much more favorable than per-block FiLM parameters (~590K) for a 292-patient dataset, providing strong regularization that prevents the clinical conditioning from overfitting to training fold quirks.

The fact that CC-LoRA (91.03%) surpasses XGBoost (88.00%) is particularly significant. XGBoost operates on the same 27 clinical features and achieves strong performance precisely because these features are directly tied to the pathological mechanisms. CC-LoRA exceeds this by integrating the spatial morphological information in the CMR images, wall thickness distributions, LV geometry, LGE patterns that clinical measurements can only approximate indirectly through derived indices like wall thickness measurements and LV mass. The combination of direct imaging evidence and structured clinical context produces a richer patient representation than either source alone.

6.2.2 The Full FT + Late Fusion Collapse

The unexpected result of this work was the overall poor performance of Full FT + Late Fusion across all three datasets, achieving approximately 10% lower AUC than Full FT image-only in each case, and 10.51% lower on Sant Pau specifically. This collapse is most dramatic on Sant Pau where balanced accuracy is 29.29% - effectively random for a 4-class problem - with 4 of 5 folds collapsing to HCMF prediction for every patient. The performance collapse suggests it is not a capacity issue, since Full FT image-only alone achieves 73.48% AUC on Sant Pau, but rather a failure in training dynamics in which the model converges to a degenerate local minimum. A plausible contributing factor is the interaction between the randomly initialized clinical branch and the fully unfrozen backbone during optimization: in early stages of training, the randomly initialized clinical MLP and fusion head contribute noisy and arbitrary gradients which, when backpropagated through the fully fine-tuned backbone, may push the image representations away from their optimal training trajectory before either branch has converged. However, the precise cause of this collapse is not established in this thesis - the convergence plots show that the model fails to train meaningfully across most folds, but a detailed analysis of the underlying training dynamics is left to future work.

This interference is magnified in our specific two-phase training regime. In Phase 1 of training (head warm up), we freeze the backbone, allowing it to rest while the image+clinical branches learn initial weights. The goal of head warm up is to allow stable weights to be found for the head which can then guide adaptation of the backbone. However, in the fused regime the head includes both image+clinical branches, and the weights for the clinical branch may be far from convergent at the start of phase 2 when backbone adaptation begins, such that backbone adaptation begins in a regime where the clinical conditioning weights are unstable and the backbone gradients become highly variable, leading to collapse. The CC-LoRA approach is able to avoid these issues entirely by keeping the backbone frozen for the entire duration of training and as such has no gradients flowing back into the backbone that can be perturbed by noise in the clinical branch's parameters. These parameters are optimized to become relevant in the context of pre-trained representations without interacting negatively with those representations, through an initial identity which acts as an identity parameterization to start them as no-ops. While designed to enable parameter efficiency, it proved crucial in our multimodal context to stabilize training.

The result of this analysis has important implications. It suggests that researchers implementing multimodal pipelines, particularly at the late fusion level, should be exceedingly wary of performing a full FT on the backbone given a small amount of data with clinical modalities added. We and others [1] [8] have found that frozen or LoRA-adapted backbones work better in low-data scenarios, and we now have evidence suggesting that adding the clinical modality adds even greater risk for full fine-tuning at the fusion head.

6.2.3 When Clinical Conditioning Fails: The Boundary Condition

The absence of CC-LoRA improvement on ACDC and M&Ms is not a failure of the method, it is a predictable consequence of the clinical feature sets available. Understanding this boundary condition is as important as demonstrating the method's success.

On ACDC, the three available clinical features - height, weight, and BMI - have no established relationship to the five pathology classes. It should also be noted that BMI is directly derived from height and weight and therefore does not constitute an independent third feature. DCM and HCM are defined by ventricular morphology, not anthropometrics. MINF is an ischaemic diagnosis. RV abnormality is a structural diagnosis. No configuration of these features can reliably predict which of these conditions a patient has, and a clinical-only model operating on height, weight, and BMI alone would be expected to perform at approximately chance level (AUC $\approx$ 50% for a binary OvR decomposition), as reflected in the estimated XGBoost clinical-only performance reported in Table 6.13. In this setting, the FiLM conditioning has no meaningful signal to learn from, and the parameters converge to near-zero values that effectively leave the forward pass unchanged from plain LoRA.

On M&Ms, the two available clinical features are age and sex. The situation is similar to ACDC: age carries some population-level correlation with cardiomyopathy risk - older patients have higher prevalence of acquired conditions - but at the level of an individual patient, age alone is insufficient to reliably distinguish DCM from HCM from NOR. Sex has limited independent discriminative value for this specific classification task, though it is known to carry meaningful signal in certain cardiovascular settings. Together, these two features provide insufficient clinical

context for the FiLM conditioning to learn meaningful patient-specific modulations, and further clinical information - such as that available in the Sant Pau dataset - would be necessary for clinical conditioning to provide benefit in this setting.

This analysis suggests a practical screening criterion for determining when to apply CC-LoRA: train a simple tabular model (XGBoost or logistic regression) on the available clinical features using the same train/test split as the imaging experiment. If the tabular model achieves substantially above-chance performance suggesting that the clinical features carry independent pathology-discriminative signal then CC-LoRA is likely to be beneficial. If not, image-only LoRA is the appropriate fine-tuning strategy. On Sant Pau, the XGBoost AUC of 88.00% clearly satisfies this criterion.

The negative M&Ms result also illustrates a subtlety of the FiLM conditioning mechanism. When clinical features are weak, the FiLM parameters do not simply learn to ignore the clinical signal and reduce to plain LoRA, instead, they may learn spurious correlations in the training set that do not generalize to the validation set, slightly hurting performance. This manifests as the -6.30% CC-LoRA deficit on M&Ms relative to LoRA image-only. A future improvement to CC-LoRA could include a regularization term that explicitly encourages the FiLM parameters toward zero when the clinical features provide no gradient signal, ensuring a graceful fallback to plain LoRA when clinical features are uninformative.

6.2.4 Training Stability and the Frozen Backbone

The consistent observation across all the datasets is the better training stability in frozen-backbone approaches (LoRA and CC-LoRA) compared to Full FT. It can also be observed clearly in the convergence curves: The ResNet-50 convergence curves are quite noisy; the full fine-tuning curves are more stable but noisy and LoRA/CC-LoRA gives smooth and fast converging curves.

In the Sant Pau dataset, the individual fold convergence curves of CC-LoRA are much tighter than any of the other methods. The standard deviation of fold-level AUC for CC-LoRA 2 blocks is 5.08%, as compared to 9.54% for Full FT and 6.05% for LoRA image-only. This provides significant clinical benefits: When applied in the production environment, cross-validation could be very expensive, therefore a stable method would require fewer evaluation folds for obtaining a reliable estimation of performance.

The observed stability of CC-LoRA is due to the combination of three effects. First, by freezing the backbone, the biggest contributor to the variance in gradients, the 87.7M backbone parameters are fully removed from the optimization problem. Second, the identity initialization provides a safe and stable starting curve similar to the normal LoRA trajectory for the first few training epochs. Third, the relatively low number of parameters (193 K) ensures that the model will not find an overfitting solution, even for the small Sant Pau dataset.

Chapter – 7

Conclusions and Future Work

The interpretation of cardiac MRI has long required the integration of two complementary sources of information: the morphological visual evidence in the scan, and the structured clinical knowledge about the patient. While multimodal approaches combining imaging and clinical data have been explored in the cardiovascular domain [12, 13], the question of how best to integrate these two sources within transformer-based vision foundation models -and specifically whether clinical context should condition the image feature extraction process itself rather than being appended at the output - remains largely open, as discussed in Section 1.3.7.

7.1 Conclusions

In this thesis, we demonstrate across three datasets and eight evaluated methods that bridging this gap at the architectural level by conditioning the image feature extraction process directly on patient clinical information produces substantially superior diagnostic classification compared to output-level clinical fusion. These findings suggest that in-transformer clinical conditioning is a promising multimodal integration strategy for cardiac MRI foundation models when informative patient-level clinical data are available.

We show that CC-LoRA can achieve an AUC of 91.03% on the Sant Pau Hospital dataset through the internal feature extraction of the transformer conditioned on the 27 clinical features of a patient and perform better than a state-of-the-art cardiac imaging foundation model (CineMA + LoRA, 74.13%) training fewer than 200,000 parameters. The approach is quick to converge, trains stably across small cohorts and scales well with respect to the number of properties on conditioned blocks which allow it to be deployed in clinical settings, where computational resources are limited and it must perform consistently across all pathology classes with a limited amount of training data. The failure of full fine-tuning with late fusion across all three datasets was an additional practically important finding that shows the architecture for clinical-image integration is far more important to the success or failure of training a multimodal cardiac imaging pipeline with a small dataset.

In addition to the performance in differentiating the etiologies of left ventricular hypertrophy, we introduce the principle that rather than injecting clinical information after visual reasoning has occurred it should do so as part of the process. A model with access to a patient's clinical features before it computes attention weights across the cardiac anatomy, should inherently do a better job of focusing on diagnostically informative regions than one that gains

access only after computing a clinically-agnostic image representation. CC-LoRA achieves this with a parameter-efficient, stably trained approach that adheres to the realities of the clinical domain. As cardiac imaging foundation models continue to evolve, and clinical datasets become more information-rich regarding patient information, utilizing a pre-trained vision foundation model in conjunction with patient-conditioned adapters is a step toward building future cardiac AI systems that reason about not only what they are imaging, but who they are imaging.

7.2 Limitations

Several limitations of this study must be taken into account when drawing conclusions from this work. The Sant Pau dataset contains 292 subjects derived from only one clinical center. While our work implemented a comprehensive 5-fold cross-validation for all evaluated methods, testing on a multicenter cohort would add crucial evidence to CC-LoRA's robustness to different scanner vendors, acquisition protocols and patient populations. Furthermore, AML is the smallest class in the dataset, which makes classification naturally difficult. Even then, CC-LoRA's considerable improvement over purely image-based methods on this class illustrates its promise even under such constraints, while the test on a larger AML cohort can be studied in the future work.

Another avenue of research involves model interpretability; although CC-LoRA succeeded in conditioning the image feature extraction with the clinical data, the specific way in which this conditionality impacts attention patterns and feature representation was not analyzed in this work and can be studied using attention map visualization or feature activations.

7.3 Future Work

Multi-centre validation. The most important next step is evaluating CC-LoRA on external LVH aetiology classification cohorts from different hospitals, scanner vendors, and patient populations, to assess whether the performance gains observed on the Sant Pau dataset generalize beyond a single-centre setting.

LDS-LoRA. A potential architectural extension is a lightweight dual-stream LoRA variant in which each conditioned transformer block maintains two parallel low-rank adaptation pathways: one modulated by the clinical FiLM conditioning signal as in CC-LoRA, and one image-only pathway that adapts independently of clinical context. Unlike a single LoRA pathway is either conditioned or not, the dual-stream approach would allow the model to simultaneously learn clinical-conditioned and clinical-agnostic feature updates within the same block, with the final representation being a learned combination of both streams. This could improve expressiveness in cases where some aspects of the image representation benefit from clinical conditioning while others are better adapted without it, potentially providing more robust performance across datasets with varying clinical feature richness.

Graceful fallback regularization. Adding a regularization term that penalizes large FiLM parameter magnitudes when clinical features are weak would ensure CC-LoRA degrades

gracefully to plain LoRA on datasets with uninformative clinical features, eliminating the – 6.30% M&Ms deficit.

Missing feature robustness. Evaluating CC-LoRA performance under partial clinical feature availability. For example, dropping individual features at inference time would characterize the method's robustness to missing data and identify which features are most critical for the conditioning effect.

Attention map analysis. Comparing self-attention patterns between plain LoRA and CC-LoRA forward passes using the same patient would provide mechanistic insight into how clinical conditioning modifies the regions of the image that the model attends to, strengthening the interpretability of the method.

Extension to segmentation and regression. CC-LoRA currently targets classification. Applying the same clinical conditioning mechanism to segmentation decoders and regression heads would test whether clinical context improves quantitative cardiac measurements such as ejection fraction and ventricular volume estimation.

Prognostic applications. The framework could be extended to prognostic tasks, predicting adverse cardiac events such as sudden cardiac death or heart failure hospitalization. A joint image-clinical model conditioned on baseline measurements could potentially outperform both pure imaging and pure clinical prognostic models.

Bibliography

1. Fu, Y., Bai, W., Yi, W., Manisty, C., Bhuva, A.N., Treibel, T.A., Moon, J.C., Clarkson, M.J., Davies, R.H., and Hu, Y. (2025). A versatile foundation model for cine cardiac magnetic resonance image analysis tasks. *arXiv preprint arXiv:2506.00679*.
2. Bommasani, R., Hudson, D.A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M.S., Bohg, J., Bosselut, A., Brunskill, E., et al. (2021). On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*.
3. He, K., Chen, X., Xie, S., Li, Y., Dollár, P., and Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16000–16009.
4. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. (2021). An image is worth 16×16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*.
5. Chen, X., Xie, S., and He, K. (2021). An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9640–9649.
6. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*.
7. Houshy, N., Giurigu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. (2019). Parameter-efficient transfer learning for NLP. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pp. 2790–2799.
8. Perramon-Llussà, G., et al. (2026). Med-DualLoRA: Local adaptation of foundation models for 3D cardiac MRI. *arXiv preprint arXiv:2603.10967*.
9. Zaken, E.B., Ravfogel, S., and Goldberg, Y. (2022). BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 1–9.
10. Lian, L., Zhou, D., Dai, B., and Darrell, T. (2022). Scaling and shifting your features: A new baseline for efficient model tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 35.

11. Perez, E., Strub, F., de Vries, H., Dumoulin, V., and Courville, A. (2018). FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence (AAAI)*, pp. 3942–3951.
12. Baltrusaitis, T., Ahuja, C., and Morency, L.P. (2019). Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(2), pp. 423–443.
13. Acosta, J.N., Falcone, G.J., Rajpurkar, P., and Topol, E.J. (2022). Multimodal biomedical AI. *Nature Medicine*, 28, pp. 1773–1784.
14. Huang, S.C., Shen, L., Lungren, M.P., and Yeung, S. (2021). GLoRIA: A multimodal global-local representation learning framework for label-efficient medical image recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3942–3951.
15. Chen, Z., Diao, S., Wang, B., Li, G., and Wan, X. (2022). Multi-modal masked autoencoders for medical vision-and-language pre-training. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 679–689.
16. Bernard, O., Lalande, A., Zotti, C., Cervenansky, F., Yang, X., Heng, P.A., Cetin, I., Lekadir, K., Camara, O., Ballester, M.A.G., et al. (2018). Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved? *IEEE Transactions on Medical Imaging*, 37(11), pp. 2514–2525.
17. Campello, V.M., Gkontra, P., Izquierdo, C., Martín-Isla, C., Sojoudi, A., Full, P.M., Maier-Hein, K., Zhang, Y., He, Z., Ma, J., et al. (2021). Multi-centre, multi-vendor and multi-disease cardiac segmentation: The M&Ms challenge. *IEEE Transactions on Medical Imaging*, 40(12), pp. 3543–3554.
18. Petersen, S.E., Matthews, P.M., Bamber, J., Bluemke, D.A., Francis, J.M., Friedrich, M.G., Leeson, P., Nagel, E., Plein, S., Rademakers, F.E., et al. (2017). Reference ranges for cardiac structure and function using cardiovascular magnetic resonance (CMR) in Caucasians from the UK Biobank population cohort. *Journal of Cardiovascular Magnetic Resonance*, 19(1), p. 18.
19. Radau, P., Lu, Y., Connelly, K., Paul, G., Dick, A.J., and Wright, G.A. (2009). Evaluation framework for algorithms segmenting short axis cardiac MRI. *The MIDAS Journal — Cardiac MR Left Ventricle Segmentation Challenge*.
20. Ronneberger, O., Fischer, P., and Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241.
21. Isensee, F., Jaeger, P.F., Kohl, S.A., Petersen, J., and Maier-Hein, K.H. (2021). nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature*

Methods, 18(2), pp. 203–211.

22. Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., Roth, H.R., and Xu, D. (2022). UNETR: Transformers for 3D medical image segmentation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 574–584.
23. Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L., and Zhou, Y. (2021). TransUNet: Transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306*.
24. Attia, Z.I., Kapa, S., Lopez-Jimenez, F., McKie, P.M., Ladewig, D.J., Satam, G., Pellikka, P.A., Enriquez-Sarano, M., Noseworthy, P.A., Munger, T.M., et al. (2019). Screening for cardiac contractile dysfunction using an artificial intelligence-enabled electrocardiogram. *Nature Medicine*, 25(1), pp. 70–74.
25. Zhang, J., Gajjala, S., Agrawal, P., Tison, G.H., Hallock, L.A., Beussink-Nelson, L., Lassen, M.H., Fan, E., Aras, M.A., Jordan, C., et al. (2018). Fully automated echocardiogram interpretation in clinical practice. *Circulation*, 138(16), pp. 1623–1635.
26. Simonyan, K., and Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR)*.
27. He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778.
28. Zhang, Y., Hager, P., Liu, C., Shit, S., Chen, C., Rueckert, D., and Pan, J. (2025). Towards cardiac MRI foundation models: Comprehensive visual-tabular representations for whole-heart assessment and beyond. *Medical Image Analysis*, 103756.
29. Jacob, A.J., Borgohain, I., Chitiboi, T., Sharma, P., Comaniciu, D., and Rueckert, D. (2023). Towards a vision foundation model for comprehensive assessment of cardiac MRI. *arXiv preprint arXiv:2307.03034*.
30. Ye, H., Gao, F., Meng, Y., Li, X., Zhao, J., Zhou, Y., and Shen, D. (2023). UniSeg: A prompt-driven universal segmentation model as well as a strong representation learner. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 508–518.
31. Tajbakhsh, N., Shin, J.Y., Gurudu, S.R., Hurst, R.T., Kendall, C.B., Gotway, M.B., and Liang, J. (2016). Convolutional neural networks for medical image analysis: Full training or fine tuning? *IEEE Transactions on Medical Imaging*, 35(5), pp. 1299–1312.
32. Raghu, M., Zhang, C., Kleinberg, J., and Bengio, S. (2019). Transfusion: Understanding transfer learning for medical imaging. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

33. Azizi, S., Mustafa, B., Ryan, F., Beaver, Z., Freyberg, J., Deaton, J., Loh, A., Chen, A.H., Kornblith, S., Chen, T., et al. (2021). Big self-supervised models advance medical image classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3478–3488.
34. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.
35. Wang, W., Xie, E., Li, X., Fan, D.P., Song, K., Liang, D., Lu, T., Luo, P., and Shao, L. (2021). Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 568–578.
36. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., and Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 10012–10022.
37. Mensah, G.A., Roth, G.A., and Fuster, V. (2019). The global burden of cardiovascular diseases and risk factors: 2020 and beyond. *Journal of the American College of Cardiology*, 74(20), pp. 2529–2532.
38. Elliott, P., Andersson, B., Arbustini, E., Bilinska, Z., Cecchi, F., Charron, P., Dubourg, O., Kühl, U., Maisch, B., McKenna, W.J., et al. (2008). Classification of the cardiomyopathies: A position statement from the European Society of Cardiology Working Group on Myocardial and Pericardial Diseases. *European Heart Journal*, 29(2), pp. 270–276.
39. Maron, B.J., Towbin, J.A., Thiene, G., Antzelevitch, C., Corrado, D., Arnett, D., Moss, A.J., Seidman, C.E., and Young, J.B. (2006). Contemporary definitions and classification of the cardiomyopathies. *Circulation*, 113(14), pp. 1807–1816.
40. Schulz-Menger, J., Bluemke, D.A., Bremerich, J., Flamm, S.D., Fogel, M.A., Friedrich, M.G., Kim, R.J., von Knobelsdorff-Brenkenhoff, F., Kramer, C.M., Pennell, D.J., et al. (2020). Standardized image interpretation and post processing in cardiovascular magnetic resonance: Society for Cardiovascular Magnetic Resonance (SCMR) Board of Trustees Task Force on Standardized Post Processing — 2020 update. *Journal of Cardiovascular Magnetic Resonance*, 22(1), p. 19.
41. Chen, T., and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.
42. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, pp. 2825–2830.
43. Loshchilov, I., and Hutter, F. (2019). Decoupled weight decay regularization. In

44. Ho, J., Jain, A., and Abbeel, P. (2020). Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33, pp. 6840–6851.
45. Requeima, J., Gordon, J., Bronskill, J., Nowozin, S., and Turner, R.E. (2019). Fast and flexible multi-task classification using conditional neural adaptive processes. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.
46. Huang, X., and Belongie, S. (2017). Arbitrary style transfer in real-time with adaptive instance normalization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 1501–1510.
47. Bannur, S., Hyland, S., Liu, Q., Pérez-García, F., Ilse, M., Castro, D.C., Boecking, B., Sharma, H., Bouzid, K., Schwaighofer, A., et al. (2026). MAIRA-2: Grounded radiology report generation. *arXiv preprint arXiv:2603.02464*.
48. Ye, Z., Wang, Y., Liu, Y., and Wen, B. (2025). TC-LoRA: Task-conditioned low-rank adaptation for continual learning of vision-language models. *arXiv preprint arXiv:2510.09561*.
49. Ntirogiannis, M., Papakonstantinou, S., Kakaroukas, A., and Gkontra, P. (2025). Adaptive parameter-efficient fine-tuning with feature modulation for medical image segmentation. *Scientific Reports*, 15, p. 29429.
50. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., and Wojna, Z. (2016). Rethinking the Inception architecture for computer vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2818–2826.
51. Chawla, N.V., Bowyer, K.W., Hall, L.O., and Kegelmeyer, W.P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, pp. 321–357.
52. Müller, R., Kornblith, S., and Hinton, G. (2019). When does label smoothing help? In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.
53. Buda, M., Maki, A., and Mazurowski, M.A. (2018). A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks*, 106, pp. 249–257.